Survey Research 2019-08-01

2019 K-Innovation ODA Program with Tanzania

Policy Consultation on Developing a National Technology Roadmap for Accelerated Industrialization

STEPI SCIENCE AND TECHNOLOGY POLICY INSTITUTE

2019 K-Innovation ODA Program with Tanzania

Title	Policy Consultation on Developing a National Technology Roadmap for Accelerated Industrialization
Institute	Science and Technology Policy Institute (STEPI)
Partner Country	United Republic of Tanzania
In Cooperation with	Ministry of Education, Science, and Technology (MoEST) Ministry of Agriculture (MoA) Commission of Science and Technology (COSTECH) Tanzania Industrial Research and Development Organization (TIRDO) Export Processing Zones Authority (EPZA)
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Policy Consultation on Developing a National Technology Roadmap for Accelerated Industrialization

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Executive Summary

Tanzania is working to transform its economy into an industrial one so as to attain middle income status characterized by high levels of industrialization, competitiveness, quality livelihood, and rule of law as well as having in place an educated, pro-learning society by the end of 2025. Thus, the government of Tanzania is currently focusing on transformation through the application of improved technologies and available resources under the Tanzania Development Vision 2025.

In this regard, STEPI organized the Invited Capacity Development Workshop (III) and Local Capacity Development Workshop (IV), which were held on April 1~5 in Seoul, Korea and August 20~22 in Zanzibar, Tanzania, respectively, to develop a national technology roadmap (NTRM). These workshops were the third and the fourth in a series linked to the previous two workshops held on August 21~24, and October 21~26, 2018, respectively.

The invited Tanzanian delegation composed of government officials, research officers, and relevant experts mostly came from the Ministry of Education, Science, and Technology (MoEST), Ministry of Agriculture (MoA), Ministry of Minerals (MoM), Commission for Science and Technology (COSTECH), Tanzanian Industrial Research and Development Organization (TIRDO), and Export Processing Zones Authority (EPZA).

During the workshops held in Korea and Tanzania, the National Technology Road Map (NTRM) for Tanzania focused on three priority sectors under the assistance of the Government of Korea through the Science and Technology Policy Institute (STEPI), namely CTA, Agriculture, and Mining (Mineral) sectors based on the Vision and Mission of Tanzania development plan and vision 2025. In the workshop, the participants underwent both theoretical and practical training by being divided into three groups according to sectors. As a result, the three groups ended up developing the guiding NTRM for Tanzania in three sectors with strategic products and services. The strategic products/services are cotton (CTA), cashew (agriculture), and gold (mining/mineral).

Therefore, in this report, the NTRM is presented based on what has been done so far from the 2019 workshops. The NTRM includes the vision, identification of drivers, analysis of opportunities/threats, identification of strategic products and key technologies, and future prospects. Moreover, the three Tanzanian experts investigated and analyzed the current status of three sectors in-depth and presented the outcomes in the report.

Overall, the Tanzanian delegation agreed to form a steering committee and a bigger group for experts to develop the NTRM for more strategic products/services and more industrial sectors. Doing so requires strong governmental commitment and more participants from both public and private sectors.

Parallel with NDV 2025, the national TRM will enable productive sectors to make important technological choices in attaining the desired industrial capability. NTRM development will also revive the initiatives that inspired the first President, the late Mwalimu Julius Kambarage Nyerere, to set up productive sector-based industries.

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CHAPTER 1

Project Overview

Policy Consultation on Developing a National Technology Roadmap for Accelerated Industrialization

Chapter 1. Project Overview

1. Introduction

The Government of Tanzania has expressed its commitment to structural transformation as reflected in its second Growth and Transformation Plan (Vision 2025), which has been operational since 1994. The plan has a long-term goal of becoming a middle-income country by 2025 with high annual growth rates. The Gross Domestic Product (GDP) in Tanzania expanded to 7.10 percent in the fourth quarter of 2018.

Tanzania is working to transform its economy into an industrial one so as to attain middle income by the end of 2025. The 1982/1992 decade of implementing economic stabilization and recovery programs did not produce satisfactory socio-economic results due to the absence of a guiding development philosophy; hence the need for new planning framework and national Vision. The preparation of the Tanzania Development Vision (TDV) 2025 started in 1994, and the Vision was launched in 1999. The idea of the Vision was that, by 2025, Tanzania should have gone through extraordinary economic transformation and development to achieve middle-income status characterized by high levels of industrialization, competitiveness, quality livelihood, and rule of law as well as having in place an educated, pro-learning society.

In view of the foregoing, the long-term vision of Science, Technology, and Innovation (STI) has been formulated to create a "knowledge-based economy and society through a globally competitive national STI system." There has been a rapid change in the STI landscape; therefore, as a country that needs to move toward an industry-driven economy, Tanzania needs to address issues that accelerate innovation like enhanced strategic STI Governance and Leadership, establishment and management of STI standards, intellectual property protection and management, information management, and removal of barriers for innovations that will lead to economic and technological development in a specific roadmap of the country. Considering the need to improve the internal capacity for science and technology, the five-year development plan requires the nation to develop a clear technology roadmap that will serve as a guide in making choices of important technologies for attaining the desired industrial capability.

The Ministry of Education, Science, and Technology (MoEST), which is responsible for technology development, is planning to select key strategic sectors that are presumed to have great impact on the societal and economic development of the country so as to develop a national technology roadmap for Tanzania.

Globally, Technology Roadmaps (TRMs) have become a prominent strategic approach capable of helping Governments understand market needs and make informed technology decisions. Therefore, under the National Development Vision (NDV, 2025) initiative, the Government of Tanzania is in the process of developing the first national TRM to support the growth of priority sectors for them to reach their potentials through industrialization. The planning exercise started in 2018 following a number of TRM capacity development workshops held by STEPI in Dar es Salaam, Zanzibar (Tanzania) and Seoul (Korea). Government officials from various sectors in Tanzania were the focal point of this exercise under the auspices of the Korean Government through STEPI and MoEST. Following the 2018 K-Innovation report on Tanzania published last year, this report focuses on the activities done in 2019.

2. Objectives

The main objective is to develop a national technology roadmap to guide the industrialization thrust in Tanzania. In order to accomplish this goal, the assembled group of Korean and Tanzania experts is expected to:

1) Conduct a scoping study to collect important data for the development of a technology roadmap
2) Practice the methodology of technology roadmapping
3) Select the priority industrial sectors and strategic product/service
4) Develop the draft of a national technology roadmap

3. Project Framework

The project aims to provide policy consultation to the Tanzanian Government. STEPI is expected to provide a team of experts working jointly with a team of local experts composed of key stakeholders from government ministries, academe, public organizations, and private sector entities in order to develop the technology roadmap.

The research team holds a capacity-development workshop in both Korea and Tanzania. The workshop is a theoretical and practical intensive course designed for Tanzanian government officials and qualified practitioners looking to broaden their knowledge in the areas of technology roadmapping by participating in the practical training. The workshop is composed of three methods: (1) lectures, (2) case studies, and (3) team projects (group presentations).

Overall, the roadmap will encompass important elements that will make its implementation easy and affordable. The Ministry expects the envisaged roadmap to be an important tool in achieving its five-year development plan and beyond.

4. Project Team

4.1 Research Team

The research team is composed of full-time experts – including a team leader (Dr. Chi Ung Song) and a STEPI researcher (Ms. So Young Cho) – and 2 external experts (Dr. Hyun Yim from KISTEP and Prof. Johng-Ihl Lee from the State University of New York, Korea (SUNY Korea)).

[Table 1-1] Research Team

Name	Organization	Position & Area of Expertise	Contact Information
Chi Ung Song	STEPI	Vice President / PM	cusong@stepi.re.kr
So Young Cho		Researcher/ Project Coordinator	sycho07@stepi.re.kr
Hyun Yim	KISTEP	Senior Research Fellow/ NTRM Methodology	hyim@kistep.re.kr
Johng-Ihl Lee	SUNY Korea	Professor / NTRM Methodology	anothermile@sunykorea.ac.kr

4.2 Local Research Team

MoEST is responsible for this project in Tanzania. The local research team works in cooperation with the STEPI research team.

[Table 1-2] Local Research Team

Name	Organization	Position	Contact Information
Maulilio John Kipanyula	Ministry of Education, Science, and Technology	Director for STI	maulilio.kipanyula@moe.go.tz
Adam Beni Swebe Mwakalobo	Ministry of Education, Science, and Technology	Director of Higher Education	adamwakalobo@yahoo.com
Rogers Alfayo Msuya	Ministry of Education, Science, and Technology	Senior Research Officer	rogers.msuya@moe.go.tz
Upendo Marko Mndeme	Ministry of Agriculture	Senior Agriculture Officer	upendo.66@gmail.com
Julius Elias Daud	Tanzania Industrial Research and Development Organisation	Research Officer	Julius.zeanimo@gmail.com

5. Project Main Activities

5.1 Activity 1: Project Proposal Review

From January to February 2019, the STEPI team and the MoEST team discussed the schedule for 2019. The plan for the project was decided and shared with the Tanzanian delegation.

5.2 Activity 2: Invited Capacity Development Workshop (III)

On April 1~5, 2019, the Tanzanian delegation of ten (10) officers from the Ministry of Education, Science, and Technology (MoEST), Ministry of Agriculture (MoA), Mineral Resources Institute, Tanzania Industrial Research and Development Organisation (TIRDO), Tanzanian Commission for Science and Technology (COSTECH), and Export Processing Zones Authority (EPZA) participated in the Capacity Development Workshop in Korea. In continuation of the previous year’s two workshops, this workshop acted as a platform for Tanzanian officials and qualified practitioners to deepen their knowledge in technology roadmapping. Moreover, they had a chance to visit organizations and places relevant to the NTRM project including KISTorium in KIST and IFEZ center, Smart City operation center, Incheon City history museum, and global campus in Songdo International City.

5.3 Activity 3: Local Capacity Development Workshop (IV)

On August 20~22, 2019, the STEPI team visited Zanzibar and conducted a “Local Capacity Development Workshop (IV).” The invited Tanzanian delegation, composed of government officials, research officers, and relevant experts, was mostly from MoEST, MoA, Ministry of Minerals (MoM), COSTECH, TIRDO, and EPZA. During the Invited Capacity Development Workshop (III) in Korea, the Tanzanian participants selected three sectors in developing the NTRM: mining (minerals/gold); cotton, textile, and apparels (cloth); and agriculture (cashew). Therefore, for this workshop, they reviewed and revised the strategic technology roadmap they developed during the previous workshop.

5.4 Activity 4: Final Report

The research team has written a final report on “Developing a National Technology Roadmap for Accelerated Industrialization” based on the data and information collected during the two workshops in Korea and Tanzania. Moreover, Dr. Rogers Alfayo Msuya of MoEST, Ms. Upendo Marko Mndeme of MoA, and Dr. Julius Elias Daud of TIRDO have written chapters on the current status of Tanzania’s three sectors (CTA, Cashew, and Mining (Minerals)) and the NTRM Tutorial Workshop for the final report.

6. Project Schedule

The entire schedule of the project is as follows:

[Table 1-3] 2019 Project Schedule

Activities	Jan. '19	Feb. '19	Mar. '19	Apr. '19	May '19	Jun. '19	Jul. '19	Aug. '19	Sep. '19	Oct. '19	Nov. '19	Dec. '19
Project Planning												
Project Proposal Review												
Implementation Plan												
Preparation of Invited Capacity Development Workshop (III)												
Invited Capacity Development Workshop (III)												
Assignment on NTRM for Tanzanian Participants												
Interim Report												
Preparation of Local Capacity Development Workshop (IV)												
Local Capacity Development Workshop (IV)												
Research on Three Industries (CTA, Minerals, and Cashew)												
Final Report												

7. Project Outputs

STEPI provided the following deliverables:

- **Deliverable 1**: Project Proposal (January-February 2019)
- **Deliverable 2**: Tentative program agenda for the invited capacity development workshop (February-March 2019)
- **Deliverable 3**: Presentations and Training Materials for the Invited Capacity Development Workshop held in Korea (April 2019)
- **Deliverable 4**: Interim Report (June 2019)
- **Deliverable 5**: Program Agenda for the Local Capacity Development Workshop (July 2019)
- **Deliverable 6**: Presentations and Training Materials for the Capacity Development Workshop held in Tanzania (August 2019)
- **Deliverable 7**: Final Report (October-December 2019)

CHAPTER 2

Current Status of the Cotton, Textile, and Apparel (CTA) Industry in Tanzania

Policy Consultation on Developing a National Technology Roadmap for Accelerated Industrialization

Chapter 2. Current Status of the Cotton, Textile, and Apparel (CTA) Industry in Tanzania

1. Introduction

With Tanzania on the verge of industrialization, the Cotton, Textile, and Apparel (CTA) sector is defined as one of the priority sectors in Tanzania toward becoming an industrialized economy. The sector plays a major role in the industrialization process as it is closely linked with the agricultural sector in terms of raw material supply (backward integration), and it taps into the huge global textiles and garment market (forward integration). Moreover, the CTA sector has different actors along the cotton production value chain (start from cotton growers to consumers or retailers of cotton lint and apparel) as well as immediate and supporting actors' value chain. Therefore, the CTA-TRM development will take advantage of the country's cotton wealth and address the shortage of cotton-based products locally and internationally by modernizing and upgrading related technologies.

2. Overview of the Cotton, Textile, and Apparel (CTA) Sector

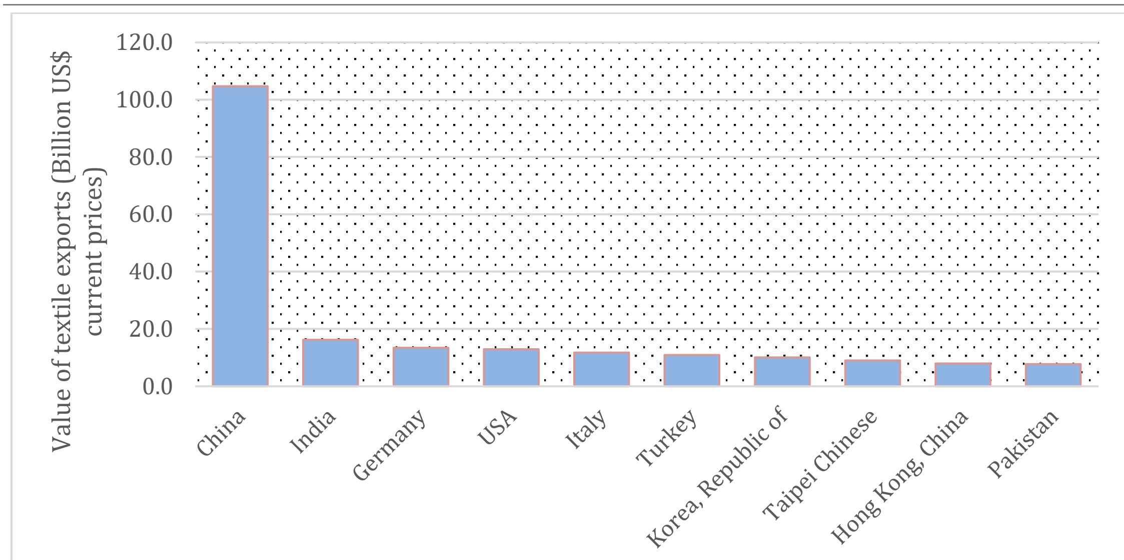
2.1 Global Trends in Cotton, Textile, and Apparel (CTA) sector

Global cotton production for the last two decades has shown an increasing trend, though with fluctuations from year to year. For instance, the total cotton production for 2018 was 26 million tons, with consumption accounting for 27.8 million tons (ICAC, 2018). The top ten cotton-producing countries worldwide are: India, China, USA, Brazil, Pakistan, Australia, Turkey, Uzbekistan, Turkmenistan, and Burkina Faso. Global exports of cotton were mostly bound for the USA with $53.7 billion in 2017, dominated as well by India, Brazil, Australia, Uzbekistan, Mali, Burkina Faso, Benin, and Turkmenistan. On the other hand, the top importing countries include, but are not limited to, Pakistan, Bangladesh, Turkey, Vietnam, China, Indonesia, Mexico, India, and Thailand. Moreover, the major producers of cotton lint consume much of what they produce, leaving only about 25 ~ 30% for the export market (ICAC, 2017).

In Africa, total production accounts for 7 million tons of cotton lint annually, which is equivalent to 6% of the global cotton production (FAO, 2017). The area under production has remained roughly constant, whereas yields, total production, and consumption have been on the rise. In some years, production surpasses consumption, whereas in others consumption is higher than production, with stocks smoothening the production and consumption cycles.

The global textile mills market was valued at $667.5 billion in 2015 with fabrics constituting 83.1% and yarns accounting for 16.9%. Asia-Pacific accounted for 54.6% of the global textile mills market in 2015, with Europe constituting 20.6% of the market. China was the largest textile exporter (Figure 2-1), accounting for over 37% of the total textile exports. Combined, the top ten exporters account for 70% of the total textile exports.

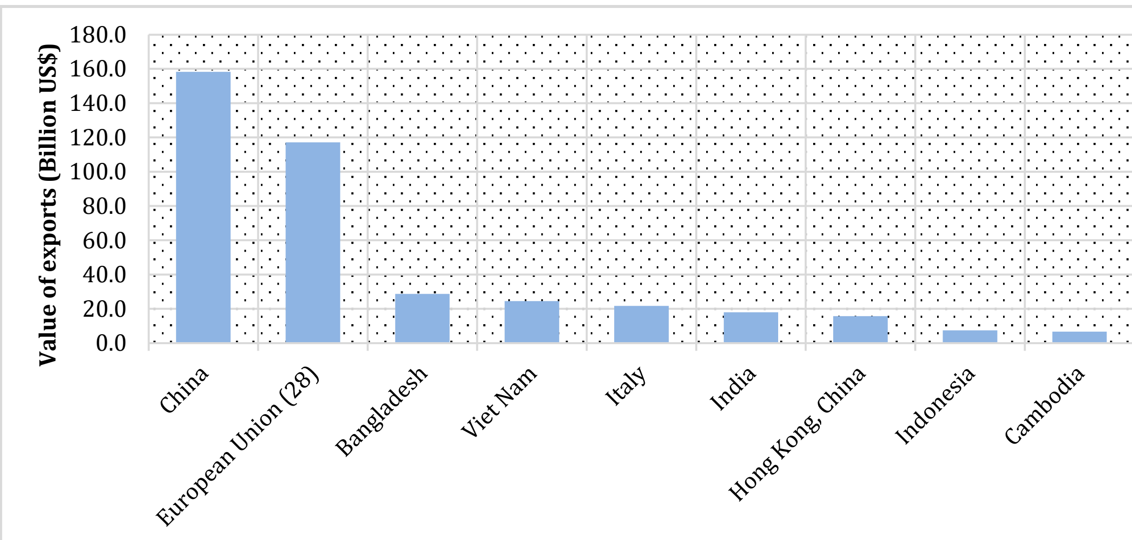
[Figure 2-1] Top 10 Exporters of Textiles in 2017



Source: WTO database

The apparel industry supply chain has a global scope and consists of millions of small, medium, and large manufacturers in every region of the world. It includes manufacturing or knitting of fabric, cutting and sewing of apparel and trim (such as buttons, zips, elastics) and their sale. The industry is composed of manufacturers that purchase fabrics and those that manufacture fabrics themselves and have fixed operational facilities. The sector employs over 60 million workers globally, creating opportunities for unskilled workers especially women at 68%. Based on WTO trade statistics, global exports in apparels (clothing) stood at U$454.5bn in 2017, with China, the European Union (EU28), Bangladesh, and Vietnam in that order as the world's top four largest exporters accounting for about 76% of the market share (Figure 2-2).

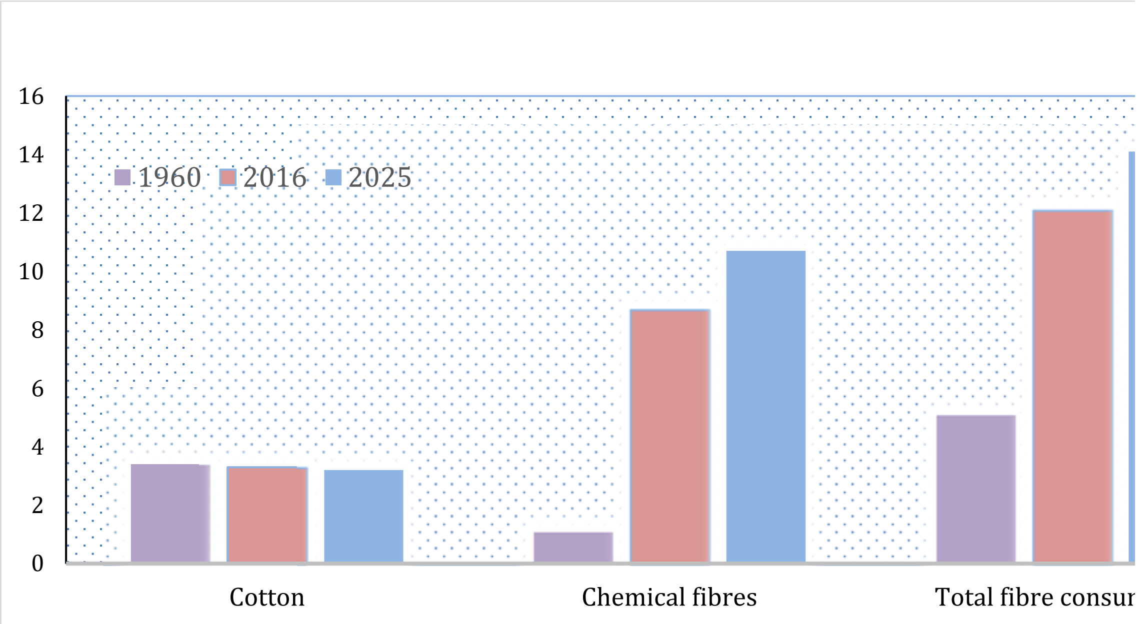
[Figure 2-2] Top 10 Exporters of Apparel in 2017



Source: WTO database

The global per capita consumption of textile fabric in 2016 was 12.1 kg, with developed countries consuming more than developing countries (Figure 2-3). Consumption by developing countries and Africa was estimated at 4.9 Kilograms and 2.5 Kilograms, respectively.

[Figure 2-3] Global per Capita Textile Consumption (Kg)



Source: ICA (2018)

2.2 EAC Trends for the Cotton, Textile, and Apparel (CTA) Sector

On average, EAC produces about 290,000 metric tons of seed cotton per year (Table 2-1). Production has been on the decline due to many challenges, the mains ones being low prices, poor agronomic practices, and declining acreage under tillage. Production is mainly by small-scale producers under rain-fed conditions and yields are low, with outputs hardly adequate to meet domestic demand by ginneries. The region produces about 85,480 metric tons of lint per year (Table 2-2), with most of it (about 85 percent) exported out of the region, whereas intra-regional trade of the same accounts for about 10 percent of the total region's cotton lint exports.

[Table 2-1] Trends in Seed Cotton Production (MT)

Country/ Year	2013	2014	2015	2016	2017	Annual Average (2013-17)	Share (%)
Burundi	2,238	2,357	2,299	2,010	1,835	2,148	1
Kenya	12,116	13,472	15,726	15,800	11,850	13,793	5
Tanzania	**352,161**	**245,831**	**202,312**	**149,767**	**132,961**	**214,450**	**74**
Uganda	41,421	49,206	48,763	66,543	89,233	59,013	20
Rwanda	-	-	-	-	-	-	
S. Sudan	-	-	-	-	-	-	-
Total	**393,237**	**293,405**	**258,088**	**101,654**	**258,667**	**288,808**	**100**

Source: Country Studies

[Table 2-2] Trends in Cotton Lint Production in EAC (2013-2017, MT)

Year	2013	2014	2015	2016	2017	Annual Average (2013-2017)	Share (%)
Uganda	18,571	14,594	17,275	20,339	27,947	19,745	20
Tanzania	**118,039**	**82,399**	**67,812**	**50,200**	**40,953**	**71,881**	**74**
Kenya	4,039	4,491	5,243	5,267	3,950	4,598	5
Rwanda	-	-	-	-	-	-	0
Burundi	946	1,011	968	854	779	912	1
S. Sudan	-	-	-	-	-	-	0
EAC	**120,717**	**88,060**	**74,087**	**67,412**	**77,122**	**85,480**	**100**

*1 bale=185 kilograms

Source: Country reports, 2018 and Source: COMTRADE database

Yields in the region are also low compared to yields in other major production areas (Table 3). In Kenya for example, only 21,000 ha is being utilized out of 400,000ha, which is less than 6 percent of potential land.

[Table 2-3] Area under Production and Yield

Country	Area under Cotton Production in 2017 (Ha)	Yields (Kg/Ha)
Burundi	3250	738
Kenya	21,000	500
Rwanda	-	-
Tanzania	**321,893**	**750**
Uganda	66,517	1,187
South Sudan	-	-
Bangladesh	17,806	3,516
Burkina Faso	844,895	999
Egypt	91,000	3,296
China	3,625,385	4,730
India	12,200,000	1,519
World	**32,979,140**	**2,255**

Source: FAO Statistics and country reports

Yields are way below the world average yields of 2,255 kg per hectare. The average yield of the commercial varieties HART 89M and KSA 81M in EAC is a far cry from their potential, which is 2500 kg/ha. Such wide yield difference is due to poor agronomic practices, low soil fertility, rainfall patterns, diseases, and pests. Notably, Tanzania grows organic cotton. Other conventional varieties have potential yield of over 4000 Kg seed cotton per hectare but need irrigation.

Cotton lint prices in the region closely reflect the international prices (taking into account the transportation costs). Notably, Tanzania's cotton lint prices are higher than those for the other EAC Partner States, probably because Tanzania grows high-quality organic seed cotton (Table 2- 4).

[Table 2-4] Recent Trends in Farm Gate and Lint Prices in Kenya and Uganda

	Seed Cotton (US$/Kg)			Lint Price (US$/Kg			
	Kenya	Uganda	Tanzania	Kenya	Uganda	Tanzania	World (Cat A)
2013	0.42	0.44	0.28	1.6	1.7	1.7	1.83
2014	0.42	0.37	0.30	1.3	1.3	2.0	1.83
2015	0.42	0.42	0.32	1.4	1.3	1.7	1.55
2016	0.42	0.44	0.34	1.4	1.5	1.6	1.64
2017	0.46	0.42	0.52	1.6	1.4	1.7	1.84

Source: Country reports

For the EAC region to achieve self-sufficiency for cotton lint while also exporting up to 40%, cotton lint production should increase to 320,000MT per year by 2019 and subsequently increase to 420,000MT by 2029. Based on the 2017 production, the region faces a deficit of about 234,520MT assuming domestic industries capture 50% of the market. Modernization of machinery, renewal of buildings, and, in some cases, power and water connection are needed to improve the performance of the ginneries. The trends in cotton lint production are shown in Table 2-5.

[Table 2-5] Trends in Cotton Lint Production in EAC (2013-2017, MT)

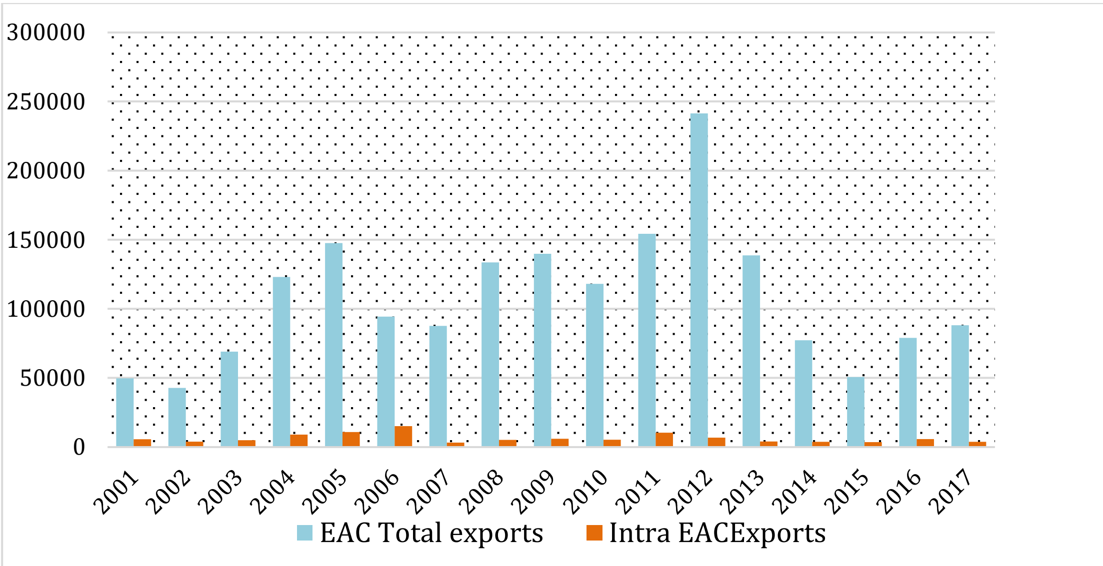
Year	2013	2014	2015	2016	2017	Annual Average (2013-2017)	Share (%)
Uganda	18,571	14,594	17,275	20,339	27,947	19,745	20
Tanzania	118,039	82,399	67,812	50,200	40,953	71,881	74
Kenya	4,039	4,491	5,243	5,267	3,950	4,598	5
Rwanda	-	-	-	-	-	-	0
Burundi	946	1,011	968	854	779	912	1
S. Sudan	-	-	-	-	-	-	0
EAC	**120,717**	**88,060**	**74,087**	**67,412**	**77,122**	**85,480**	**100**

**1 bale=185 kilograms*

Source: Country reports

Most of the cotton lint (fiber) produced in the region is exported out of EAC, with that traded intra-EAC accounting for less than 10 percent of the total cotton production (Figure 2-4). There is limited consumption of cotton lint in the EAC region due to weak capacities in the textile mills and subdued demand in the apparel's segments.

[Figure 2-4] Intra- and Extra-EAC Exports of Cotton Fiber



Source: COMTRADE database

Textile mills and garment manufacturers range from micro to large enterprises, with the sector mainly dominated by imports. The main production mills are based in EPZ and/or SEZs, and they produce for export markets. The integrated textile and garment mills in the region do not operate at full capacity due to poor and outdated technology, which adversely affects their efficiency and the quality of textiles produced (Table 2-6).

[Table 2-6] Textile and Garment Manufacturing and Capacities in EAC

Country	Number of Operational Integrated Textile Mills	Mills in Spinning Fabrics	Mills in Weaving Fabrics	Mills in Fabrics Knitting & Printing	Madeups Companies	Textile/ Garment Mills' Average Capacity Utilization (%)
Burundi	1	1	1	1	1	50%
Kenya	52	8	17	17	22 for EPZ 82 for local	50-60% for mills 100% for apparel
Rwanda	1	1	1	0	2 operational 2 planned to open 2 not operational	10%
S. Sudan	-	-	-	-	-	-
Tanzania	**5**	**10**	**6**	**5**	**6 operational** **2 planned to open**	**3 at 100%** **7 at 40-60%**
Uganda	2	2	1	2	2	61-65%

Source: Country consultations, 2017

EAC trade in textiles, apparels, and madeups accounted for an average of 3.7% of the total EAC trade for the period 2013-2017. Exports of CTA in EAC averaged at US$ 680 million per year, accounting for about 4.9% of the total region's exports. On other hand, imports averaged at US$ 2.7 billion per year and accounted for about 3.3 of the total region's imports. Trends in exports of textiles and madeups by EAC Partner States for the period 2013-2017 show a decrease from US$ 719 million in 2013 to about 645 million in 2017. Cotton textile exports fell from US$ 153.25 million in 2013 to US$ 106.22 million in 2017. Other textile exports increased marginally from US$ 97.7 million in 2013 to US$ 115.03 million in 2017, with madeup exports also increasing marginally from about US$ 420 million in 2013 to about US$ 502 million in 2017.

Imports of textiles and madeups by EAC Partner States grew slightly from US$ 2.1 billion in 2013 to US$ 2.5 billion in 2017. Cotton textile imports decreased from US$ 245 million to 70 million; imports of other textiles grew from US$ 78 to 86 million, whereas madeup imports increased from US$ 404 million to US$ 491 million. Madeups account for the largest share, followed by other textiles and cotton textiles as shown in Table 2-7.

[Table 2-7] Structure of Cotton Textile Exports and Imports (2017)

Product	Exports		Imports	
	Value in US'000	Proportion (%)	Value in US'000	Proportion (%)
Cotton, not carded or combed	41069.9	42.9	1247.6	2.2
Cotton waste (incl. yarn waste)	619.4	0.6	16.0	0.0
Cotton, carded or combed	36913.1	38.6	3.0	0.0
Cotton sewing thread	197.9	0.2	360.4	0.6
Cotton yarn, with >=85% cotton, not put up for retail sale	5534.3	5.8	158.3	0.3
Cotton yarn (excl. sewing), put up	9266.2	9.7	101.0	0.2
Woven fabrics of cotton, with >=85% cotton weighing not more than 200g/m^2	1469.4	1.5	42777.6	76.5
Woven fabrics of cotton, with >=85% cotton, weighing more than 200g/m^2	140.5	0.1	7976.1	14.3
Woven cotton fabrics with man-made fibers with <85% cotton, weighing more than 200 g/m^2	4.0	0.0	2450.7	4.4
Woven cotton fabrics with man-made fibers with <85% cotton, weighing not more than 200 g/m^2	469.1	0.5	357.6	0.6
Other woven fabrics of cotton, nes	15.1	0.0	500.2	0.9

Source: Computed from the COMTRADE database

India, Singapore, Indonesia, and China were the top export markets in 2017, accounting for about 58% of the export market for EAC (Table 2-8). China and India are the top import markets, with China alone accounting for 84 percent of EAC cotton textiles import.

[Table 2-8] Exports and Import Markets for EAC Cotton Textile (2017)

Exports			Imports		
Market	Value	Market Share (%)	Market	Value	Market Share (%)
India	17606.2	18.4	China	47153	84
Singapore	17393.9	18.2	India	2585	5
Indonesia	11171.4	11.7	Uganda	1381	2
China	9306.6	9.7	Pakistan	1243	2
Bangladesh	7588.6	7.9	Kenya	935	2
Switzerland	5963.0	6.2	United Arab Emirates	563	1
Kenya	5236.4	5.5	Hong Kong, China	455	1
United Kingdom	3509.9	3.7	Tanzania	396	1
Vietnam	3112.6	3.3	Congo, Dem. Rep.	260	0
Other countries	14810.5	15.5	Other countries	976	2

Source: Computed from the COMTRADE database

Imported apparels and madeups for the same period were valued at US$1.6 billion per year on average. Madeup knitted or crocheted articles accounted for about 30%, whereas madeups not knitted or crocheted and other madeups (clothing and clothing accessories; blankets and traveling rugs; bed linen, table linen, toilet linen, and kitchen linen; furnishing articles except carpets; and footwear and headgear of any material other than asbestos) each constituted 35%. On average, the region imported worn clothing and other worn articles worth US$ 268 million per year, accounting for about 16% of imports of all apparels and madeups. Top import markets in 2017 were China, Canada, U.S, and United Kingdom, accounting for 24, 10.3, 10.1, and 9.1 percent, respectively, of the worn clothes and other worn articles import market. The U.S constituted 78% of the export markets in 2017. Interestingly, EAC Partner States are among the top ten export market destinations for worn/used clothes, with Kenya, Uganda, Burundi, and Tanzania combined accounting for about 8.3% share of the region’s export market. China and India top the list of the import markets (Table 2-9).

[Table 2-9] EAC Export and Import Markets for Madeups (2017)

Exports			Imports		
Market	US$ '000	Market Share (%)	Market	US$ '000	Market Share (%)
United States	392584.4	78.1	China	986443.1	64.4
Kenya	13787.1	2.7	India	185595.3	12.1
Uganda	12678.5	2.5	EU	72435.1	4.7
Burundi	11345.6	2.3	Tanzania	39805.1	2.6
European Union	10926.6	2.2	Canada	26129.4	1.7
Zambia	9437.7	1.9	Pakistan	24516.7	1.6
South Africa	8549.2	1.7	United States	23176.1	1.5
Germany	7265.2	1.4	Korea, Rep.	20585.9	1.3
Canada	6364.2	1.3	UK	19446.1	1.3
Tanzania	4218.6	0.8	Poland	17176.7	1.1

Source: UNCOMTRADE database

The combined total intra-EAC textiles and madeups exports grew from US$ 95 million in 2013 to US$ 222 million in 2015, only to drop to US $ 91 million in 2017 (Table 10). Intra-EAC exports of textiles and madeups account for about 7.3 percent of the total regional exports of the same products during the period 2013-2017. This suggests that EAC exports most of her textiles and madeups to non-EAC countries.

[Table 2-10] Intra-EAC textiles and madeup exports (2013- 2017; US$ ‘000)

Exporting Country	US ‘000						Share (%)
	2013	2014	2015	2016	2017	Total (2013-2017)	
Burundi	409.3	94.1	32.0	67.1	130.6	733.1	0.3
Kenya	17479.3	22710.8	22743.5	21316.5	17067.1	101317.2	39.4
Rwanda	677.9	969.7	437.9	449.1	426.3	2960.9	1.2
S. Sudan	0	0.1	0.2	4.2	27.1	31.6	0.0
Tanzania	38722.3	17945.9	7443.5	14222.7	40899.2	119233.7	46.4
Uganda	5096.1	3430.5	3398.3	5061.2	5593.3	22579.3	8.8

Source: UNCOMTRADE database

Intra-EAC imports of textiles and madeups account for 4.3% of the region's total imports of the same products for the period 2013-2017. Intra-regional imports decreased from US$ 94 million in 2013 to US$ 34 million in 2015, thereafter increasing to US$ 64 million in 2017 (Table 2-11).

[Table 2-11] Intra-EAC Imports of Textiles and Madeups

Importing Country	Value of Imports (US$ '000)					
	2013	2014	2015	2016	2017	Share (%)
Burundi	5345.6	5212.6	1622.0	2664.1	16339.8	5.3
Kenya	33217.0	107874.0	212822.2	29429.0	32003.7	70.8
Rwanda	8462.0	2951.6	2247.1	2790.9	6948.4	4.0
S. Sudan	4327.5	1349.2	480.8	1105.6	5279.7	2.1
Tanzania	15645.5	239.6	34.9	2246.9	8341.7	4.5
Uganda	28970.8	13883.1	4903.6	7771.5	22447.7	13.3

Source: UNCOMTRADE database

2.3 Trend of the CTA Sector in Tanzania

2.3.1 Cotton Production

Tanzania produced about 222,725 metric tons of seed cotton in the year 2018/19 (Table 2-12) with yield of about 750kg/ha of seed cotton or 250kg/ha of lint cotton. The cotton-cultivated land area takes up 1.5 hectares of land, which is considered smallholder farming, or commonly referred to as peasant farming. Smallholder farmers use limited amount of inputs including seeds and pesticides, with majority of them using simple hoe (80%), oxen hoes (52%), and tractors 44.2%) as common technologies for land tilling.

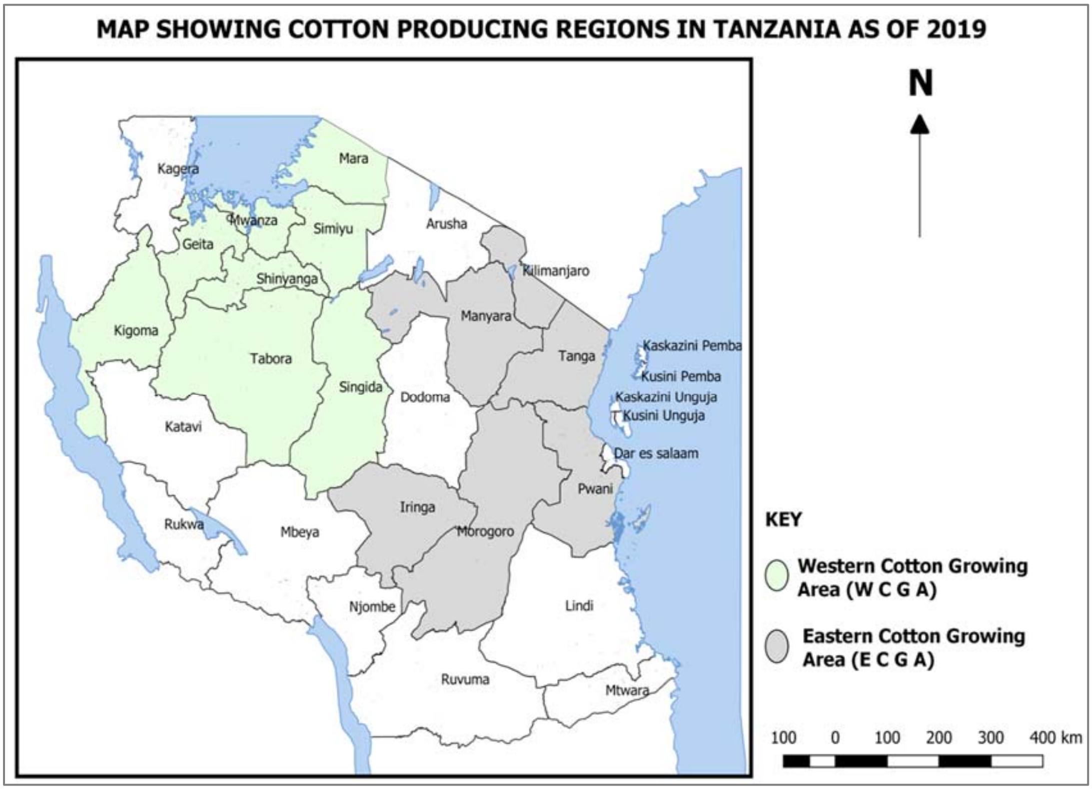
[Table 2-12] Production Trend of Seed Cotton in Tanzania for the Last 5 Years

REGIONS	SEED COTTON PRODUCTION IN TONS				
	WESTERN ZONE				
	2014/15	2015/16	2016/17	2017/18	2018/19
Sinyanga	31,398	19,804	18,062	14,776	23,156
Simiyu	100,603	69,898	65,515	71,347	112,915
Mwanza	17,538	11,558	4,985	11,739	17,377
Geita	16,665	18,767	11,993	12,866	19,565
Mara	14,948	8,863	7,557	6,691	22,624
Kagera	154	506	658	984	1,842
Tabora	19,139	17,439	11,797	12,327	20,213
Kigoma	54	139	86	52	316
Singida	997	2,498	783	1,713	2,536
Katavi	0	0	0	0	519
TOTAL	**201,496**	**149,472**	**121,436**	**132,496**	**221,065**
%	**99.6**	**99.7**	**99.2**	**99.7**	**99.3**
	EASTERN ZONE				
Manyara	424	200	495	157	571
Morogoro	201	118	251	105	784
Kilimanjaro	79	66	139	16	117
Pwani	9	12	28	67	38
Tanga	90	44	11	40	104
Iringa	13	1	2	0	10
Dodoma	0	0	0	49	37
TOTAL	**816**	**441**	**926**	**433**	**1,661**
%	**0.4**	**0.3**	**0.8**	**0.3**	**0.7**
GRAND TOTAL	**202,312**	**149,913**	**112,362**	**132,929**	**222,725**

Source: UNCOMTRADE database

Majority of Tanzania's cotton (99%) is grown in the Western Cotton Growing Area (WCGA), and the remaining 1%, in the Eastern Cotton Growing Area (ECGA). WCGA includes Simiyu, Shinyanga, Mwanza, Geita, Mara, Tabora, Kagera, Singida, and Kigoma whereas ECGA consists of Pwani, Morogoro, Iringa, Tanga, Manyara, and Kilimanjaro (Figure 2-5). As with most countries in Africa, the bulk of Tanzania's cotton production (in excess of 70%) is exported as lint (Table 2-13), which is a key concern of the government at present.

[Figure 2-5] Tanzania's Cotton Production Zones



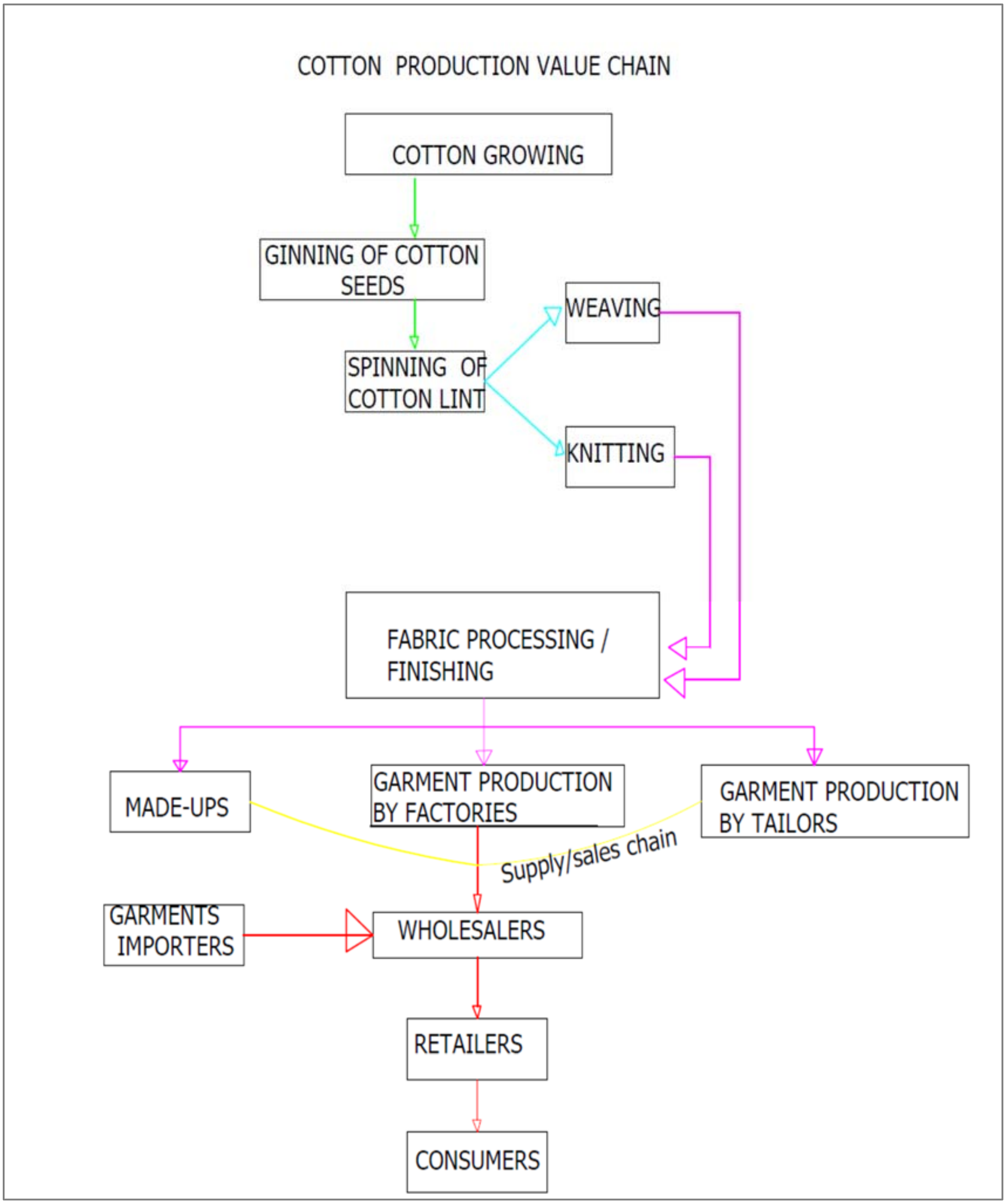
[Table 2-13] Lint Exported by Country of Origin (in bales, value in USD)

DESTINATION	2014/2015		2015/2016		2016/2017		2017/2018		2018/2019	
	BALES	VALUE IN US$	BALES	VALUE IN US$	BALES	VALUE IN US$	BALES	VALUE IN US$	BALES	VALUE IN US$
BANGLADESH	15,467	4,681,151	28,528	7,307,619	18,227	5,743,416	26,100	8,822,455	49,492	17,960,343
BELGIUM	-	-	-	-	-	-	-	-	500	118,206
CHINA	10,000	3,302,731	260	74,615	-	-	500	62,500	-	-
CZECK REPUBLIC	-	-	-	-	-	-	-	-	-	-
DJIBOUT	-	-	-	-	-	-	-	-	-	-
ERITREA	100	30,423	100	26,881	123	36,616	300	99,758	-	-
FRANCE	100	37,821	-	-	-	-	700	264,545	1,390	393,529
GERMANY	2,300	840,787	3,300	1,108,810	200	68,895	1,500	612,882	1,587	1,009,488
INDIA	16,445	5,555,565	5,881	2,159,203	34,321	10,921,370	11,706	4,611,285	66,469	22,771,380
INDONESIA	12,760	3,685,232	12,877	3,193,367	13,835	4,406,808	11,670	3,995,682	28,387	9,277,792
ITALY	-	-	-	-	-	-	-	-	-	-
JAPAN	-	-	100	40,436	-	-	49	21,319	150	37,611
KENYA	4,500	1,610,199	4,400	1,215,616	-	-	1,592	617,812	-	-
MALAYSIA	27,398	9,362,910	1,375	357,738	13,260	4,106,479	30,499	10,290,578	4,328	1,613,567
MAURITIUS	4,148	1,218,236	5,015	1,414,803	4,550	1,563,802	4,000	1,406,182	4,500	1,677,409
MOZAMBIQUE	-	-	600	185,451	-	-	3,600	1,287,496	1,000	417,106
MOROCCO	-	-	-	-	-	-	-	-	400	141,302
NETHERLAND	200	78,348	95	35,648	-	-	-	-	100	41,880
PAKISTAN	-	-	2,000	574,083	-	-	500	170,403	-	-
PORTUGAL	243	86,189	1,700	474,198	1,000	271,643	5,255	1,884,253	5,824	1,432,867
SINGAPORE	19,244	6,332,268	-	-	1,000	315,846	1,000	359,041		
SWITZERLAND	-	-	-	-	-	-	1,000	358,912	4,399	1,620,480
SPAIN	1,500	390,398	-	-	-	-	200	72,015		
TAIWAN	1,000	241,802	3,400	998,363	500	170,902	1,628	579,935	806	291,588
THAILAND	9,600	3,139,083	5,585	1,417,718	5,280	1,694,018	5,400	1,640,882	28,212	9,598,097
TURKEY	958	277,061	7,712	2,429,028	1,026	383,216	3,700	1,323,921	805	399,435
UAE	3,670	1,149,404	900	238,714	-	-	1,000	363,564		
VIETINAM	20,500	6,510,747	38,992	9,621,441	11,473	3,581,358	4,946	1,729,288	1,790	760,338
UGANDA	-	-	-	-	-	-	-	-	1,475	626,535.56
TOTAL	150,133	48,530,353	120,500	36,544,957	104,795	33,264,368	116,845	40,574,707	201,614	70,188,953

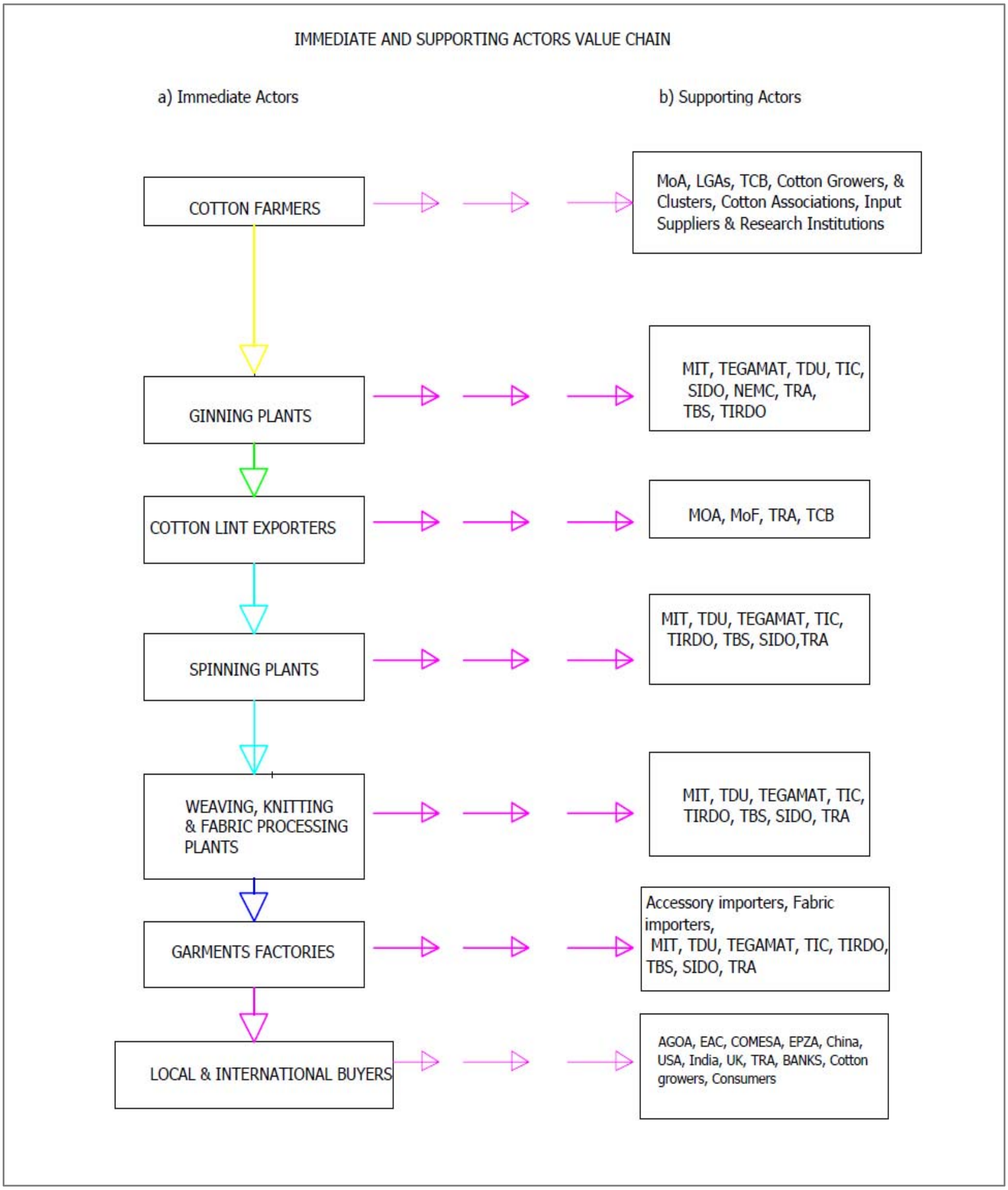
2.3.2 Current Value Chain of the CTA Sector in Tanzania

The Tanzanian Government has defined the textile sector in its National Development Vision 2025 as one of the priority sectors toward becoming an industrialized economy. The sector is playing a major role in the industrialization process as it is closely linked with the agricultural sector in terms of raw material supply (backward integration) and it taps into the huge global textiles and garment market (forward integration). In that respect, the sector has different actors along the cotton production value chain (start from cotton growers to consumers or retailers of cotton lint and apparel) as well as immediate and supporting actors' value chain (Figures 2-6 & 2-7).

[Figure 2-6] Cotton Production Value Chain



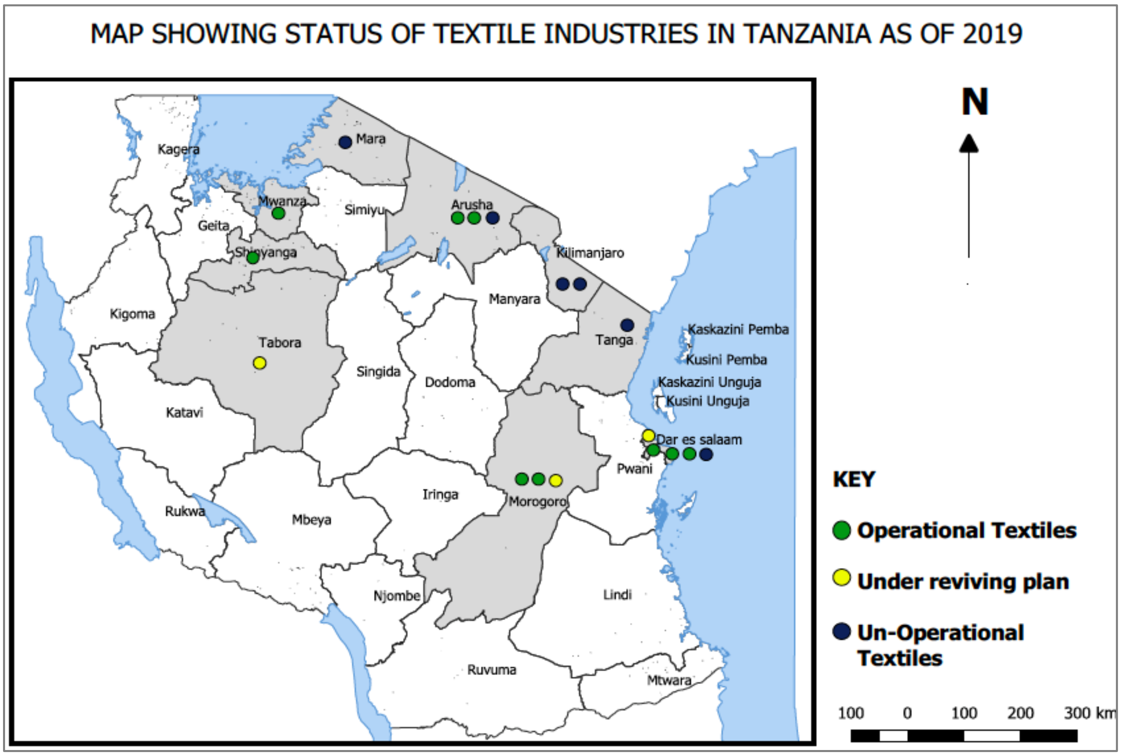
[Figure 2-7] Immediate and Supporting Actors' Value Chain



2.3.3 Textile, Apparel Manufacturing Industries

Like many African countries, Tanzania is a net importer of textiles and clothing and a net exporter of cotton lint. Textile manufacturing industries in Tanzania include: ginning mills, seed processing mills, spinning mills, weaving mills, knitting mills, dyeing mills, and madeup (apparel (or designing and sewing)) makers, for the purpose of converting seed cotton (fiber) into yarn or fabrics and garment. According to the Ministry of Industry and Trade 2004 report on the Status of Textile Industries in Tanzania, 50 textile industries were established by the year 2002 by the government and private companies. Note, however, that only 23 (46%) of the established industries involved in dyeing, spinning, wearing, printing Kanga, bedsheets, garments, knitting, woven blankets, and socks started operating. Till the year 2017, many textile mills in Tanzania were not operational (Figure 2-8), and most of those that are operational are operating below production capacities at 25% ~ 40% (TDU Industrial Survey report, 2017). The use of poor and outdated technologies has adversely affected the efficiency (production capacities) and quality of textiles produced.

[Figure 2-8] Map of Tanzania Showing the Location of the Textile Industries



2.3.3.1 Ginning

Over 55 out of 80 ginneries are currently in operation in the country, with the other 25 ginneries not operational. Locally produced cotton is used by a heterogeneous mix of ginneries that include five integrated ginning and spinning mills (Nida, 21st Century, Musoma, Jambo, and Mwatex). In addition, there are 67 ginneries in WCGA, 21 of which are old cooperatives and 46 are new private enterprises. In ECGA, there are only 13 ginneries (7 old enterprises and 3 new companies). The ginning segment has a total capacity of 300,000 bales per month, but utilization is only 30~40% (ginning outturn is ~34 %). The installed capacity is not being utilized fully as only a few of the ginneries are in operation due to limited cotton and ginning technology and seasonal nature of cotton production. Additionally, the cooperative ginneries are in a poor state of maintenance, following many episodes of mismanagement and high operation costs. Modernization of machinery, renewal of buildings, and, in some cases, power and water connection are needed to improve the performance of the ginneries.

2.3.3.2 Seed Processing

Only 32 ginneries in Tanzania have installed oil mills that only use local cottonseeds. They produce oil (for both edible and industrial uses, such as soap and margarine), linters (in negligible quantities), hulls for animal feed, and cotton seed cake. These cottonseed mills have processing capacity of about 110,000 tons, with utilization capacity of less than 20%. While some of the processed cottonseed products are sold in the local markets, a portion is also exported mainly to Switzerland (cottonseed oil), China (linters), and India (cottonseed oil).

2.3.3.3 Spinning

Spinning mills in Tanzania include: integrated ginning and spinning mills, two stand-alone spinning factories (Dahong Spinning Mill and Tabora Textiles Spinning Mill), 21 semi-integrated mills, two integrated fabric and/or garment enterprises, and a number of integrated textile and clothing companies in industrial zones. The spinning mills have total capacity of 0.4 million ring spindles and 15,000 open-end rotors, for a 40% utilization rate capacity. The yarns produced are exported mainly to China, South Africa, Kenya, the Netherlands, Japan, and Switzerland, whereas the remainder is used by the local knitting and weaving mills.

2.3.3.4 Weaving, Knitting, Dyeing, and Finishing

This segment relies largely on yarns' imports from India, China, and Republic of Korea. In Tanzania, there is one stand-alone home textile company, one cottage sector enterprise (Marvelous Flotea Co. Ltd), and five stand-alone fabric-producing companies. Together, they have a capacity of 1,200 shuttleless looms and 5,000 shuttle looms. There are no stand-alone dyeing or finishing plants or services. These activities are instead integrated into the textile mills.

The semi-integrated mills produce woven printed women's kanga and kitenge as well as yarn dyed woven kikoyi fabrics, bed linen, and home textiles. Meanwhile, the integrated fabric or garment enterprises produce kanga, kitenge, African prints, Maasai-wear fabric, mosquito nets, agronets, knitted fabric, bed linens, and home textiles.

Finished fabrics are exported to Kenya, Uganda, and Bangladesh, whereas greige fabrics are exported to Kenya and Uganda. A portion of the fabrics are used by domestic garment manufacturers.

2.3.3.5 Design and Sewing

Garment manufacturers range from micro to large enterprises, with the sector dominated mainly by imports (Table 2-14). The garment mills include: one stand-alone company dedicated to uniforms and workwear (Open Sanit Enterprise); two sportswear companies (Mazava Fabrics and Production Ltd., WINDS Group); and two diversified, ready-made garment manufacturers (Bema Authentic Fashion & Design, Tanzania Tooku Garments Co., Ltd). Garment makers producing for the export markets depend on imported cotton, yarn, and man-made fibers, which is usually a requirement of the global apparel buyer who controls the market. They mostly rely on fabric imports from China and South Africa and very few of the domestically produced textiles. The integrated fabric or garment mills produce uniforms and workwear, polo shirts, shirts, lingerie, and various other garments. The segment's total capacity is 100,000 tons, but its capacity utilization is only 20%. The garment mills face significant competition from imported clothing (including second-hand clothing), which comes mainly from China, the United States, and South Africa.

The main production companies are based in EPZ / SEZs, and they produce for export markets. The biggest market is the United States, which receives 50% of Tanzanian garment exports through mass merchandising chain department stores, specialty stores, factory outlets, and off-price and mail order outlets. Meanwhile, 23 % of exports are sold to South

Africa. Other important markets include Kenya (10 %), Uganda (8 %), the United Kingdom (6 %), Congo (1 %), Mozambique (1 %), and Madagascar (1 %).
The major challenges faced by the textiles sector include: poor and outdated technologies; cost of power and access to finance; skilled labor; contrabands and used clothes; and access to trims.

[Table 2-14] Textile and Garment Manufacturing Industries in Tanzania

S/N	Mill	Description	Employment	Products	Remarks
1.	A to Z Ltd., Arusha	• Working well; knitting using yarn from SUNFLAG and outside the country. • Its garment section is also working well.	• Employs 1,800 workers in the textile sector.	• Knitted T-shirts, polo shirts, under-wears, and mosquito nets.	• Selling its products in the domestic and foreign markets.
2.	SUNFLAG (T) Ltd., Arusha	• Working well. • Has working weaving, knitting, and garment sections.	• Employs 2,700 workers.	• Cotton and cotton-blended yarn and fabrics, bedsheets, khanga, kitenge, Maasai kikoi, shirting and suiting fabrics, combat fabrics, T-shirts, polo shirts, mosquito nets, organic products – Yarn, fabrics, and garments.	• Selling its products in the domestic and foreign markets.
3.	AFRITEX Ltd., Tanga	• Member of the MeTL group of companies; Opened in late 2018.		• Produce grey khanga and kitenge fabrics from the 21st century for local sales.	
4.	MUTEX Ltd., Mara	• Member of the MeTL group of companies; closed	-	• Used to process grey khanga and kitenge fabrics	• Will re-open once the market improves.

S/N	Mill	Description	Employment	Products	Remarks
				from the 21st century for local sales.	
5.	21ST CENTURY Ltd., Morogoro	• Member of the MeTL group of companies; working well. • Produces woven fabrics; it has started knitting as well. • It has a garments unit utilizing its knitted fabrics.	• Employs 2,050 workers.	• Cotton and cotton-blended yarn and fabrics, bedsheets, khanga, kitenge, Maasai kikoi, imitated batik, shirting and suiting fabrics, combat fabrics, mercerized fabrics for tie and dye and batik printing, stitched garments.	• Produce for domestic and foreign markets.
6.	TABOTEX Ltd., Tabora	• Owned by the Rajan family; closed	-	• Used to spin 100% carded cotton yarn for local and export sales.	• Owner contemplating on reopening the factory.
7.	FRIENDSHIP TEXTILES Ltd., D'Salaam	• Working	• Employs 790 workers.	• Khanga, kitenge, khaki fabrics, bedsheets.	• Exploring opportunities to expand production. • Produces for the domestic market.
8.	MWANZA TEXTILES Ltd., Mwanza	• Owned by the Ladhen family. • Produces cotton woven fabrics.	• Employs 400 workers.	• Khanga, kitenge, bedsheets and mercerized cotton fabrics for tie and dye, batik printing.	• Produces for the domestic market.
9.	KARIBU TEXTILES Ltd., D'Salaam	• Owned by the Jetha family; closed	-	• Used to process local and imported grey	• The young owner (heir) is not making any

S/N	Mill	Description	Employment	Products	Remarks
				khanga and kitenge fabrics.	efforts to reopen it.
10.	NAMERA Ltd., D'Salaam	• Working well. • Produces cotton woven fabric for its sister mill, NIDA Ltd.	• Employs 606 workers.	• Cotton woven fabrics	• Hasa plan to expand into knitting.
11.	NIDA Ltd., D'Salaam	• Working well. • Processes grey fabric from NAMERA and blended imported grey fabrics as well.	• Employs 98 workers.	• Khanga, kitenge, bedsheets, mercerized fabrics for tie and dye, batik printing,	• Produces for domestic and foreign markets.
12.	MAZAVA GARMENTS Ltd., Morogoro	• Working well. • Stitches 100% polyester knitted garments from imported fabrics. • Operates in rented premises.	• Employs 2,400 workers.	• Sportswear.	• Has acquired 20,000 m^2 at the STAR CITY zone, Morogoro to establish its own production premises. • Expansion plans in progress. • Produces solely for export.
13.	TOOKU GARMENTS Ltd., D'Salaam	• Working well. • Stitches woven men's jeans from imported fabric. • Operates in rented EPZA premises	• Employs 2500 workers.	• Men's jeans.	• Expansion plans in progress. • Looking for 50 acres of land to expand. • Produces solely for export.
14.	MOROGORO CANVAS Ltd., Morogoro	• Ownership has been transferred to CRDB Bank Plc. • The factory remains closed.		• Used to produce canvas fabrics and tarpaulins for local and export sales.	• CRDB Bank Plc is looking for a buyer. • Social Security Funds has shown interest in the factory.

S/N	Mill	Description	Employment	Products	Remarks
15.	JAMBO SPINNING Ltd., Arusha	• Owned by the proprietor of the Jambo group of Shinyanga; closed.		• Used to spin 100% carded cotton yarn for local and export sales.	• Owner wants to sell the factory.
16.	KILIMANJARO BLANKETS Ltd., Tanga	• Has been taken over by the MeTL group based on court auction; closed.		• Used to produce blankets for local sales.	• Former owners – the Hirji family – has appealed. • Awaits judgment on the court appeal to reopen.
17.	TANGANYIKA BLANKETS Ltd., Pwani (Coastal)	• Owned by JTK Investments Ltd.; closed.		• Used to produce blankets for local sales	• Looking for big orders to reopen.
18.	MOSHI TEXTILES Ltd., Moshi (Kilimanjaro)	• Owned by the Joshi family; closed.		• Used to weave and knit fabrics, mosquito nets, and medical bandages for local sales.	• Owner looking for joint venture partner to reopen.
19.	JOC Textiles (T) Ltd.	• Owned by Jiangsu Overseas Corporation of China	• Employs 120 workers	• Producing open-end yarn for export to China.	• 10 000 tons of yarn

2.4 Potential for CTA and Madeups Sector in Tanzania and/or EAC

According to TEGAMAT estimates, the current per capita consumption of clothing by various countries is as follows: USA 39 Kg; EU 25 Kg; China 15 Kg; India 6 Kg; and Africa 2.5 Kg. The Africa per capita clothing consumption of 2.5 Kg is representative of EAC. Moreover, the population of the 6 EAC Partner States in 2019 is around 196.5 million people, which is expected to increase to 256.9 million people by 2029 (United Nations, DESA-Population Division, 2018). Based on per capita clothing of 2.5 Kg, this translates into regional demand of 491.3 million Kg or 491,300 MT of clothing, which is likely to increase to 642,450 MT by 2029. Similarly, three assumptions are made to estimate the amount of lint required in EAC per year. These include: (i) Clothing of a ratio of 40% cotton and 60% other fibers, (ii) 86% conversion of cotton lint into fabric, and (iii) Intra -EAC trade in cotton lint increases by 30%; thus, EAC region exports of cotton lint to non-EAC member countries have been reduced from the current level of 70 ~ 40%. For the EAC region to meet the self-sufficiency level of lint while also exporting 40% of its lint production, it will need to produce about 320,000MT of lint per year in 2019, which is likely to increase to 420,000MT of lint by 2029. The region produced a combined total 85,480 MT of cotton lint in 2017, so it is facing a deficit of about 234,520MT of cotton lint if is to meet her regional market demand for clothing sufficiently and export 40% of its total lint production.

Assuming that, after the ginning processes, cotton lint accounts for 38% of the seed cotton harvested, then to meet the EAC market demand and export requirements, the EAC Partner States need to produce at least a total of 842,000 MT of seed cotton per year, which is likely to increase to about 1,105,000 by 2029. For the period 2013-2017, the region produced an average of about 290,000 MT per year. Therefore, the region faces a huge deficit of about 552,000 MT of seed cotton assuming EAC lint self- sufficiency and 40% export of the produced lint.

In terms of regional fabric requirements, the consumption per capita of 2.5 Kg in EAC is equivalent to 14 square meters of fabric per person per year, which translates to about 2.75 billion square meters of fabric per year. Assuming 50% of this fabric is supplied domestically, i.e., 1.35 billion square meters of fabric and weaving in EAC garments accounts for 60 % and knitting constitutes 40%, then the region needs to produce 810 million square meters of fabric. To estimate the number of factories needed for woven fabric, we assume weaving capacity of 70 looms and production of 3,300 square meters of fabric per day. In one year, the production capacity is 8.7 million square meters. Thus, to produce 810 million square meters of woven fabric per year, the EAC region requires at least 93 factories producing woven fabrics alone.

2.5 Potential Investment Opportunities Based on Demand

Numerous investment opportunities exist in EAC textiles, apparels, and madeups:

a) Investments in seed cotton production

Investment in conventional seed cotton growing through irrigation and organic cotton in the cotton growing areas, which are largely underutilized, has huge potential. The region has substantial land suitable for cotton growing, which is not utilized, e.g., in Kenya, only 11% of the land suitable for cotton growing is utilized. As of 2019, EAC Partner States need 842,000 MT of seed cotton, which is likely to increase to about 1,105,000 per year by 2029 for the region to be self-sufficient in apparels production while also exporting about 40% of the produced lint. For the period 2013-2017, the region produced an average of about 290,000 MT per year, and it faces a deficit of 552, 000 MT per year for it to be self-sufficient in apparels production while also exporting about 40% of the produced; hence the huge investment potential in green or environment-friendly cotton production.

b) Investment in production of other natural fibers

The production of other natural fibers such as protein-based fibers including silk and wool as well as vegetable-based fibers, e.g., sisal, coconut, industrial hemp, and banana, presents an investment opportunity.

c) Investment in the production of synthetic fibers including nylon, polypropylene, viscose, and polyester and dyes

Synthetic fibers are important in fabrics manufacture since most of the fabrics are a blend of cotton and synthetic fibers. Additionally, investment in dyes, starch, and accessories for incorporation in garments manufacture has potential, including the manufacture of spare parts, as the bulk of these inputs are imported into the region.

d) Investment in modern and state-of-the-art equipment in textile and madeup value chain

Analysis has shown that, based on the current existing technology, EAC needs at least 93 mills for making woven fabrics to supply 50% of the region's demand. Technology upgrading is needed in the existing equipment in the textiles and madeups value chain, from farm production to fiber processing, fabrics, and fashion garments as well as integration into the international market for garments. There are also opportunities for the revitalization of dormant textile mills whose operations have been adversely affected by stiff competition from secondhand and new cheap clothing following the onset of liberalization in the region from the 1990s. There are also opportunities for refurbishment of mills that produce below capacity, based on the availability of cotton lint for fiber and spinning as necessary to facilitate accreditation by major garment brands buyers/retailers. The revitalization and refurbishment plans should be supported by a well-laid strategy to facilitate access to cotton lint, modern technologies, skilled manpower, production of internationally fashion-driven, niche, and high-value fabrics/garments products, local and external markets, availability of affordable and sufficient utilities (power and water), and efficient logistics services for clearing imported inputs and exports at the region's entry/exit ports.

e) Investment in trims and other textiles and apparels industry inputs

Garment accessories such as trims, buttons, elastics, zippers, labels, and badges among others are estimated to contribute significant added value to the garment. There is limited capacity in the region to make such accessories, however.

f) Investment in apparels and other madeups

EAC has a population of 196.5 million, and COMESA has over 560 million people, with the whole region depending on imported garments. Additionally, AfCFTA presents an additional market for EAC garments. There lies an opportunity to make garments with less stringent standards for the vast majority who are not high-income earners in EAC and African region. In the wake of the phaseout of used clothes in EAC, along with the high youth population and sense of fashion, there lies an opportunity for both SMEs and FDIs in the garmenting sector to dress the region, particularly in the areas of knitting and garmenting, for which members of this generation are high consumers.

Similarly, access to the USA and European markets under AGOA and EPAs presents an opportunity for Tanzania, EAC, and Africa to produce madeups for the high-end markets. In addition, it is expected that there will be major changes in global consumption patterns of textiles and madeups by year 2025, with China and India seen to focus more on producing garments for their own domestic markets, which are growing fast, in addition to increasing demand as a result of increase in income; thus giving a huge opportunity for Tanzania- and Africa-based apparels manufacturers to produce for developed and developing countries, which have largely depended on China and India madeups supply.

g) Investment in the manufacture of absorbent cotton for medical and cosmetic needs

The Tanzania, EAC, and Africa markets have huge opportunity for these products in the health institutions, which largely rely on import markets as the country's/region's absorbent cotton producers operate below capacity. In Uganda for example, the 6 producers of this product operate at an average of 34% of their installed capacity.

h) Investment in CTA skills training facility

The region is limited in terms of facilities to provide skills development in textiles and madeups. Relevant short courses in fashion and graphics design, machinery and equipment maintenance, and skills upgrade in the whole textiles and madeups industry value chain are urgently needed.

i) Investment in natural fibers research

Due to the current low cotton productivity levels and limited production of other fibers including silk, wool, and vegetable fibers (sisal, coconut, banana, hemp), there is a need to invest in research to enable the development of innovations and regional sharing of knowledge on modern production technologies. Controversial topics such as organic and Bt cotton production will also be considered.

CHAPTER 3

Current Status of the Cashew Industry in Tanzania

Policy Consultation on Developing a National Technology Roadmap for Accelerated Industrialization

Chapter 3. Current Status of the Cashew Industry in Tanzania

1. Overview of the Cashew Sector

1.1 Background

Tanzania has a total land mass of about 44 million hectares of arable land. The country's potential irrigable land area is estimated to be 29.4 million hectares, with about 475,012 developed. There are seven (7) agro-ecological zones suitable for producing a wide range of crops from temperate, sub-tropical to tropical climate.

The country's economy is dominated by agriculture, contributing 28.7 of the GDP, with the crop subsector contributing 16.58. Agriculture provides 65.5 percent of jobs and 65 percent of industrial raw materials. It also plays an important in making the country's food self-sufficient with FSSR of more than 100.

The current focus of the country is on transformation into an industrial economy so as to attain middle income by 2025 through the application of improved technologies and available resources. The country is implementing Agricultural Sector Development Program Phase II (ASDP II), aiming at the transformation of the Agricultural Sector to higher productivity, commercialization level and increase of small holders' income for improved livelihood through the increased use of improved technologies. The program has four pillars: sustainable water and land use management; enhanced productivity and profitability; rural commercialization and value addition; and strengthening of Sector enablers.

Cashew (*Anacardium occidentale*) is a native of Northeastern Brazil and was introduced in Easter Africa by the Portuguese in the 16th Century. In Tanzania, cashew is among the six strategic cash crops with major contribution to foreign earnings; it also provides employment, raises farmers' income, and improves nutrition. Improving the performance of the cashew industry will efficiently respond to the market demand of various products at all levels; thus contributing significantly to the growth of the national economy and improving the livelihood of farmers and value chain actors.

1.2 Achievement of the Cashew industry

The country implemented a series of activities aiming at increasing production, productivity, and overall performance of the industry through:

- Improved technologies along the value chain by the National Research Institute, where 54 improved varieties were developed;
- Post-harvest and processing technologies were developed and evaluated;
- Increased utilization of agricultural inputs such as improved seeds, better crop husbandry, protection and post-harvest techniques, and strong technical support from the government;
- Increased number of processing groups and processing plants that can be rehabilitated and improved to increase the production of cashew kernels;
- Tanzania is the 1st country in Africa to breed and register improved varieties under the (UPOV) protocol.
- Use of tissue culture technology to multiply cashew seed wherein one seed can produce over 100 seedlings;
- Ongoing tissue culture research on using tender shoots-scions, leaves, and flowers to produce seedling;
- Increased area under cashew production (20 regions are cultivating the crop)

1.3 Challenges Faced by the Cashew Industry

Despite its achievements, the subsector is facing a number of challenges in production, productivity, processing capacity, quality of produce, and market. Low cashew production and productivity are attributable to a number of factors such as aging of cashew trees, attack by insect-pests and diseases, and inadequate agricultural extension services.

1.3.1 Low Production and Productivity

The presence of large number of old plants, inadequate management practices, disease outbreak and erratic weather conditions are some of the factors.

- **Old (traditional) Cashew Trees**

This is due to the fact that most of the cashew trees are either old that they cannot produce the optimal amount of at least 20kg per tree or are genetically low-yielding with poor nut quality.

- **Inadequate Cashew Management Practices**

Poor cashew husbandry practices including non-planting of new varieties, poor crop protection practices, poor farm management practices (ploughing, harrowing, intercropping, chemical fumigation, fertilization, soil acidity control, pruning), inadequate rehabilitation practices, and lack of upgrading of existing cashew farms have generally affected cashew production in Tanzania.

- **Attack by Diseases and Insect-Pests**

The decline in cashew production in the country was reported to be due the abandonment of cashew farms, outbreak of powdery mildew disease and, to a certain extent attack by insects-pests. Other diseases include blight, diebacks, and anthracnose. Nonetheless, efforts to address the situation have been done by releasing disease-tolerant varieties and dissemination control technologies for the improvement of revenues and standard livelihood of cashew farmers.

1.3.2 Unreliable Funding of Cashew Research Program

The research activities require investment in terms of cash and human capital. Scarce resources lead to inadequate dissemination of improved technologies, capacity building, capacity development and infrastructure development/improvement, and technology transfer mechanisms.

1.3.3 Inadequate Processing Technologies

Due to inadequate availability, accessibility, and low use of improved technology, raw nuts are exported, and this in turn contributes to unemployment and less income. The processors use old or outdated technology of equipment and machinery in cashew processing due to inadequate capital for equipment and machinery.

1.3.4 Inadequate Quality Storage Facilities

The number and quality of storage facilities for RCN do not match the current production. Nonetheless, the existing facilities can store 61.92% of the current production capacity.

1.3.5 Inadequate Business Skills and Market Information

Farmers have low business skill, so they fail to plan and cope with market trends worldwide. The huge cost of product development made cashew kernels less competitive in the domestic market.

1.3.6 Inadequate Understanding of Cashew Laws and Regulations

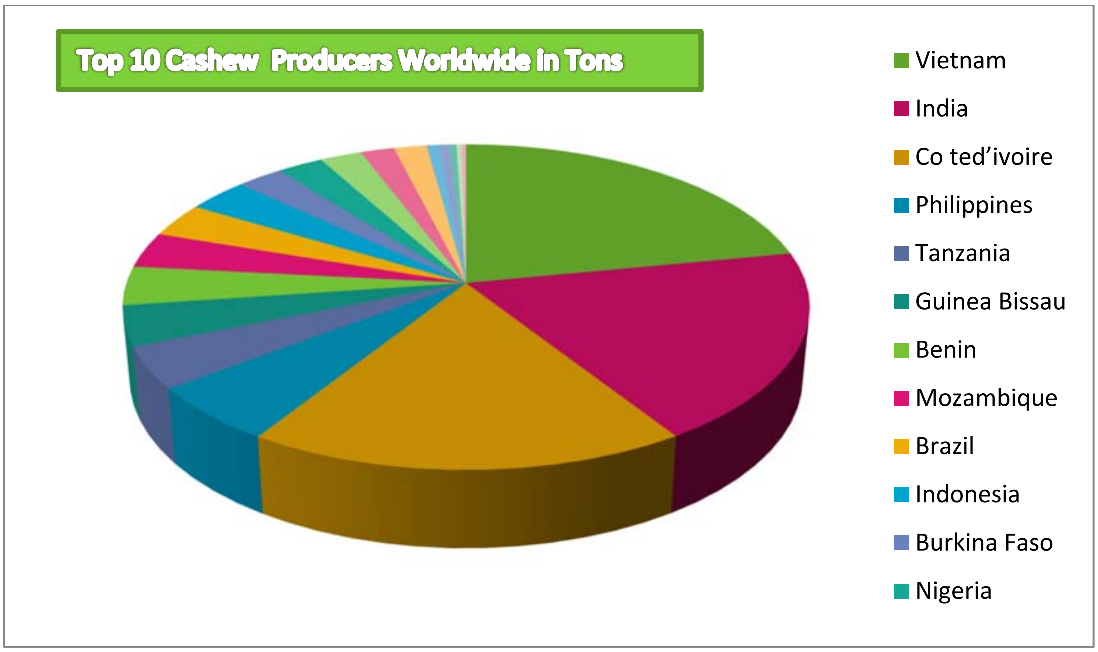
The laws and regulations governing the cashew subsector are not understood well by users, and some do not meet the current customer requirements.

2. Cashew Technology and Industry

2.1 Overview of World Cashew Production

Cashew production is done by more than 34 countries worldwide. In 2017, Tanzania ranked fifth (5th) among the top 10 producing countries after Vietnam (22%), India (19%), Cote d'Ivoire (18%), and the Philippines with 4.14% share of the world production (FAOSTAT). Others are Guinea Bissau, Benin, Mozambique, Brazil, and Indonesia. Nonetheless, data from the Cashewnut Board Indicate that Tanzania produced 265,000 for the year 2016/2017, which was higher than what was reported and the volume produced by the Philippines (Figure 3-1 & Appendix 1).

[Figure 3-1] World Production of Cashew



Source: FAOSTAT DATA

The world cashew kernel production has been fluctuating for the past 11 years. It increased from 488,530 (2007/2008) to 789,050 (2017/2018), with the lowest production recorded in the year 2010/2011 (Table 3-1).

[Table 3-1] World Production of Cashew Kernels

S/N	YEAR	PRODUCTION (MT)	RATE
1	2007/2008	488,530	10
2	2008/2009	538,400	8
3	2009/2010	522,327	9
4	2010/2011	469,079	11
5	2011/2012	576,431	6
6	2012/2013	549,692	7
7	2013/2014	601,642	5
8	2014/2015	716,682	4
9	2015/2016	724,556	3
10	2016/2017	783,994	2
11	2018/2018	789,050	1

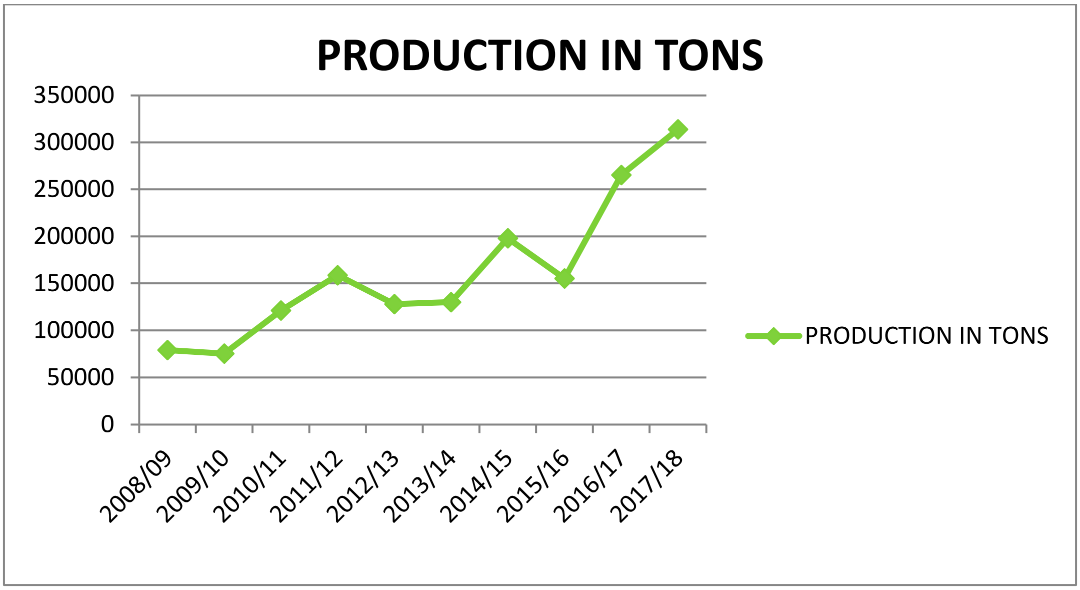
Source: FAOSTAT DATA

2.2 Cashew Production in Tanzania

Cashews are grown by small-scale farmers and, to a lesser extent, large-scale farmers. Approximately 314,000 farmers engage in cashew production (Cashewnut Board, 2019).

Cashew production started far back in the 1960s with its introduction by the Portuguese in the 16th century. Tanzania used to produce more than 20 percent of global production in the 1970s, producing about 145,000 metric tons in 1974. Generally, cashew production fluctuates, and production dropped to 16,500 tons in the year 1986/87. Nonetheless, cashew production increased from 79,069 tons in 2008/09 to 313,826 in 2017/2018 (see Figure 3-2, Appendix 4). The slight increase in cashew production is attributable, among others to the use of improved technologies such as use of quality seed and pesticide application, adherence to agronomic packaging, and expansion of area under cashew production.

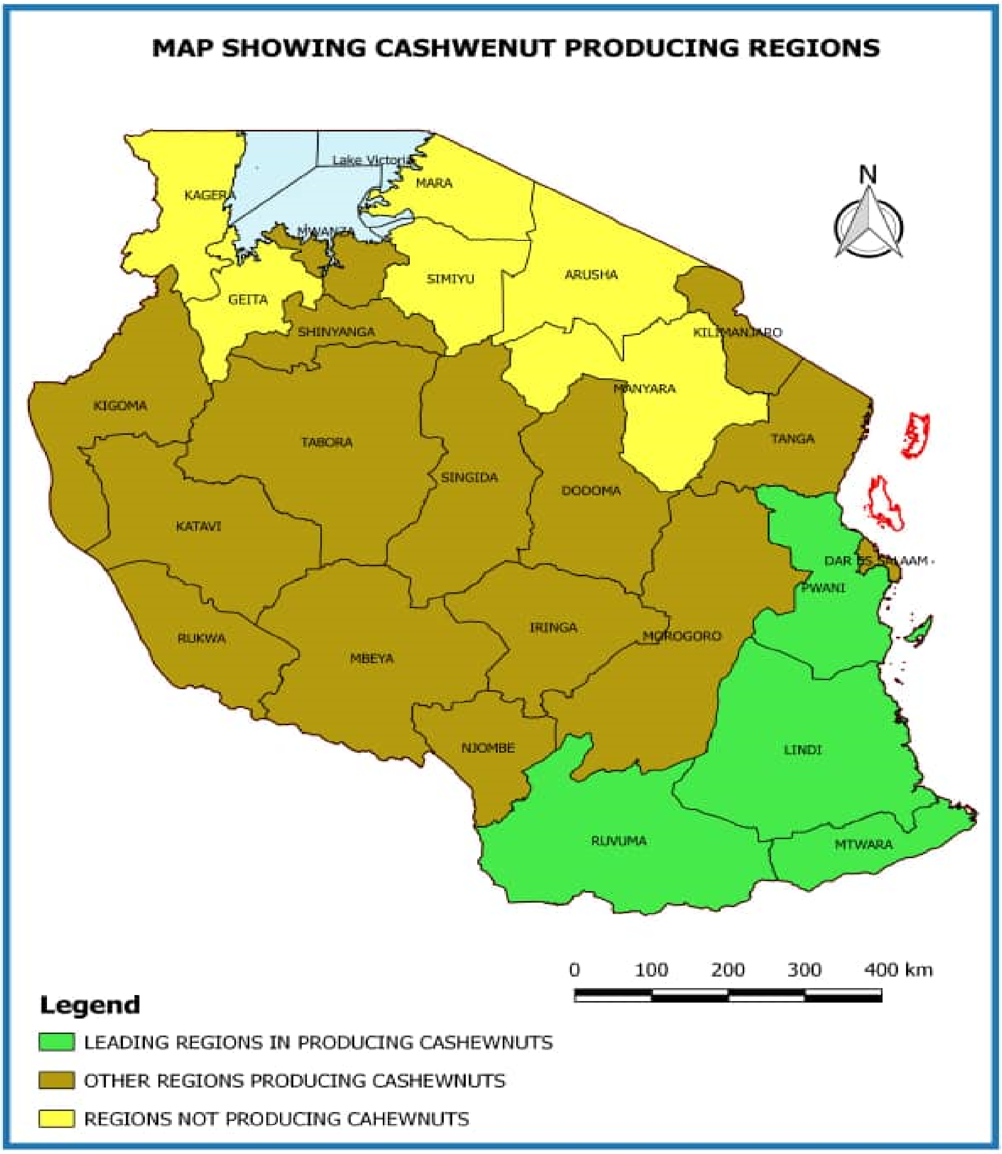
[Figure 3-2] Cashew Production Trend for Ten-Year Period



There are about 50 million cashew trees with potential yield of 5-8 kg/tree. The low productivity is due to the existence of a large number of aged trees, use of indigenous variety with low yielding potential, low use of agricultural input, pest and disease, and unpredictable weather, among others. Use of improved technology such as quality seedling can increase the yield up to 15 ~ 20 kilograms per tree. There are 54 improved cashew varieties.

The crop is grown in 90 districts in twenty (20) regions such as Dar es Salaam, Mtwara, Lindi, Ruvuma, Pwani, Tanga, Dodoma, Singida, Mbeya, Morogoro, Njombe, Tabora, Katavi, Kigoma, Iringa, Songwe, Rukwa, Kilimanjaro, Shinyanga, and Mwanza (Figure 3-3). Nonetheless, the major producing regions of what was produced in the year 2017/2018 are Mtwara (62%), Lindi (24%), Ruvuma (6.7%), and Pwani (6.4%) (Figure 3-3).

[Figure 3-3] Potential Cashew-Producing Areas



2.3 Cashew Research in Tanzania

Cashew research is conducted by the Tanzania Research Institute Naliendele (TARI Naliendele) located in Mtwara region. Research activities started back in the 1970s with a mandate to conduct agricultural research that addresses the needs and aspirations of the farmers, particularly improved crop productivity and quality. TARI coordinates Cashew and

Oil Seed research at the National level. As a center of excellence for cashew research in Africa, the center is working in collaboration with other research stations including international organizations. The aim of agricultural research is to promote sustainable income generation, employment growth, and export enhancement through the development and dissemination of appropriate, environment-friendly technology packages. Nonetheless, the amount of fund allocated for research is not sufficient. The core functions include:

- To operate a breeding program for the development of cashew varieties with traits suitable for end users
- Develop technologies to combat pest and diseases
- To develop agronomic packages
- Technology transfer
- Research on soil suitability for cashew production
- Evaluate and develop pre- and post-harvest technologies
- To research on biological control for cashew pests and develop a package on how it can be utilized beneficially
- Research on cashew product diversification and value addition
- Capacity building for cashew experts from within and outside the country

In addition, achievements include:

- Development of 54 improved varieties, 16 of which are polyclonal, 22 are Hybrid varieties, and 16 are Dwarf varieties. These varieties can produce about 15 per tree at the age of 8 years compared to local varieties, with yielding potential of 4-8 kg per tree. The developed varieties are also disease-tolerant, with superior processing quality, high turnout, and ease of peel removal.
- Tanzania is the 1st country in Africa to breed and register improved varieties under the (UPOV) protocol.
- Use of tissue culture technology to multiply cashew seed wherein one seed can produce over 100 seedlings.
- Ongoing tissue culture research on using tender shoots-scions, leaves, and flowers to produce seedling.
- Increased area under cashew production

2.4 Processing of Cashews in Tanzania

2.4.1 Processing

RCN after harvesting is graded into two grades -- standard grade and Under Grade -- and each grade fetches different prices. Cashew processing in Tanzania began in the early 1960s when a private company known as African Cashew Processors Company Ltd. established a simple manual processing plant in Dar es Salaam; in 1965, the first mechanical processing factory incorporating Italian technology was installed. Starting from 1968 to the late 1970s, the Government of Tanzania built 12 cashew-processing factories with accumulated capacity of around 100,000 Mt, with government parastatal Cashew Nut Authority of Tanzania (CATA) as the owner. In the 1980s, the rapid decline in the production of cashew resulted in the closure of all 12 factories. All factories were large-scale mechanized types using either Italian (Oltemare) or Japanese (Cashco) technology.

Cashew Nut Act No. 18 of 2009 and its regulations of 2010 categorized the processing industry into three categories: small-scale (1 ton/year), medium (1-5 tons/year), and large-scale (above 5 tons/year).

Currently, there are 27 cashew processing plants with total installed capacity of 133,000 tons per year and utilized capacity of 59,500 tons. Out of 27 processing plants, 13 operate under capacity, and the rest are out of order (Appendix 2a&b). In addition to the processing plant, there are 90 processing groups that process small quantities to supply the local markets. The current utilized capacity can process only 22% of 224,000 tons of cashew produced in 2018/2019.

The closure of the processing plant has contributed to unemployment particularly to the community around production regions. For instance, the Korosho Africa processing factory at Tunduru with capacity of 3,500 tons per year employs about 825 individuals, 90% of whom are women.

2.4.2 Value Addition for Other Products

Cashew byproducts (apples, shells, CNSL) processing has not been adequately exploited as additional source of income and efficiency in the processing industry. For example, cashew apples serve as a viable raw material for value-added products such as beverages, snacks, jam, and ethanol production. The current raw cashew production volume of around 224,000 Mt in the year 2018/2019 would yield an estimated 2.24 million Mt of apple fruits, which could be processed further. A very small proportion of fresh cashew apples are used in making locally fermented drinks in the villages. The Research Institute at Naliendele is working on the development of technologies to promote the use of untapped opportunity in cashew byproducts such as:

- Juice, Syrup, Jam, Pickle, and Wine from cashew apple
- Production of cashew butter from kernels
- Extraction of cashew shell liquid (CNSL)

Currently, there is no commercial production of the aforesaid products. Thus, it is an opportunity for increased income for cashew growers and entrepreneurs.

2.4.3 Warehouse

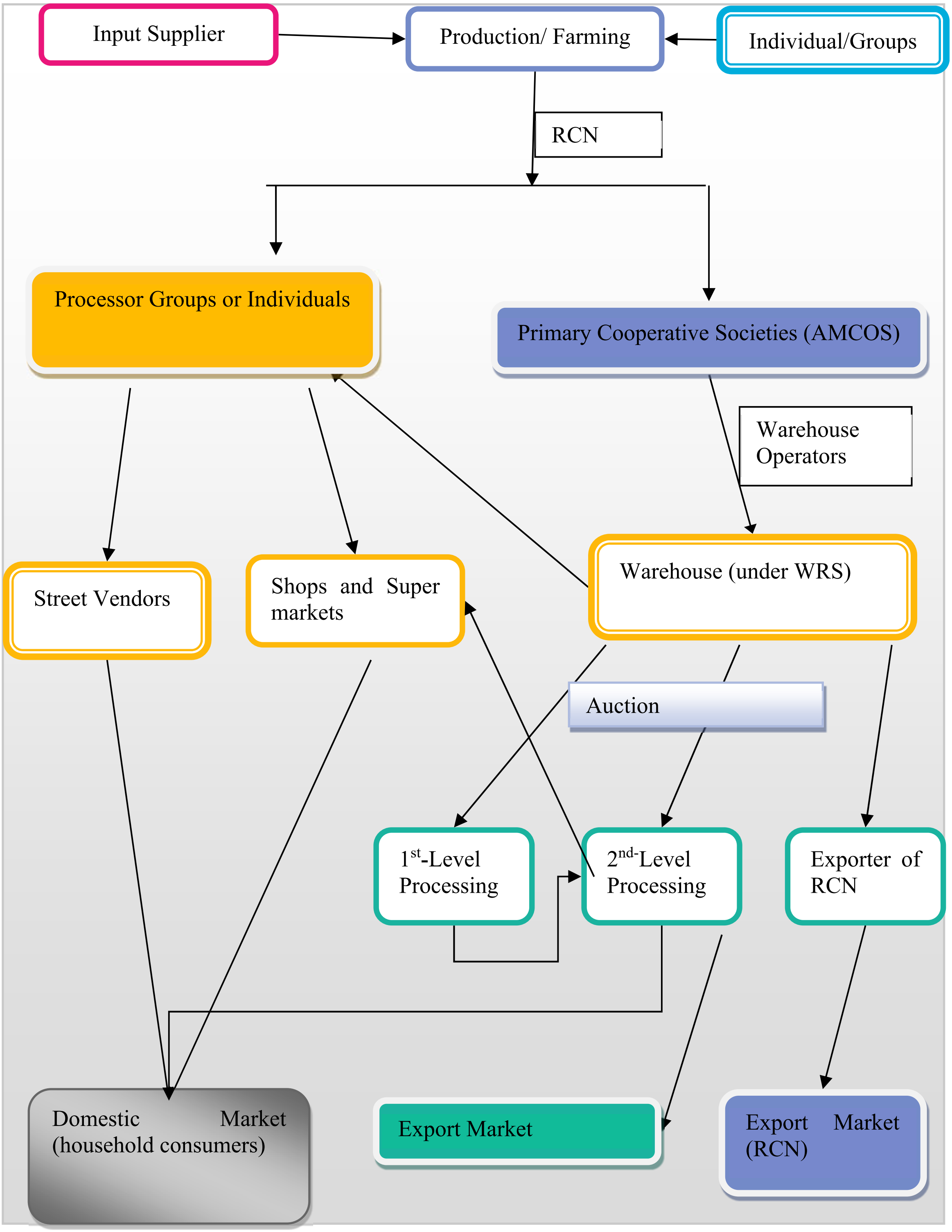
All raw cashews by law have to be transported to certified warehouses where they will be stocked in the designated system to enable traceability. There are a total of twenty-four (24) warehouses with the capacity to store 138,700 tons of RCN at a time. The existing capacity can store only 61.92% of the 224,000 tons produced in 2018/2019. Note, however, that the qualities of these storage facilities do not support long-term storage. The stored RCN will then be auctioned to buyers in a process wherein the warehouses are not involved. Moreover, there are inadequate quality storage facilities for cashew kernels.

2.5 Cashew Value Chain and Actors

2.5.1 Cashew Value Chain

The cashew value chain involves production, input suppliers, storage facilities, processing, marketing, and end users as illustrated in the diagram below.

[Figure 3-4] Schematic Illustration for the Cashew Value Chain

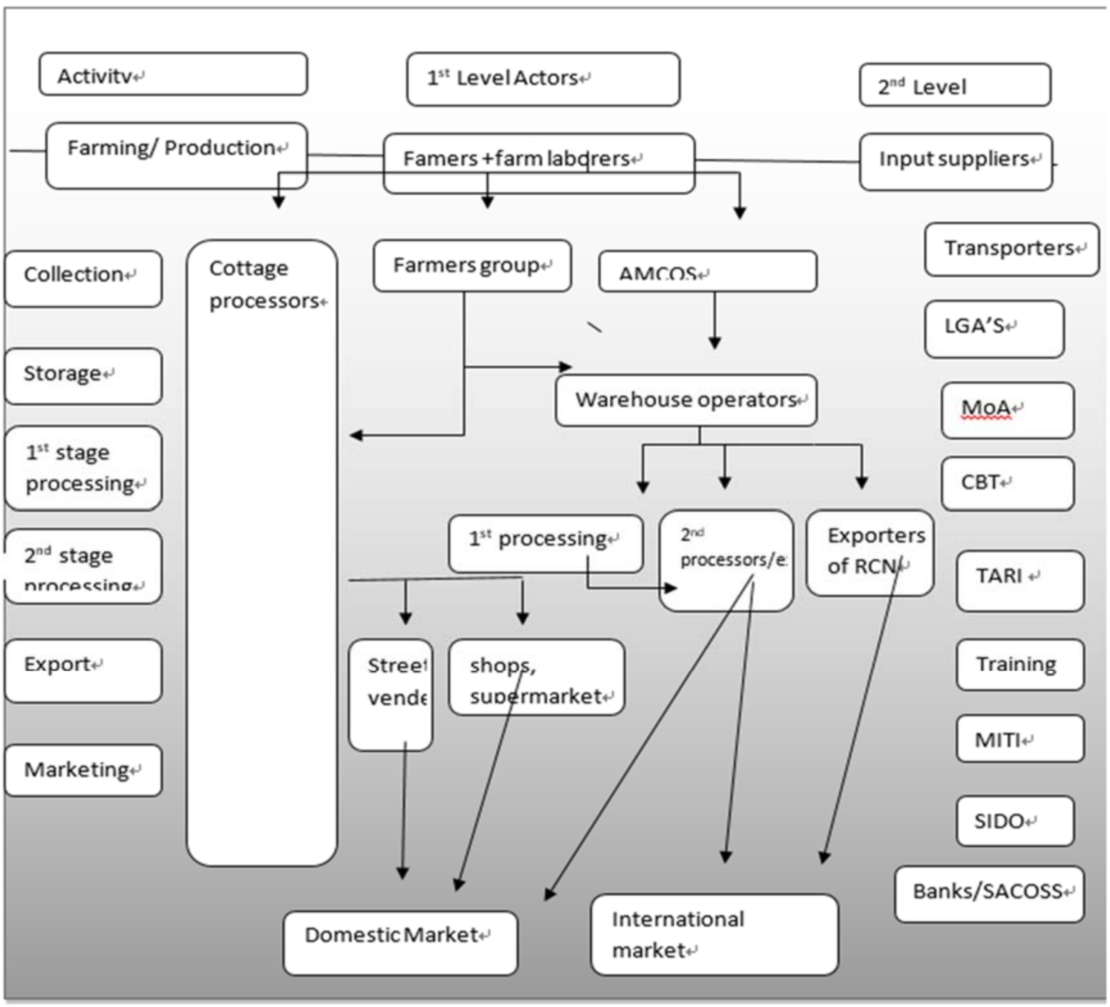


Source: Cashewnut Board and Own modification

2.5.2 Cashew Value Chain Actors

The cashew value chain actors are crop growers (farmers), Primary Cooperative Societies (AMCOS), cooperative unions, processors, traders (retailers, shops, and vendors), importers, and exporters. Other actors are service providers at various levels along the value chain such as enterprises providing agricultural inputs, research institutions, training institutions, Ministry of Agriculture, Cashewnut Board Tanzania, SIDO, warehouse operators, Ministry of Industry, TANTRADE and Trader, LGAs, transporters, and investors. Each actor has a role to play for the industry to operate effectively.

[Figure 3-5] Value Chain Actors



- **Farmers**

According to the Cashewnut Board of Tanzania there are approximately 318,000 farmers, about 273,000 of whom have been registered up to August 2019. The number does not include farm operators and laborers. The size of hired labor may extend the number of farmers, making the cashew sector an important source of income for a large number of Tanzanian households.

- **Primary Cooperative Societies (AMCOS)**

AMCOS provides inputs, mainly pesticides, and procures supplies in bulk such as farm equipment, fertilizers, sprayers, and gunny bags. The cooperative collects the cashews and sells the raw cashews to buyers via the warehouse receipt system. In addition, AMCOS carries out other services such as savings and revolving credit products for cashew investments.

- **Cooperative Unions**

Cooperative Societies procure materials in larger quantities such as gunny bags and sisal ropes, acquire loans, transfer money to primary society offices, and identify and register authorized transporters of raw nuts to the warehouse and also interact with the warehouse marketing system. They also prepare the sales catalogue for every consignment in the warehouse based on the information provided weekly by the warehouse operator to the union.

- **Farmers' groups and individuals**

They produce cashew on a larger scale in block farms; others engage in small-scale production but are organized in such a way that they are able to share experience, orders their input, and services together.

- **Processing groups**

About 90 farmers groups have ventured into the processing of cashews locally, and they supply the cashew kernel to the domestic market.

- **Warehouse operators**

Raw cashews have to be transported to certified warehouses where they will be stocked in designated lots separately for each cooperative. The warehouses provide a receipt for the reception of the goods. The lots will then be auctioned to buyers in a process wherein the warehouses are not involved. The buyer pays the amount to a bank, which will then apportion payment to the various parties involved. This system is intended to eliminate or minimize the number of middle players, limiting marketing for operation by the receipt system only.

- **Cottage processors**

These de-shell, peel, and roast cashews manually in backyards. The products are sold by small shops and street vendors.

- **First-level processors**

At this level, the processing of cashews is up to the level of deshelling before peeling. This type of processing can be outsourced to certain operators. On the other hand, 2nd-level processing requires the more rigorous application of hygienic standards.

- **Second-level processors**

In general, processors can be categorized into small-, medium-, and large-scale processors. The processing kernels are for local, regional, and international markets. It starts with the peeling of cashews till sorting and packing.

- **Exporters**

Exporters as well as large-scale processors seek and sell the raw cashew and processed kernels to international markets.

- **Street vendors**

Small-scale cashew kernels are sold along the roadside, on the streets, and at traffic lights. These sellers often operate on an individual basis. Sometimes they work for middlemen that send them out and get paid after the products have been sold, leaving them with only a very small commission for selling the product.

- **Shops, minimarts, and supermarkets**

Cashews are mainly available in the cashew growing regions and some big cities at shop outlets, roadside stands, minimarts, and larger supermarkets. The current local market price for one kilogram of kernels is approximately 30,000Tsh, so they are less competitive compared with other nuts.

- **Service providers**

These include all institutions, individuals who are technology developers, those who provide technical support, regulatory organization, and financial institutions.

3. Market Trends and Projections

There are a number of products marketed worldwide. These include raw cashews, raw cashew kernel, roasted cashew kernels, cashew apple, and cashew shell oil. These products are marketed at different levels for different uses.

3.1 World Market Overview

The cashew market was valued at USD 9.9 million in 2018 and is projected to be worth up to USD 12.7 million dollars by 2024. Among all cashew-producing nations, African countries contributed 56.5% of the global production in 2018, whereas Asian countries constituted 44%. In the last eight years, Vietnam had increased its export presence to over 85 countries worldwide. It is the largest supplier of cashew to the United States, China, Australia, Canada, and New Zealand. The United States was the largest importer of cashew kernel in the global market, with imports valued at USD 1.6 million in 2018.

3.2 Markets for Raw and Processed Nuts in Tanzania.

Cashew is one of the six strategic cash crops in Tanzania. It provides income to farmers, traders, and service providers, as well as the nation at large. The raw cashew price in the country increased from 1,402 Tsh. in 2013/2014 to 4,128 Tsh. in the 2017/2018 cropping season.

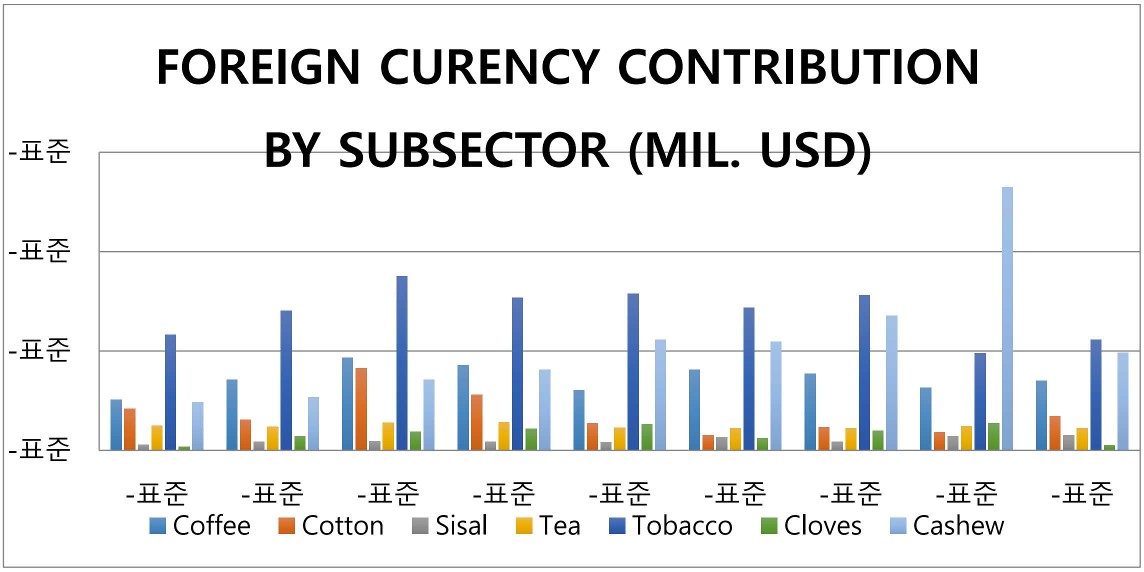
[Table 3-2] Raw Cashew Price

Season	2013/2014	2014/2015	2015/2016	2016/2017	2017/2018
Price/Kg (Tsh.)	1,402 -1,723	1,440 -2,200	1,820- 2,900	1,800 – 4,000	1,700 - 4128

Source: CBT

The foreign currency contribution for the past five years increased from 96.9 (2010) to 529.6 (2017) million USD. In 2017, cashew contributed about 529.6 million USD or Tsh. 1.187 Trillion, which is more than sum of the foreign currency earnings from other cash crop such as Tobacco (195.8 USD), Coffee (126.1 USD), Cotton (36.6 USD), Tea (49.0 USD), Cloves (54.4 USD), and Sisal (28.7 USD), which, together, contributed about 490.6 million USD (BOT, 2017). The foreign earnings trend for the past nine (10) years is shown in Figure 3-6 below.

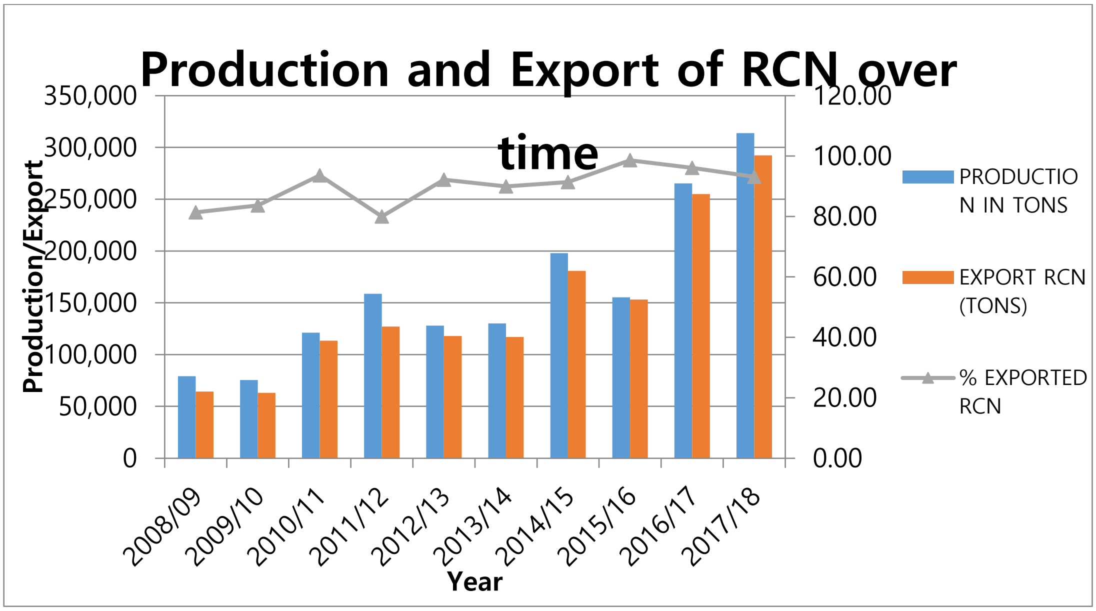
[Figure 3-6] Foreign Currency Contribution by Subsector



Source: BOT

Most of the cashews produced in Tanzania are exported as raw cashew nut in shell (RCN). For the past 10 years, the percentage of RCN exported ranged from 80.01 % to 98%, with the lower percentage recorded in 2011/2012 and the highest being 2015/2016. For the last three years, however, the percentage RCN export decreased from 98% (2015/2016) to 93.2 (2017/2018). Figure 3-7 below shows the quantity harvested and exported and the percentage.

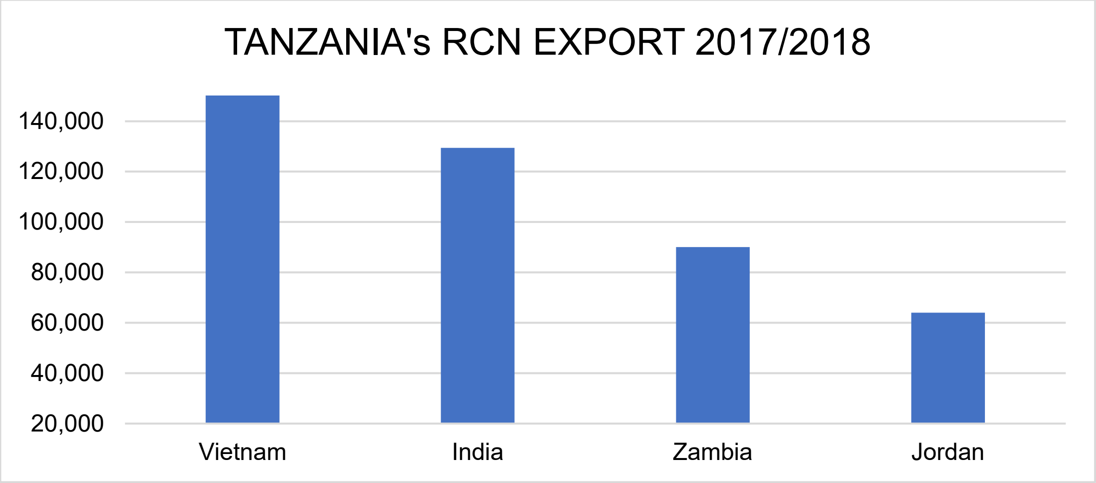
[Figure 3-7] Cashew Production and Export of RCN



Source: BOT

It is estimated that about 90% of Tanzania's exported raw cashews are destined for India; in 2017/2018, however, the trend was different as a large quantity of RCN was exported to Vietnam (Figure 3-8).

[Figure 3-8] Export of RCN 2017/2018



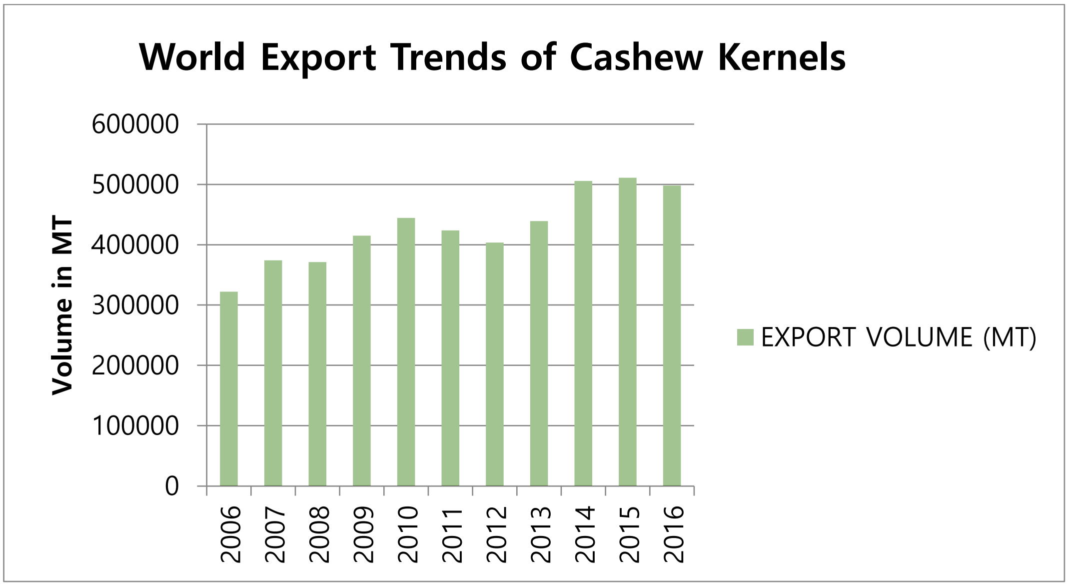
Source: BOT

Even with the government of Tanzania discouraging the export of raw nut by imposing 15% FOB value or USD 160 per metric ton, whichever is higher, on the export levy on raw nut and encouraging the export of cashew kernel by removing the export levy (0%), the export of raw nut persists.

3.3 Export of Cashew Kernel

Cashew kernel export worldwide increased from 320,144 metric tons in 2006 to 496,019 in 2016, for a 55 % increase (INC Statistical Yearbook, 2017/2018). In 2016, the leading exporting countries include Vietnam (276,368 MT-56%), India (85,097 MT), the Netherlands (32,504 MT), Brazil (15,600MT), Germany (10,657 MT), Indonesia (9,631MT), Cote d'Ivoire (5,587 MT), and others (60,495). India and Vietnam are processing countries, so they are also the leading importing country of RCN. In the same year, India imported 73% and 24 % of raw cashew from West Africa and East Africa, respectively, whereas Vietnam imported 55% from West Africa and 27% from East Africa mainly Tanzania. Figure 3-9 below shows the trend of world export of cashew kernels.

[Figure 3-9] World Export Trends of Cashew Kernels



Source: BOT

3.4 Cashew Kernel Consumption

It is assumed that, worldwide, more than half of cashew kernels are consumed in the form of snacks, with the rest included in confectionery (UNIDO, 2011). The cashew competes in the same market as other edible nuts including almonds, hazelnuts, walnuts, pecans, macadamias, pistachios, and peanuts. India is the top cashew consumer with 0.228 kg Per Capital followed by the United States, whose consumption increased from 109,448 metric tons in 2012 to 148,256 metric tons in 2016. Germany consumed about 0.97 pounds per capita in 2016 with consumption of 35,930 MT, for 37% growth. It is also the third importer of cashew from Vietnam, Benin, Cote d'lvoire, India, Nigeria, and Netherlands. (Table 3-3)

[Table 3-3] Leading Cashew Kernel-Consuming Countries

RANK	COUNTRY	CONSUMPTION MT
1.	India	301,719
2.	United States	143,256
3.	Germany	35,930
4	Netherlands	17,236
5	United Kingdom	16,772
6	Canada	14,267
7	France	8,649
8	Japan	8,040
9	Saudi Arabia	7,854

Source: BOT

The domestic consumption of cashew kernels in Tanzania varies with time and lifestyle. A change in food habits following an improvement in living standards and lifestyle in urban areas has led to increased usage of cashew by the food industry, especially in bakery and confectionery products. Besides, the use of value-added cashew products as snacks is also on the rise among the middle, upper-middle, and high-income groups. As a result, Tanzania's domestic consumption of cashew kernels increased from an estimated 20,000 tons in 2010 to around 25,000 metric tons in 2012. The increasing domestic consumption of cashew kernels is attributable to improved distribution systems through street vendors, shops, minimarts, and supermarkets.

4. National Technology Roadmap Deployment

Tanzania is working to transform its economy into an industrial one so as to attain middle income by the end of 2025. The agriculture sector and cashew subsector are the major source of raw material for agro-industries and employment creation. Therefore, appropriate technologies along the value chain are of crucial importance for the effective and efficient performance of the subsector to meet the intended goals.

4.1 Outline of the Roadmap

The technology roadmap is a guiding document that provides the future direction and prospects for the cashew subsector for a period of ten (10) years. It tends to analyze the technology candidate that will enable the sector to meet its goals and vision. The development of technology roadmap (TRM) involves four phases: planning and preparatory phase, visioning phase, development phase, and implementation and evaluation phase. These phases involve a number of stakeholder consultations to carry out situation analysis (SWOT) on various aspects such as social, technological, economic, environment, and political issues. Driver analysis is a basic tool used to identify the products and product prioritization based on various criteria.

4.2 Future Prospects of the Cashew Subsector

The technology roadmap is expected to increase production, productivity, volume of processed cashew (Kernel), export volume of cashew kernel, and value-added products from cashew and cashew byproducts. Its implementation will help the country attain medium income through employment, selling of products, and foreign currency contribution. The table below shows the projected production and processing by 2023/2024.

[Table 3-4] Current and Projected Production of RCN and Processing by 2024

	2018/19	2019/20	2020/21	2021/22	2022/23	2023/24
Production (MT)	244,000	350,000	450,000	600,000	800,000	1,000,000
Processing						
Small-scale	1,300	11,800	16400	26,000	95,800	120,600
Medium-scale	16,200	46,200	91,200	142,200	202,200	268,200
Large-scale	42,000	82,000	111,000	162,000	182,000	212,000
Total Processed	**59,500**	**140,000**	**218,600**	**330,200**	**480,000**	**600,800**
Percentage	24%	40%	49%	55%	60%	60%

Source: Cashewnut Board

4.3 Technology Drivers

The technology drivers can be grouped into four categories such as social, technological, economic, and political as explained below.

4.3.1 Social Drivers

The increase in population within and outside the country together with various product requirements based on cultural practices and lifestyle provides an opportunity for the agricultural products market. In addition, more than 65 percent of Tanzania's population is engaged in agriculture, where youths are emerging as the group that currently has interest in crop and livestock production. Therefore, use of improved technology is crucial for increased production, quality, and reduced cost of production.

4.3.2 Technological Drivers

In and outside the country are a number of technologies that can be exploited for increasing the production and quality of the products. The existing cashew technologies such as processing technology are manual and semi-automated, so they are less efficient in supporting the processing of large volume and quality produce. There are inadequate technologies for cashew product diversification such cashew apple (juice, jam, wine, and ethanol), cashew butter, and extraction of cashew shell liquid. Similarly, improved storage technologies for both RCN and finished products are inadequate.

Nonetheless, research institutions have developed improved technologies such as improved seeds, processing and value addition technologies, crop management, seedling production technologies (TC, grafting) and others, which need to be upscaled. It is important to protect the developed technologies (Intellectual Property - IP) and introduce, evaluate, and adopt appropriate technologies along the value chain. Cashew stakeholders need to be capacitated to enable participation in the uptake of improved technology and consequently take part in technology transfer.

4.3.3 Economic Drivers

The growing demand for cashew products worldwide, growing interest of various stakeholders to invest in the cashew subsector, expansion of production areas, current government efforts to export processed cashew, government commitment, and existence of traders, warehouse operators, and large-scale block farms make the future prospects of the cashew industry promising. For cashew to contribute efficiently to economic growth, the use of improved cost-effective technologies along the value chain is inevitable.

4.3.4 Environmental Drivers

Tanzania has seven (7) agro-ecologies namely Northern, Eastern, Central, Southern, Southern Highland, Western, and Lake Zones, and they are suitable for the production of various crops. The climate ranges from arid, semi-arid, humid, temperate, and tropical; hence its suitability for growing different types of crops. The seasonality also provides Tanzania a cooperative advantage in increasing the production of cashew as it differs from other major producing countries (Appendix 3). Additionally, the presence of land suitability map is a guiding tool for the design and implementation of eco-friendly agriculture (green technology) practices. Cashew production, processing, and value addition practices with minimal negative effect on the environment need to be adhered to for sustainability. This can be achieved by ensuring that all key cashew stakeholders are fully engaged from the planning to the implementation phase.

4.3.5 Political Driver

The country has stable and political will to support crop subsector development. It has established an investment incentive scheme and a blueprint aimed at facilitating investment in agriculture and other disciplines. The Tanzania Investment Center (TIC) plays a coordination role for smooth investment for local and external investors. Moreover, the country has strong national, regional, and international relationships and legal and regulatory frameworks. The political stability and commitment to support cashew development serve as opportunities that will enable the transformation of the cashew subsector into a middle-income industry when optimally utilized.

4.4 Opportunities and Threats

There are a number of opportunities and threats associated with the cashew subsector, influencing the subsector's performance.

4.4.1 Opportunities

- Availability of suitable land for cashew production
- Available indigenous technology
- Seasonality
- Availability of agri-institutions (R&D, universities) for technology development
- Availability of improved technologies (varieties, value addition technologies)
- Growing demand for agri-produce including cashew in the local, regional, and international markets
- Industrialization drive
- Growing population worldwide
- Available policy statement and programs
- Regional and international cooperation (on agreements)
- Political will through government commitment
- Road networking

- Presence of Tanzania Agricultural Development Bank (TADB)

4.4.2 Threats

- Limited number of extension officers, with low adaptation to change
- Inadequate research studies and research facilities
- Low purchasing power of farmers (agro-machinery)
- Low quality and quantity (inadequate knowledge of GAP and GMP)
- Change of government priority
- Regional conflict
- Weak, outdated policies and laws
- Inadequate processing technologies
- Price fluctuation
- Inadequate quality storage facilities

4.5 Macro Technology Roadmap for Strategic Products

4.5.1 Identification of Products and Key Technologies

Product Identification is done based on the stipulated criteria such as economic importance, potential for success, and strategic importance. The listed key products were analyzed and ranked as shown in the table below. The selection was based on deskwork done during the third workshop in Korea in April 2019. The scores of the identified key products (crops) are as shown in Table 3-5.

[Table 3-5] Product Identification

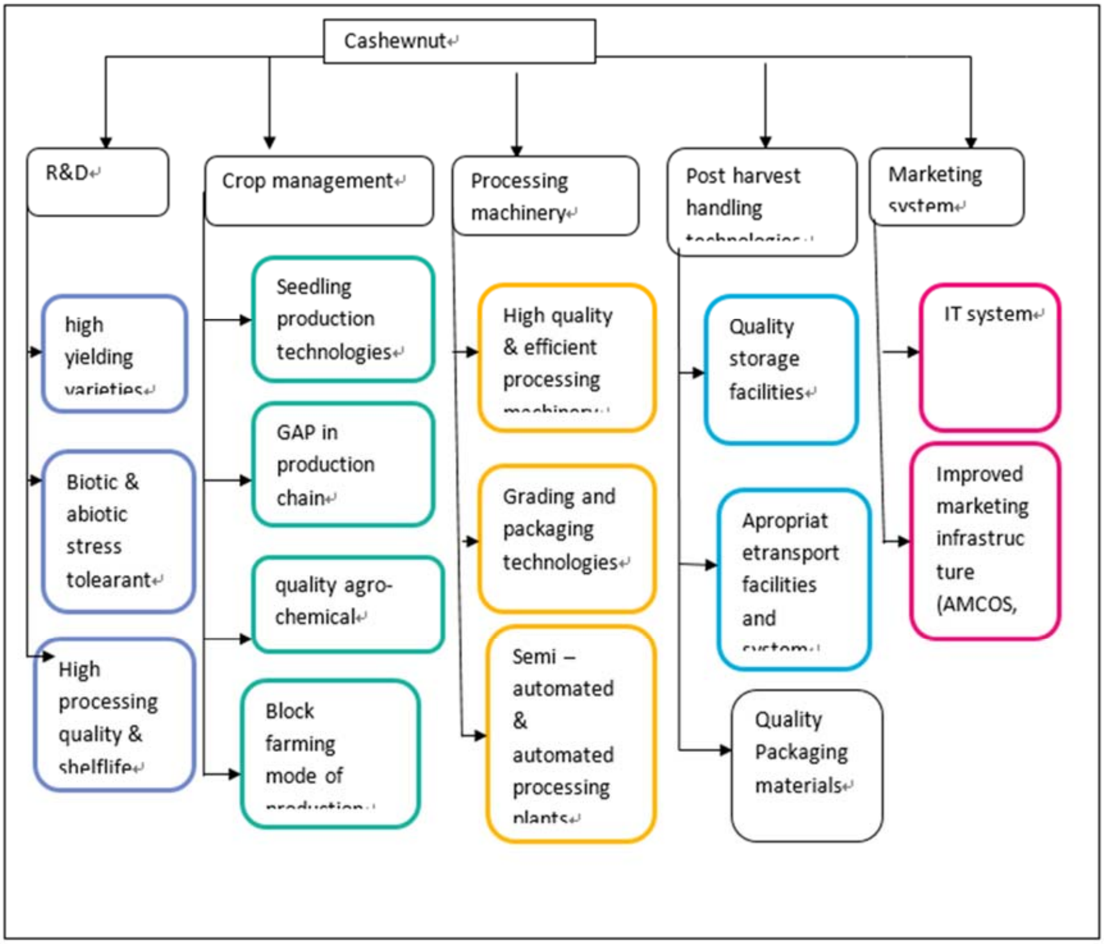
Crop	Product ranking (1-5; 1 very low, 2 low, 3 medium, 4 high, and 5 very high), Criteria used and weight			Total	Rank
Crops	Economic impact (2)	Potential for success (1)	Strategic importance (3)		
P1: Cotton	4	5	3	12	2
P2: Coffee	3	4	3	10	4
P3: Cashew	5	5	3	13	1
P4: Cereals (maize, grains)	3	5	2	10	4
P5: Cassava	2	4	3	9	5
P6: Sugarcane	3	5	2	10	4
P7. Oil seeds sunflower, palm	2	4	3	9	5
P8. Sisal	3	3	2	8	6
P9. Spices	3	5	3	11	3

The result showed that the cashew subsector is a product identified based on the given criteria.

4.5.2 Key Technologies Identified for Cashew

After selecting the product, key technologies were identified as shown in the technology trees below.

[Figure 3-10] Technology Trees



[Table 3-6] Macro Technology Roadmap for Cashew (Nut)

Vision	To transform the agriculture sector through vibrant and sustainable practices that assure food security, commercialization, and contribution to industrialization by 2028									
Goal	To enhance agriculture productivity and quality products through the selection, adoption, and development of appropriate technologies, infrastructure, and support R&D for market-driven products									
Future prospects	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
	Drivers Market expansion, technical know-how,			Drivers Market expansion, agro-processing industries, volume of production, growing population				Drivers: Technical know-how, economic value		
Cashew	1st performance target			2nd performance target				3rd performance target		
R&D	High yielding (10%), Biotic and abiotic stress-tolerant (5%), High processing quality & shelf life (10)			High yielding (30%), Biotic & stress-tolerant (25%), High processing quality & shelf life (20)				High yielding (20%), Biotic & abiotic stress-tolerant (25%), High processing quality & shelf life (20%)		
Crop management	Seedling production technologies (20%), GAP (20), Quality agro-chemical (10%), block farming model (5)			Seedling production technologies (30%), GAP (30), Quality agro-chemical (20%), block farming model (20)				Seedling production technologies (30%), GAP (30), Quality agro-chemical 30%), block farming model (40)		
Processing machinery	High-quality & efficient processing machinery (10%), Grading and packaging technologies (15%), semi-automated and automated processing plant (5%)			High-quality & efficient processing machinery (20%), Grading and packaging technologies (20%), semi-automated and automated processing plant (15%)				High-quality & efficient processing machinery (25%), Grading and packaging technologies (20 %), semi-automated and automated processing plant (15)		
Post-harvest handling technologies	Quality storage facilities (20), Appropriate transport facilities (10), quality packaging materials (5%)			Quality storage facilities (30), Appropriate transport facilities (20), quality packaging materials (15%)				Quality storage facilities (30), Appropriate transport facilities (20), quality packaging materials (20%)		
Marketing system	Effective IT system (10%), improved market infrastructure (20%)			Effective IT system (15%), improved market infrastructure (30)				Effective IT system (20%), improved market infrastructure (20)		

[Table 3-7] Technologies for Cashew Value Chain Improvement

Technology alternatives	Key technologies	Importance of technology	Technology acquisition strategy	Technology development stage	Final target %	Accomplishment period		
						Short	Medium	Long
R&D	High yielding varieties	High	R&D	Applied	60	10	30	20
	Biotic and abiotic stress-tolerant	High	R&D	Applied	55	5	25	25
	High processing quality and shelf life	High	R&D International collaboration	Applied	50	10	20	20
Crop management	Seedling production technologies (grafting, TC, direct seed)	High	R&D	Applied	80	20	30	30
	GAP (planting density, tree rehabilitation)	High	R&D	Basic	80	20	30	30
	High-quality Agro-chemical	High	Outsourcing	Applied	60	10	20	30
	Block farming model	Medium	Adoption & development	Applied	55	5	20	30
Processing machinery	High-quality and efficient processing machinery	High	International collaboration	Development	55	10	20	25
	Quality grading and packaging technologies	High	Licensing	Applied	55	15	20	20
	Semi-automated & automated	Medium	International collaboration	Development	55	5	15	25

Technology alternatives	Key technologies	Importance of technology	Technology acquisition strategy	Technology development stage	Final target %	Accomplishment period		
						Short	Medium	Long
	processing plant							
Post-harvest handling	High-quality storage facilities (equipped warehouses for RCN, kernels)	High	Licensing	Applied	80	20	30	30
	Appropriate transport facilities	High	Outsourcing	Applied	45	5	20	20
	Quality packaging materials	Medium	R&D	Applied	40	5	15	20
Marketing system	Effective IT system	Medium	Outsourcing		45	10	15	20
	Improved marketing infrastructure (warehouses, AMCOS, Cooperative unions)	High	Licensing	Applied	70	20	20	30

5. Conclusions

The implementation of the key technologies identified along the cashew value chain will help attain the national goal of becoming a middle-income economy. The increased production and productivity as well as rehabilitation and establishment of efficient processing technologies will enable the country to become a leading producer and exporter of processed cashew kernel in East Africa and beyond.

There will be strategic implementation of each technology so that its impact could be realized at a specified time and in a cost-effective manner. The processing plant will be established in a strategic area with high production potential. Additionally, the research findings and recommendations will be adhered to, and all key actors will be fully engaged at all levels to ensure successful implementation.

CHAPTER 4

Current Status of the Mineral Industry in Tanzania

Policy Consultation on Developing a National Technology Roadmap for Accelerated Industrialization

Chapter 4. Current Status of the Mineral Industry in Tanzania

1. Mineral Policy Objectives

The Mineral Policy of 2009 seeks to address the challenges faced by the mineral sector in Tanzania. The Government continued to attract and enable the private sector to take the lead in exploration, mining, mineral beneficiation, and marketing. Its purpose is to increase the mineral sector's contribution to the GDP and alleviate poverty by integrating the mining industry with the rest of the economy. Other objectives include:

a) To improve the economic environment in order to attract and sustain local and international private investment in the mineral sector;

b) To promote economic integration between the mineral sector and other sectors of the economy so as to maximize the contribution of the mineral sector to the economy;

c) To strengthen the legal and regulatory framework for the mineral sector and enhance the capacity for monitoring and enforcement;

d) To strengthen the institutional capacity for effective administration and monitoring of the mineral sector;

e) To participate strategically in viable mining projects and establish an enabling environment for Tanzanians to participate in the ownership of medium and large-scale mines;

f) To support and promote the development of small-scale mining so as to increase its contribution to the economy;

g) To facilitate, support, and promote the increased participation of Tanzanians in gemstone mining;
h) To establish transparent and adequate land compensation, relocation, and re-settlement schemes in mining operations;
i) To strengthen the involvement and participation of local communities in mining projects and encourage mining companies to expand their corporate social responsibilities;
j) To promote and develop a marketing system of minerals to ensure that miners get the right values of minerals traded in formal markets;
k) To promote and facilitate value addition activities within the country to increase income and employment opportunities;
l) To promote research & development and training as required in the mineral sector and encourage its utilization;
m) To develop a local base for technical capacity;
n) To improve communication on the mineral sector to the public through education and provision of accurate and timely information;
o) To strengthen the institutional capacity of the Geological Survey of Tanzania (GST) to enable it to perform its roles and functions effectively and efficiently;
p) To strengthen cooperation with the regional and international bodies to take advantage of the facilities, resources, and information provided by the organizations;
q) To promote safety and maintain hygiene conditions and protect the environment in the mining areas; and
r) To encourage and promote women's participation in mining activities and strengthen enforcement of laws and regulations against child labor in mining activities.

1.1 Vision and Mission

The Mineral Sector Policy of the United Republic of Tanzania provides a written declaration of the framework of policy objectives and statements that will guide the Government and stakeholders in the management of the mineral sector on a sustainable basis. Accordingly, the policy is based on the following Vision and Mission:

1.1.1 Vision

Effective mineral sector contributing significantly to the acceleration of socio-economic development through sustainable development and utilization of mineral resources in Tanzania by 2025.

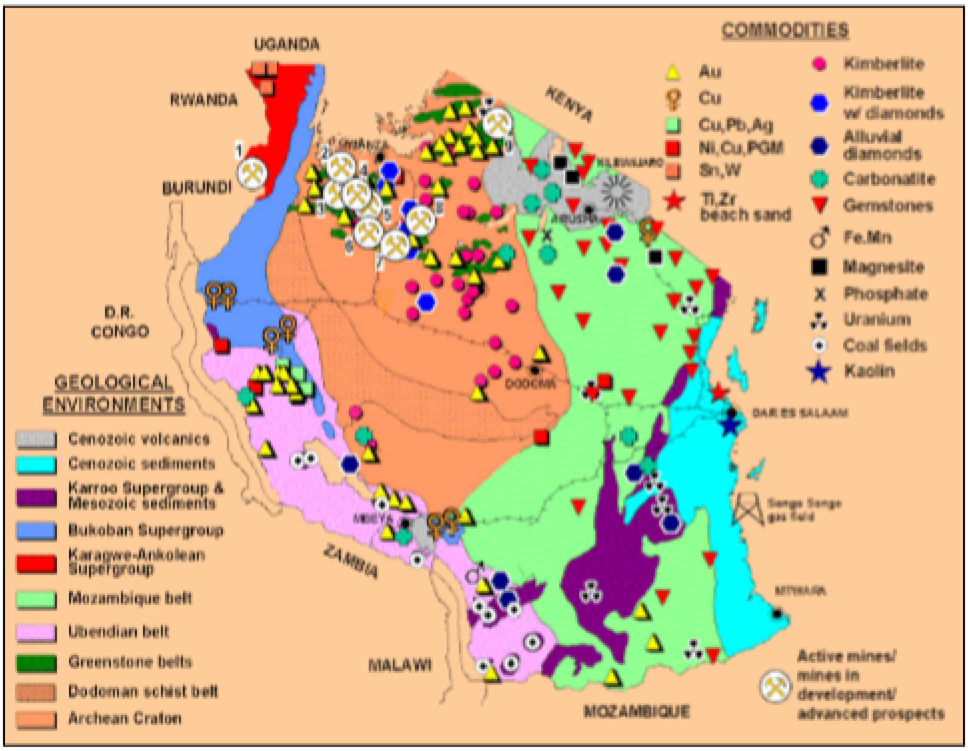
1.1.2 Mission

To set policies, strategies, and laws; regulate mineral exploration, production, trading, value addition, and mineral actors for the sustainable development of mineral resources; and integrate with other sectors of the economy.

1.1.3 Mining Sector in Tanzania

At present, the mining sector is engaged in mineral exploration, license provision, and mining of few mineral commodities and quarrying of rocks for different purposes. The country is endowed with a variety of mineral deposits with high economic potential in the following geological systems: Craton, Nyanzian, Proterozoic, Ubendian, Usagaran, Karagwe Ankolean, Bukoban, Karroo, and Cenozoic sediments and Volcanics (Figure 4-1).

[Figure 4-1] Export Geological Map and Mineral Occurrences in Tanzania



Source: Tanzania Mineral Policy 2009

The minerals found in Tanzania include: metallic minerals such as gold, iron, silver, copper, platinum, nickel, and tin; gemstones such as diamonds, tanzanite, ruby, garnet, emerald, alexandrite, and sapphire; industrial minerals such as kaolin, phosphate, lime, gypsum, diatomite, bentonite, vermiculite, salt, and beach sand; building materials such as stone aggregates and sand; and energy minerals such as coal and uranium. These minerals occur in different areas (Fig. 4-1) as follows:

a. Gold in the greenstone belt south, east, and west of Lake Victoria;
b. Diamonds in kimberlite pipes in Central and South Tanzania and southern part of Lake Victoria Goldfield;
c. Nickel, cobalt, copper, tin, and tungsten minerals in Northwestern Tanzania;
d. Titanium, vanadium, and iron southwest of Tanzania;
e. Coal southwest of Tanzania;
f. Uranium in Central and Southern Tanzania;

g. Soda ash, salt, gypsum, travertine, and trona (evaporites) in the rift valley and along the coast; and
h. Kaolin, mica, phosphate, magnesite, beach sand, diatomite, stone aggregates, dimension stone, and sand in different parts of Tanzania.

In addition to the vast mineral endowment, Tanzania has the following competitive advantages:

- Considerable unexploited mineral potential;
- Accessible geological data and information;
- Transparent process of granting mineral rights;
- Peace and political stability.

The intelligent exploitations of these minerals would play a critical role in the earning of foreign exchange and create employment opportunities in the primary and secondary industries in the country. Mining constitutes more than 50% of the country's total exports, a large part of which comes from gold. The country has gold reserves of 45 million ounces, generating revenue of over a billion USD (Mining Industry in Tanzania: An Overview. Retrieved 16 February 2018). Diamond is also found in significant amounts. Since it was opened in 1940, the Williamson diamond mine has produced 19 million carats (3,800 kg) of diamonds. Gemstones, nickel, copper, uranium, kaolin, titanium, cobalt, and platinum are also mined in Tanzania. Illegal mining and corruption are ongoing problems. In 2017, the government passed a series of bills aimed at increasing revenue from minerals and controlling illegal mining.

The ongoing exploration works has resulted in the discovery of resources in excess of 45 million ounces of gold between 1998 and 2006, with 1.5 million tons of nickel and 50 million carats of tanzanite. The economic reforms introduced in the mid-1980s reopened doors to private investment in the mineral sector. This led to the revival of large-scale mineral exploration and mining in Tanzania. The country has six gold mines, two of which are world-class deposits, and two nickel projects are underway. Annual gold production from the six gold mines is about 50 tons, ranking Tanzania among the leading gold-producing countries in Africa.

2. Mineral Exploration and Mining Opportunities

2.1 Gold

The Lake Victoria goldfields (greenstone belt) is still promising as an area of investment in gold mining. There are prospects such as Kitongo, Nyakafuru, Miyari, and Sekenke, which are not yet developed. Many others do exist following exploration in this belt. The Lupa goldfields in the southwestern part of Tanzania and Mpanda area are promising for gold mineralization with which other base metals are associated.

2.2 Diamond

Over 300 kimberlites are known in Tanzania, 20% of which are diamondiferous. Some 600 dipolar magnetic anomalies with similar geophysical characteristics are known kimberlite pipes that have been recorded during recent geophysical surveys. Also of relevance are the pseudo kimberlites or para-kimberlites along the young craters where diamonds have been discovered. Alluvial diamonds have been recorded, but a large deposit of economic exploitation has not yet been found. Locating shallow buried superficial deposits using airborne infrared surveys may prove useful. Areas in the Tabora and Singida regions are worthwhile for detailed work.

2.3 Ferrous and non-ferrous metals

Ore bodies for iron, nickel, copper, cobalt, chromium, and Platinum Group Metals (PGM) are associated with ultramafic intrusions, with tin and tungsten related to granitic intrusions. None of these metals have been mined in Tanzania, although there are advanced projects such as Kabanga nickel.

2.4 Iron

Numerous iron ore bodies have been identified in the Proterozoic rocks. Titaniferous magnetic bodies associated with anorthositic gabbro occur at Liganga in Southwest Tanzania, and they are is in close proximity (80 km) to the coal resources of Ketewaka-Mchuchuma. Shallow drilling established a resource of 45 million tons with 52 percent Fe. Titanium minerals are also known to occur in beach sands along the coast.

2.5 Platinum Group Metals (PGMs – platinum, palladium, rhodium, rhenium, osmium, and iridium).

These minerals occur in the form of layered mafic igneous intrusive such as gabbros and anorthosites and ultramafic rocks such as peridotite, dunite, and serpentine. It involves the concentration of molten sulfide droplets or oxide crystals in mafic or ultramafic magma. Localities are Kabanga, Kapalagulu, and Zanzui. Others are Kabulyanwele, Mwahanza Hill, Garauja-Basuto, Twamba, Nkenza, Itiso, Haneti, and Uluguru mountains.

2.5.1 Gemstones

Varieties of gemstones are found in the Proterozoic rock formations mainly east of the Archaean Craton. Scattered areas where gemstones are known to occur are the west and south of the Craton. The gemstones include: ruby, tanzanite, garnets, tourmaline, sapphires, spinel, topaz, scapolite, emeralds, chrysoprase, and alexandrite.

2.5.2 Carbonatites

Carbonatites are associated with the rift valley system, and they occur in the northern, southern, and central parts of the country. Alteration zones up to 1.5 km wide surround the carbonatites. Minerals hosted in carbonatites include Rare Earth Metals (thorium cerium, lanthanum, neodymium, lanthanum, and praseodymium).

2.5.3 Coal

The country possesses coal resource of as high as 1.5 billion tons, with ash content ranging from 14.2% to 45%. Coal resources occur in Karoo rock formations in the southwestern part of Tanzania. Reserves in order of 1000 million tons of coal have been proved by drilling in all the coal fields, and only 40% can be extracted by surface mining methods. Currently, coal is being exploited on a small scale at Kiwira Coal Mine. Coalfields with the highest potential are Katewaka–Mchuchuma in the Ruhuhu basin and Songwe–Kiwira. The coals vary in rank from sub-bituminous C to medium-volatile bituminous (mvb). The ash content of 15~20% go as high as 40%, with generally low sulfur content (less than 2%). The Songwe-Kiwira coalfield is the only coalfield producing coal using the mechanized underground coal mine at Kiwira.

2.5.4 Evaporites

Evaporites such salt, gypsum, and soda ash occur in the rift valley and younger rock formations along the coastal belt.

3. Geo Information

For the past 80 years, Geo Information Tanzania (GST) has being conducting geo-surveys and research activities in the country, generating a substantial amount of geoscientific data and information on the Earth's crust and natural resources. The available information includes the following:

- Vast geological, geophysical, and geochemical information, which can readily assist investors in the mineral sector in selecting prospective areas for the search of minerals.
- Geological maps covering 85% of the 322 Quarter Degree Sheets (QSD) of Tanzania. Twenty-nine percent (29%) of the maps are at 1:100,000 scale, 48% are at 1:125,000 scale, 3% are at 1:250,000 scale, and 5% of the geologically mapped QDS are not yet published.
- Other geological maps include: Block maps of the Lake Victoria Goldfields, Kigoma-Mpanda, and Lake Rukwa areas at a scale of 1:500,000.
- Geological map of Tanzania showing the geology and mineral occurrences.
- Countrywide airborne geophysical map coverage of magnetic and radiometric surveys.
- High-resolution airborne geophysical survey data for selected areas around Lake Victoria Goldfield (Kahama, Biharamulo, and Mara) and Mpanda Mineral Field.

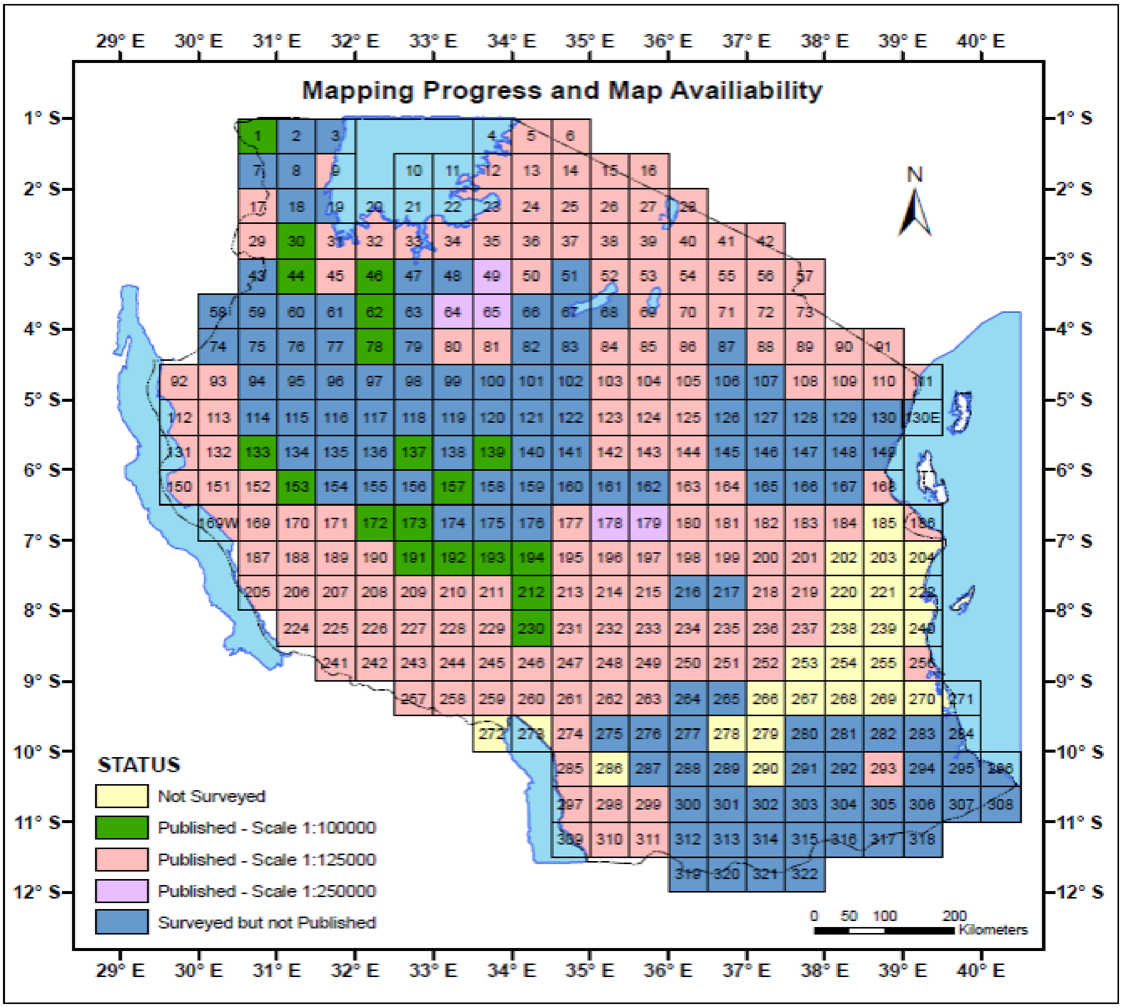
3.1 Geological Data Acquisition and Survey

The Geological Survey has made notable achievements since its establishment about 80 years ago. The achievements include:

a. 94% of the entire country has been geologically mapped, and published geological maps as quarter degree sheets (QDS) are available at various scales. This has generated basic geological and mineralogical data and information used by investors in the exploration activities. The discovery of a substantial number of mineral occurrences, some of which developed into world-class mines (e.g., Kahama and Geita Gold Mines) was achieved.

b. Increased contribution of the Mineral Sector to the country's GDP from about 0.3 % in the 1980s to about 4.3 % at the turn of the century. Such growth was facilitated by the various economic policy changes made by the Government during the late 1980s and the 1990s.
c. 100% of the entire country has low-resolution airborne geophysical data at 1 km line spacing and 120m ground clearance.
d. 16% of the entire country has high-resolution airborne geophysical data (200m – 250m line spacing and ground clearance of 45m – 70m).
e. 19% of the entire country has geochemical data at various sampling densities and sampling media.
f. Availability of modern laboratory equipment for chemical, mineralogical, geotechnical & mineral processing laboratory.
g. Availability of modern and reliable ICT infrastructure (fast Internet, intranet, high-storage capacity servers, e-mail, Public database web portal, website, fax, etc.).
h. Availability of centralized and standardized modern Database Management System (Geological and Mineral Information System-GMIS).
i. Existence of up-to-date countrywide mineral occurrences database and its associated geological metadata, which attracts further investment; thus broadening the GST market and programs.
j. Existence of Core and rock library & Geological Museum.
k. Existence of archive with a wide range of geoscientific information (geotechnical reports, geological, geophysical, and geochemical data in digital and hard copies); Possession of modern field geological gear and equipment (vehicles, tents, generators, GPS, tough books, etc.).
l. Availability of Modern geophysical equipment and facilities
m. Geological Mapping
n. The current regional geological mapping programs are focused on the thematic coverage of the country, project-oriented data capture, interpretation, and product delivery to various stakeholders (Figure. 4-2).

[Figure 4-2] Progress in Geological Mapping in Tanzania 2017-2018



Source: CAG25, Dar es salaam - Tanzania

3.2 Geoscientific Research

The Geological Survey of Tanzania (GST) is the Government agency responsible for the acquisition and storage of geoscientific data and information used in the Mining Sector. GST is one of the leading geoscientific organizations in Tanzania conducting research in various disciplines of earth science. The expertise is focused on Precambrian geology, Proterozoic geology, Paleozoic geology, quaternary geology, and natural earth resources. GST has experienced scientists specializing in various aspects of geology, environmental research, geophysics, geochemistry, and information technology. The research staff is supported by a well-equipped geo-laboratory as well as modern IT infrastructure.

The geological earth resources of strategic and economic importance – such as precious and base metals, industrial minerals, aggregates, dimension stones, and energy sources (coal and geothermal) – continue to be the core of GST's research mission. Research on studies on geohazards and environment are also core activities. Future research priorities will include areas like assessment of mineral resources, technologies for the sustainable use of minerals, and advanced multidimensional geological modeling. GST's main focus is applied science and research.

3.2.1 Laboratory

As one of the leading establishments in the country, the Geological Survey of Tanzania (GST) laboratory undertakes chemical, mineralogical, and petrological analyses of mineral, soil, rock, and water samples. It also provides services on environmental studies and mineral beneficiation tests. The GST Laboratory has developed a Quality Assurance System (QAS) that complies with the ISO/IEC 17025-General Requirements for the Competence of Testing and Calibration Laboratories. Some of the services offered in these laboratories include:

3.2.2 Chemical analysis

Services offered in this laboratory include sample preparation, fire assay, and wet chemistry analysis. The chemical laboratory performs a wide range of determinations, including:

- Major elements (SiO_2, Al_2O_3, TiO_2, P_2O_5, CaO, Fe_2O_3, K_2O, MgO, MnO, and Na_2O.
- Base metals (Pb, Zn, Ni, Cu).
- Trace elements (Ag, As, CO, Cr, Cu, MO, Ni, Pb, Zn, Au, Hg).
- Precious metals (Au, Ag).
- Water analysis (both clean and waste) for various components such as TDS, sulfates, conductivity, salinity, pH, turbidity, color, and cations is done in water color.
- Pollutant elements (metals) such as mercury, lead, zinc, cadmium, arsenic, and selenium.

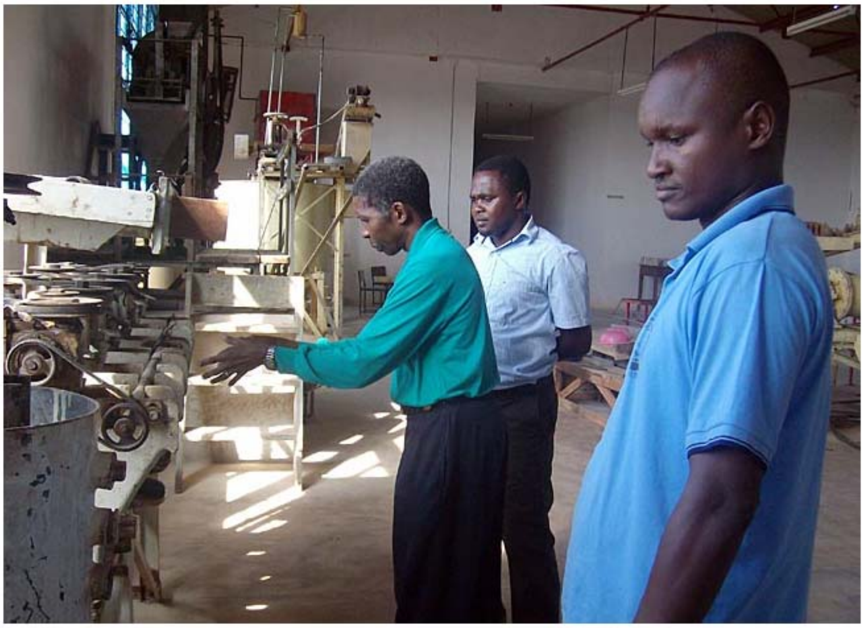
3.2.3 Petrophysical Laboratory

The Geological Survey of Tanzania (GST) has a petrophysical laboratory for measuring the petrophysical properties of rock and ore samples. Petrophysical parameters can be used for the quantification of geophysical interpretations, and they also have direct geological applications. Petrophysical laboratory measurements include density, porosity, magnetic susceptibility, and remnant magnetization and galvanic or inductive resistivity. The GST laboratory equipment has been developed for the rapid measurement of many samples with high accuracy. The measured data are currently stored in a database, and data analyses and summaries are available for scientific and commercial purposes.

3.2.4 Mineral Processing Laboratory

The minerals processing laboratory incorporates facilities for technology testing in mineral separation using magnetic, electrostatic, gravitational, froth flotation, and gold cyanidation techniques. Upgrading tests on industrial minerals are also conducted.

[Figure 4-3] Mineral Processing Laboratory



Source: GST

3.2.5 Petrological and Mineralogy Laboratory

The mineralogy and petrology laboratory is equipped with modern facilities for sample preparation, analysis of different minerals and rocks, and gemstone identification. The laboratory offers services on rock cutting, polishing, and preparation of thin and polished sections. It uses Digital Polarized Microscopes and X-ray machines in identifying rock types and behaviors.

3.2.6 Engineering Geology Laboratory

The GST lab has recently extended its services to Engineering/Geotechnical Services. The Geotechnical lab was established in 2014 to serve various stakeholders especially in the Construction Sector. The lab is capable of characterizing the physical behavior of pebble/rock, bim (concrete cube), brick, and clay for various applications including the construction of infrastructures such as buildings, roads, and dams.

The services offered by the Engineering Geology Laboratory are as follows:

- Soil testing, e.g., triaxial, direct/residual test, particle size distribution, permeability, atterberg limits consolidation;
- Aggregate test, e.g., abrasion, impact value, ten percent fines, flakiness index, elongation index, crushing value, particle density, and water absorption;
- Rocks mechanics testing, e.g., permeability of core, determination of porosity, uniaxial and triaxial compressive strength, tensile strength of material, point load strength point, determination of Young's modulus and fish's ratio;
- Bim and brick tests (Concrete Cubes and Brick Test);
- Provide guidance to various stakeholders in identifying the best possible areas for development activities such as geo-technical investigations;
- Slope stability analysis for geo-hazards studies.

3.2.7 Chemical Laboratory

The chemical laboratory's main functions are the determination of elements in rocks, soils, plants, and water samples for mineral exploration and Environmental Monitoring applications.

Some of the services offered by the chemical laboratory include:

- Sample preparation (grinding of rocks) for further laboratory investigations;
- Analysis of Base Metals and Trace Elements, precious metals and environmental pollutants such as mercury, zinc, pollutant elements, e.g., Hg, Pb, Zn, Cd, Ar in rocks, soils, water, and biological materials;
- Analysis of environmental samples (water, plants, and soil) for assessing environmental impacts associated with mining activities.

3.2.7.1 Some Equipment Available in the Chemical Laboratory

This laboratory has modern equipment applying the current technology. These include several X-ray Radiation Machines (XRF- Epsilon5 & Minipal4), ICP-OES, GFAA, FAAS, CS-Analyzer, UV-Vis Spectrometers, AAS Fitted with Graphite Furnace, AAS – 50 & 55 B – Atomic Absorption Spectrometer, Centrifuge (5702 & 46933) and others.

3.3 Library Services

The library of the Geological Survey of Tanzania is the main library for geosciences information in Tanzania. The library and archive services feature published and unpublished material such as reports, field notes, and observation data from studies carried out by GST and other geoscientific institutions collected since 1885. The GST library and archive hold more than fourteen thousand (14,000) items including more than seven hundred (700) geologic maps and 2,800 geophysical maps.

3.4 Geological Museum at GST as a Center of Learning

GST has a museum with rock and mineral specimens from all over the country, fossil specimens, and geological models. The museum has a variety of rocks, minerals, and fossils collected from within the country and abroad. The Geological Museum has an exchange program with other Geological Museums in the world, wherein exchanges are made for geological samples and other materials including meteorites and fossils for similar samples from other parts of the world.

Visitors from Tanzania and outside the country visit the museum to see and learn from the rich geological collections of Tanzania and some of the products of GST.

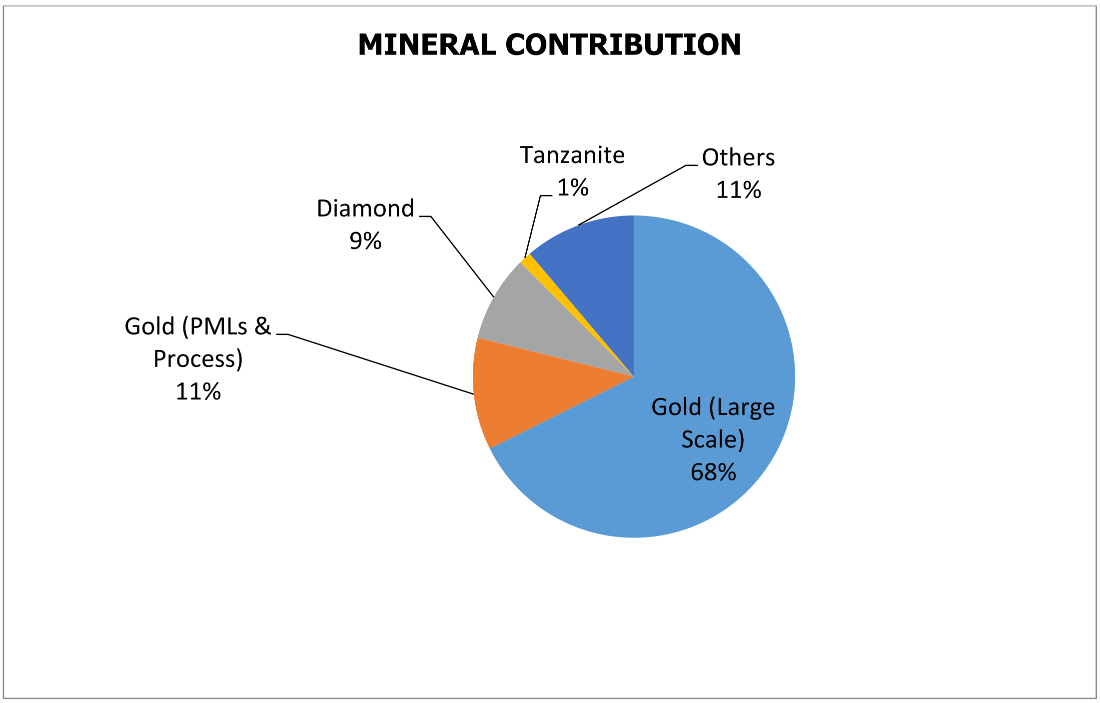
[Figure 4-4] Geological Museum at GST

Source: GST

4. Contribution of the Mineral Sector to the National Economy

The mining industry of Tanzania has experienced a dramatic development during the last twelve years. It has attracted over USD 2.5 billion Foreign Direct Investment since 1998. Within a period of over ten years, Tanzania has risen from an insignificant gold producer to become Africa's third largest gold-producing country. Gold production in 2003 was about 1.3 million ounces; the country's mineral exports rose from $312 million in 2001 to $ 1,032 million in 2007. Tanzania has experienced a significant rise in the contribution of the mineral sector to the economy. Export values from minerals have increased from 0.07% in 1995 to 42.4% in 2005, and mining contribution to GDP has gone up from 1.4% in 1995 to 2.7 % in 2007. For the Month of March 2019, the contributions per mineral type are shown in Figure. 4-5. Obviously, gold accounted for a big share of contributions followed by Diamond and Tanzanite.

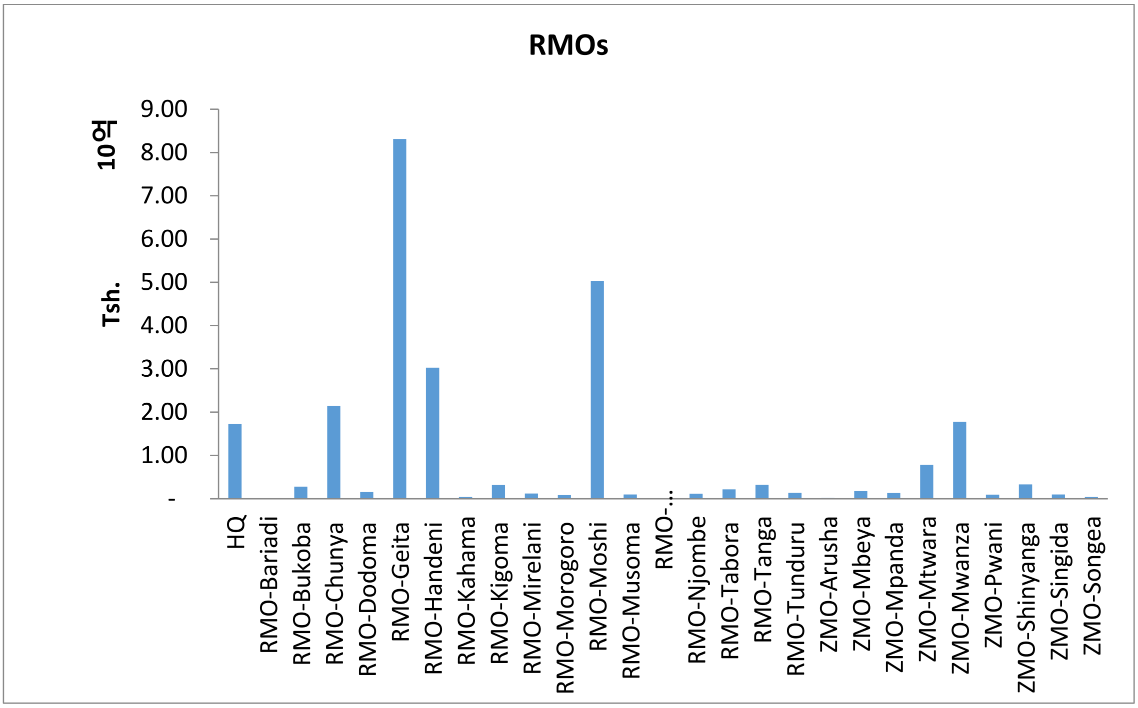
[Figure 4-5] Minerals' Contributions to the Tanzanian Economy for March 2019



Source: Tanzania Monthly Report for Minerals

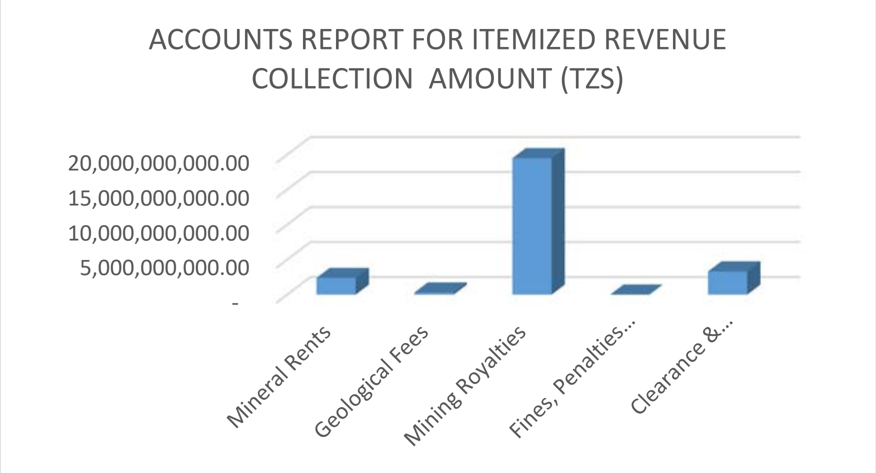
The availability of minerals per regions is shown in Figure. 4-6. It can be seen from the figure that the Geita Region contributes a big portion of government revenue as it produces a lot of minerals specifically Gold, followed by Moshi, Handeni, and Chunya. Note that RMO stands for Resident Mining Officer, and ZMO, Zonal Mining Officer. This is where the report has been collected. The report is normally produced monthly.

[Figure 4-6] Minerals Production and Sales



Source: Tanzania Monthly Report for Minerals

[Figure 4-7] Revenue Collections by the Government



Source: Tanzania Monthly Report for Minerals

The itemized revenue collections for Tanzania in January 2019 show that the Mining Royalties contributions are very high compared to fine penalties and forfeitures.

[Table 4-1] Revenue Collection for January 2019

Type of Mineral	Value (TZS)	Type of Mineral	Value (TZS)
Bricks (Mud)	263274936.5	Ruby	0
Bricks (Stones)	0	Kaolin	24766666.67
Gold (Large- & Medium-Scale)	3.03597E+11	Coal	9440051370
Silver	1297006072	Feldspar	16666666.67
Gold (PML & PCLs)	6290342055	Dolomite	2100000
Gold (Dealers)	0	Other Gemstones	279890195.6
Gold (VAT Leaching)	480904868.5	Sapphire	106054300
Diamond (Large-Scale & Dealers)	40163286256	Green Garnet	79387383.11
Tanzanite (Rough)	1587477360	Murrum	763518333.3
Tanzanite (Reject)	48490624.17	Pozzolana	1200000
Tanzanite (Cut&Polished)	80401993.7	Marble	122065760.3

Type of Mineral	Value (TZS)	Type of Mineral	Value (TZS)
Limestone	10352030467	Magnesite	56426000
Lime	0	Stones	1714616947
Salt	608234088.7	Copper Ore	708333.3333
Clay	588827014	Iron	13950416.67
Sand	6652791583	Building Materials	438306420.1
Gypsum	1805616755	CO_2 (ToL)	1449773118
Tin (Cassiterite)	0	Other Minerals	573504516.1
Aggregates	12845322499	Granite Blocks	0
Graphite	2202980000		

Source: Tanzania Monthly Report for Minerals

[Figure 4-8] Revenue Collection for January 2019 by Share Mineral Type

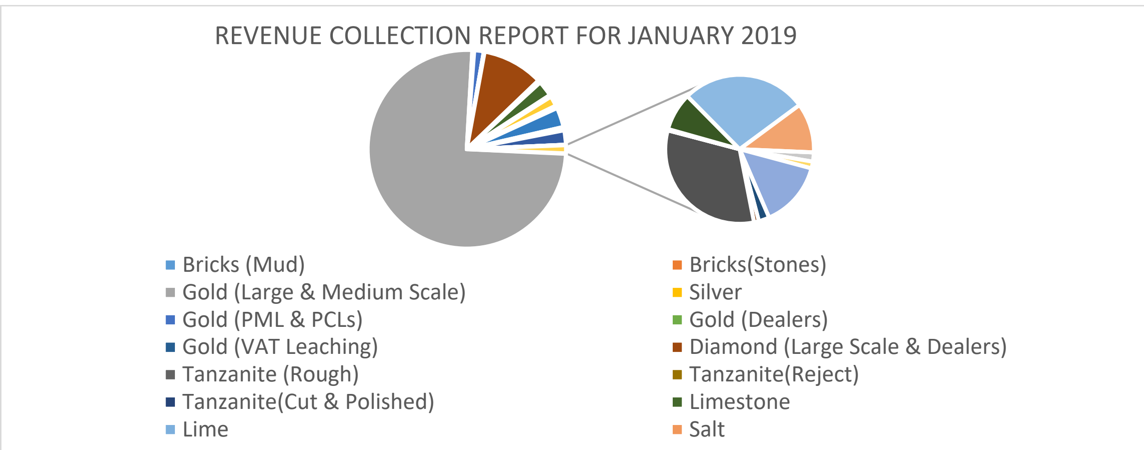


Table 4-1 and Figure 4-8 almost show the same thing -- revenue collection for January 2019 -- although Fig. 5 also shows the share of the Mineral Type making bigger contribution than the other, which is Gold followed by Diamond. It should be noted that the other types of Minerals are not yet mined on a large scale like Uranium, etc. The sector has the task of developing an effective mineral prospecting system and looking for technologies to shorten the associated processes and enhance added value .

4.1 Current and Future Prospects of the Mineral Industry of Tanzania

In the financial year 2017-2018, the mining sector led the growth of the country with approximately 18 percent compared to the financial year 2016-2017 wherein the sector posted a 12 percent growth. In financial year 2015/2016, revenue collections stood at TZS 210,185,986,188.33, increasing to TZS 214,506,024,960.99 in 2016/2017. In 2017/18, the Ministry planned to collect a total of Tsh. 194,633,498,763 but exceeded the target by 155% by collecting TZS 301.6 billion. In year 2018/2019, revenues have continued to grow; as of February 2019 alone, a total of TZS 218,450,606,093 was already accumulated, which is 106% of the target for the period. Increasing revenue collections has enabled the mining sector to contribute to GDP by 4.8% for the Financial Year 2016/2017 and 2017/2018.

In addition, after the Government completed the construction of a wall around the Tanzanite quarry, Mirerani, established other infrastructure such as the Mining Trade Hall (Brokers House), the One-Stop Center for Mineral Business Services, and installed electronic infrastructure and digital protection systems (CCTV security systems), productivity has increased. Mining has increased Tanzanite production from Kg. 147 (in 2017) to Kg. 781 (in 2018). The government has also spent 11.9 billion TZS to assist the miners by building seven centers of excellence and three centers of demonstration centers and conducting training for small-scale miners.

Accordingly, the development of National Technology Roadmap (NTRM) for the Mining Sector is very critical in utilizing the mineral resources of the country for sustainable economic and social economic development. Therefore, this guiding NTRM for Tanzania was developed by a combination of technical experts from Tanzania who attended the capacity development workshops in Korea with the assistance of STEPI and funding by the Government of Korea. The developed NTRM will be the guiding document for the country to develop many related documents to cover the different technology sectors in Tanzania.

4.2 Scope and Boundary Conditions of the Roadmapping Effort

The development of the technology roadmap for the mineral sector involved the submission of a project proposal to STEPI by MoEST and request for support in the capacity building on NTRM development. As a starting point of the NTRM development, three sectors were selected -- Mining, Industries (Cotton), and Agriculture (Crops) -- as previously mentioned. The STEPI experts conducted training and guided the development of the guiding NTRM document for Tanzania.

CHAPTER 5

Capacity Development Process

Policy Consultation on Developing a National Technology Roadmap for Accelerated Industrialization

Chapter 5. Capacity Development Process

1. Invited Capacity Development Workshop in Korea

1.1 Introduction

1.1.1 Objectives

This workshop served as a platform for Tanzanian officials and qualified practitioners to deepen their knowledge in technology roadmapping. Moreover, they had a chance to visit organizations and places relevant to the NTRM project.

1.1.2 Research Team

[Table 5-1] Research Team

Name	Organization	Position & Area of Expertise	Contact Information
Chi Ung Song	STEPI	Vice President/ Project Manager	cusong@stepi.re.kr
So Young Cho	STEPI	Researcher/ Project Coordinator	sycho07@stepi.re.kr
Hyun Yim	KISTEP	Senior Research Fellow	hyim@kistep.re.kr
Johng-Ihl Lee	SUNY Korea	Professor	anothermile@sunykorea.ac.kr
Sehee Park	SUNY Korea	Researcher	sehee.park@stonybrook.edu

1.1.3 Tanzanian Delegation

[Table 5-2] Tanzanian Delegation

Organization	Name	Position
Ministry of Education, Science, and Technology (MoEST)	Margareth Misheck Komba	Assistant Director for STI
	Rogers Alfayo Msuya	Senior Research Officer
Ministry of Agriculture (MoA)	Upendo Marko Mndeme	Senior Agriculture Officer
Mineral Resources Institute (MRI)	Fredrick Martin Mangasini	Acting Principal
Tanzania Industrial Research and Development Organisation (TIRDO)	Julius Elias Daud	Research Officer
Commission for Science and Technology (COSTECH)	Amos Muhunda Samwel Nungu	Director General
	Gerald Majella Kafuku	Principal Research Officer
	Judith Francis Kadege	Principal Research Officer
	Mashuhuri Mwinyihamisi Mushi	Principal Research Officer
Export Processing Zones Authority (EPZA)	Innocent John Umbulla	Senior Officer
	Veronica Gideon Ngerageza	Senior IT Officer

1.1.4 Schedule

[Table 5-3] Workshop Schedule

Date	Time		Place	Key Activity
3/31 (Sun.)	16:45	23:20	Julius Nyerere Int'l Airport and Dubai Int'l Airport	Departure from Julius Nyerere Int'l Airport (Emirates, EK726)
4/1 (Mon.)	03:40	16:55	Dubai Int'l Airport and Incheon Int'l Airport	Arrival at Incheon Airport (Emirates, EK322)
	18:00	19:30	Seoul	Move to Seoul & check-in (Novotel Ambassador Dongdaemun, Seoul)
	20:00	21:00	Seoul	Dinner
4/2 (Tues.)	09:00	11:00	Seoul	**[Lecture 1]** - Introduction to Technology Roadmap - Korea's experience: National Technology Roadmap
	11:00	13:00	Seoul	**[Lecture 2]** - Introduction to the Roadmaking Pilot Project (hands-on project) STEP 1: Vision Building
	13:00	14:00	Seoul	- Luncheon
	14:00	18:00	Seoul	**[Lecture 3]** - STEP 2: Environment Analysis - STEP 3: Strategic Products/Services
	18:00	19:00	Seoul	Dinner
4/3 (Wed.)	09:00	13:00	Seoul	**[Lecture 4]** - STEP 4: Key Technologies - STEP 5: Technology Roadmapping
	13:00	14:00	Seoul	Luncheon
	14:00	18:00	Seoul	**[Presentation]** - Group Presentations

Date	Time		Place	Key Activity
	18:00	19:00	Seoul	Dinner
4/4 (Thurs.)	09:00	10:00	Seoul → Songdo	Move to Songdo (SUNY Korea)
	10:00	12:00	SUNY Korea	**[Lecture 5]** - Industrial Technology Roadmap - Wrap-up and follow-up
	12:00	13:00	Songdo	Luncheon
	14:00	17:00	Songdo	Incheon Free Economic Zone Tour
	17:30	19:00	Songdo	Dinner
	19:00	20:00	Songdo → Seoul	Move to Seoul
4/5 (Fri.)	10:00	11:00	Seoul	**[Organization Visit]** Korea Institute of Science and Technology (KIST)
	12:00	13:00	Seoul	Luncheon
	13:00	18:00	Seoul	Field Trip
	18:00	19:00	Seoul	Dinner
	23:55	04:25	Incheon Int'l Airport → Dubai Int'l Airport	Departure from Incheon (Emirates, EK323)
4/6 (Sat.)	10:15	14:40	Dubai Int'l Airport → Julius Nyerere Int'l Airport	Arrival at Julius Nyerere Int'l Airport (Emirates, EK725)

1.2 Invited Capacity Development Workshop

STEPI held the Capacity Development Workshop (III) on the development of a national technology roadmap (NTRM) on April 1- 5 in Seoul, Korea. This workshop was the third in a series linked to the previous two workshops held on August 21-24, and October 21-26, 2018. The Tanzanian Delegation from the Ministry of Education, Science, and Technology, Ministry of Agriculture, Mineral Resources Institute, Tanzania Industrial Research and Development

Organisation, Tanzanian Commission for Science and Technology, and Export Processing Zone Authority was invited to Seoul, where they deepened their knowledge in technology roadmapping in order to participate in a tutorial workshop for developing the NTRM for three industries (Cotton, Agriculture, and Mining).

1.2.1 1st-day Lecture and Tutorial Workshop

a) Introduction to TRM

STEPI researchers along with Dr. Hyun Yim (KISTEP) and Dr. Ki-ha Hwang (KISTEP) described the concept of technical roadmap and its development methodology and shared Korean case studies.

b) NTRM Procedure

Dr. Yim explained the five steps of NTRM Procedure: (1) Vision Building, (2) Environment Analysis, (3) Strategic Products/Services, (4) Key Technologies, and (5) Technology Roadmapping.

c) NTRM Tutorial Workshop

The ten participants were split into three groups. Group A chose cotton, Group B chose agriculture, and Group C chose mining as strategic industries. Moreover, strategic products/services were prioritized by evaluating the economic impact, strategic importance, and potential for success. As a result, Group A identified cloth and cotton lint, Group B selected cashew, cotton, cereals, and spices, and Group C chose gold, graphite, and iron as strategic products.

1.2.2 2nd-Day Lecture and Tutorial Workshop

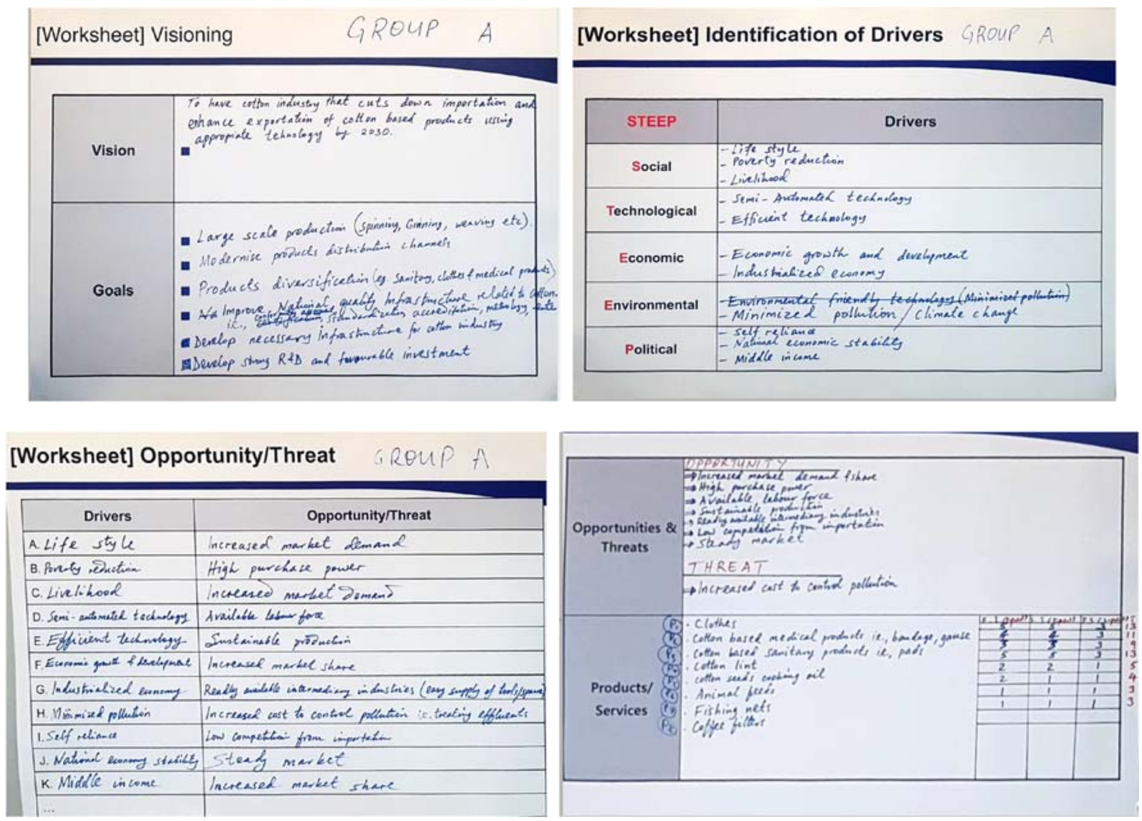
a) NTRM Tutorial Workshop II

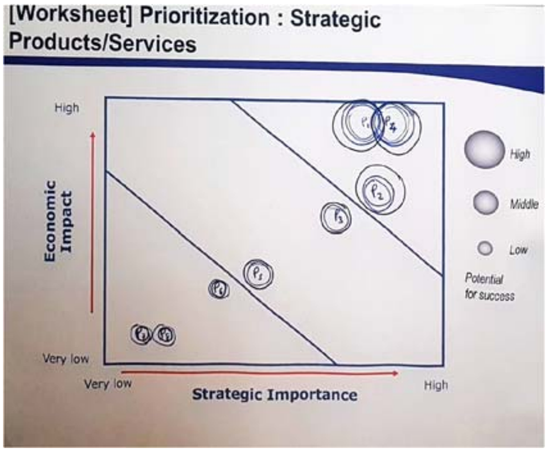
The three groups identified a strategic product, derived key technologies, and completed the technology roadmap. Group A selected cloth as strategic product, identified a core technology necessary for the commercialization of cloth, and developed a virtual technology roadmap. Group B selected cashew nut as a strategic product for agricultural cultivation, identified a core technology group necessary for commercialization, such as processing, and created a virtual technology roadmap. Likewise, Group C selected gold as a strategic product for the mining industry, identified a core technology for transportation and energy procurement, and created a virtual technology roadmap.

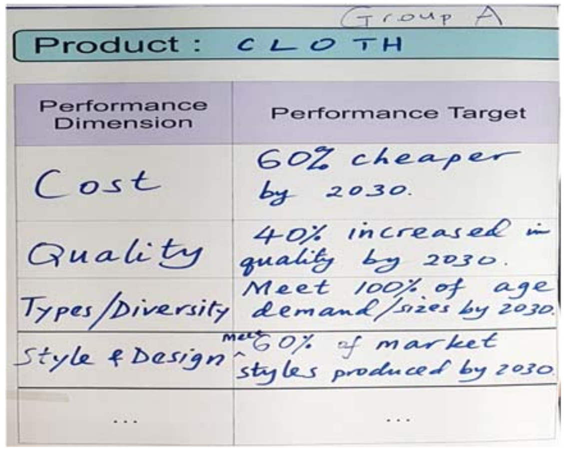
1.2.2.1 Group Exercises at the Tutorial Workshop

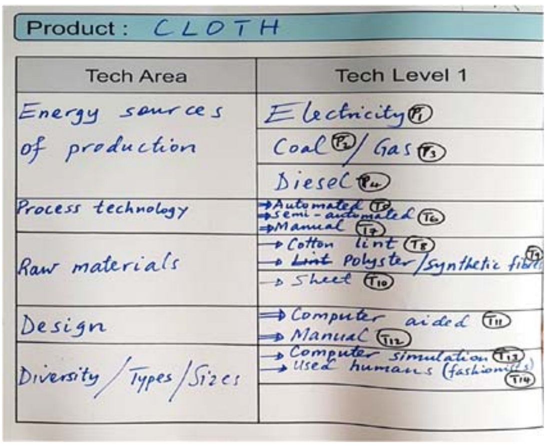
These are the results of the group exercises at the workshop:

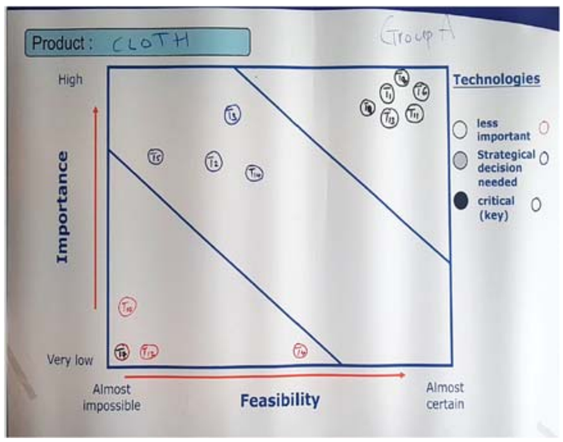
[Figure 5-1] Group A –CTA (Cotton)

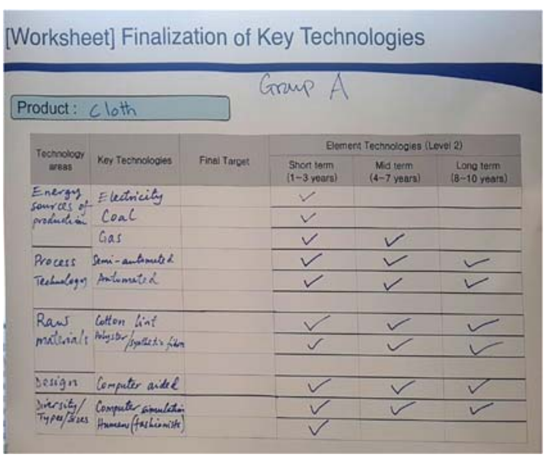


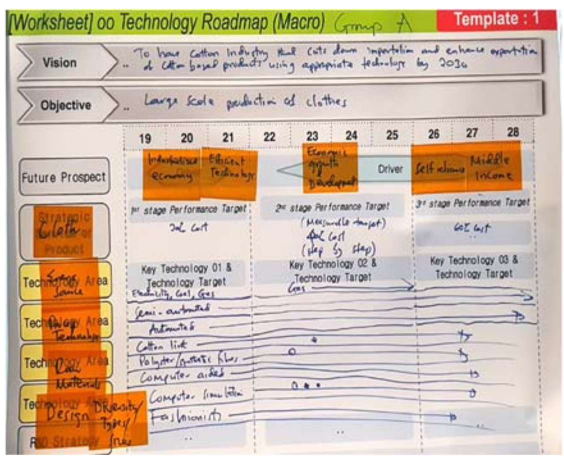

[Worksheet] Prioritization : Strategic Products/Services
High
Economic Impact
Very low
Very low
Strategic Importance
High
High
Middle
Low
Potential for success


Group A
Product : CLOTH
Performance Dimension
Performance Target
Cost
60% cheaper by 2030.
Quality
40% increased in quality by 2030.
Types/Diversity
Meet 100% of age demand/sizes by 2030.
Style & Design
Meet 60% of market styles produced by 2030
...
...

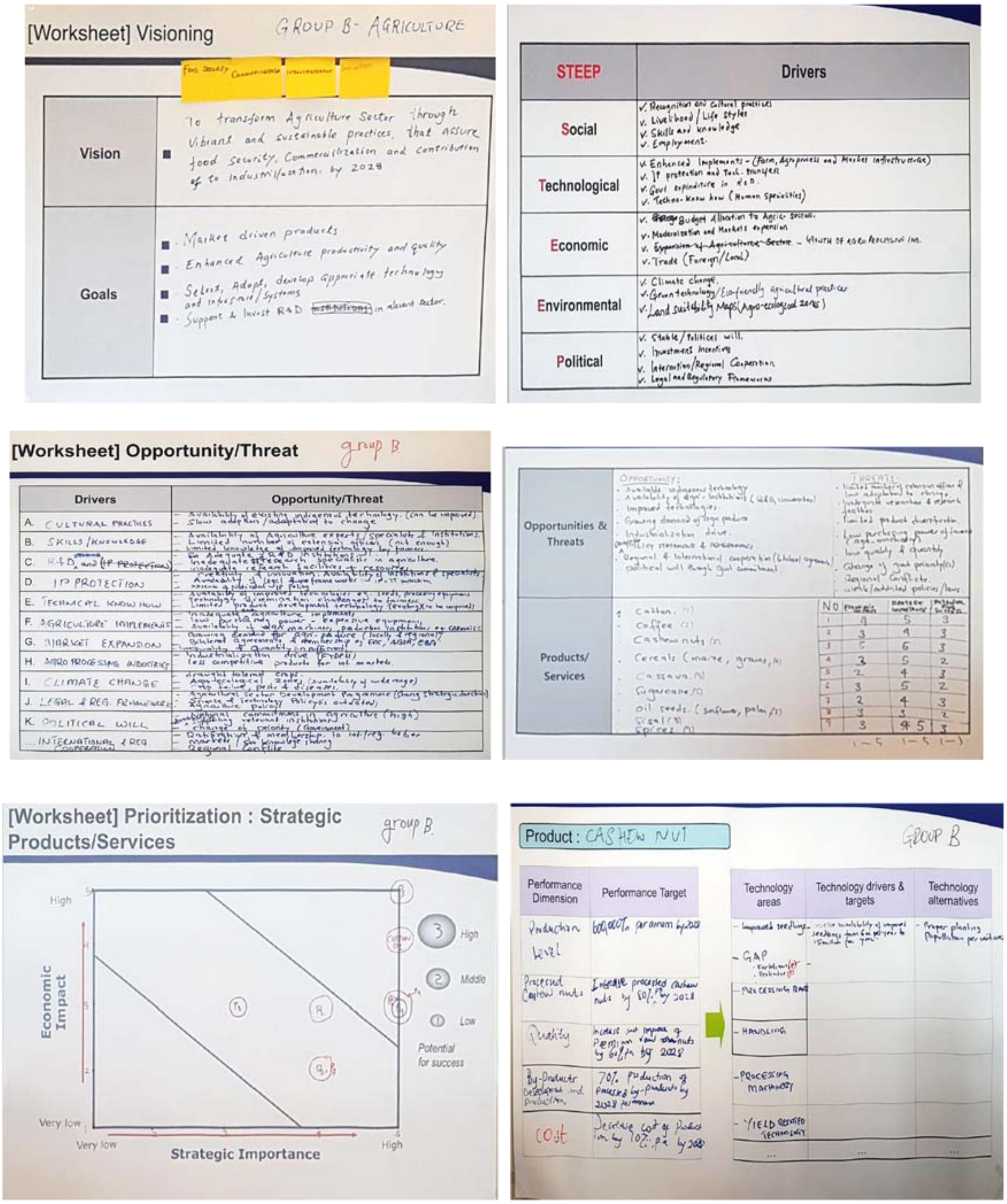

Product : CLOTH
Tech Area
Tech Level 1
Energy sources of production
Electricity
Coal / Gas
Diesel
Process technology
Automated
Semi-automated
Manual
Raw materials
Cotton lint
Polyster/synthetic fibre
Sheet
Design
Computer aided
Manual
Diversity/Types/Sizes
Computer simulation
Used humans (fashionists)

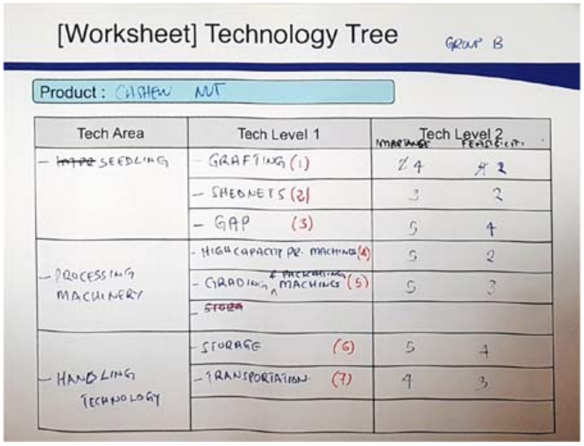

Product : CLOTH
Group A
High
Importance
Very low
Almost impossible
Feasibility
Almost certain
Technologies
less important
Strategical decision needed
critical (key)

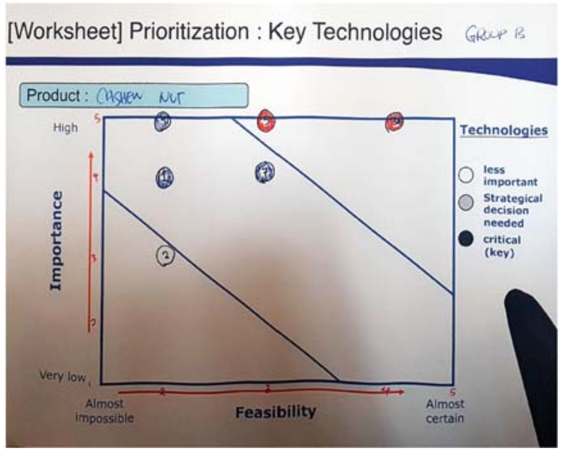

[Worksheet] Finalization of Key Technologies
Group A
Product : Cloth
Technology areas
Key Technologies
Final Target
Element Technologies (Level 2)
Short term (1~3 years)
Mid term (4~7 years)
Long term (8~10 years)
Energy sources of production
Electricity
Coal
Gas
Process Technology
Semi-automated
Automated
Raw materials
Cotton lint
Polyster/synthetic fibre
Design
Computer aided
Diversity/Types/Sizes
Computer simulation
Humans (fashionists)

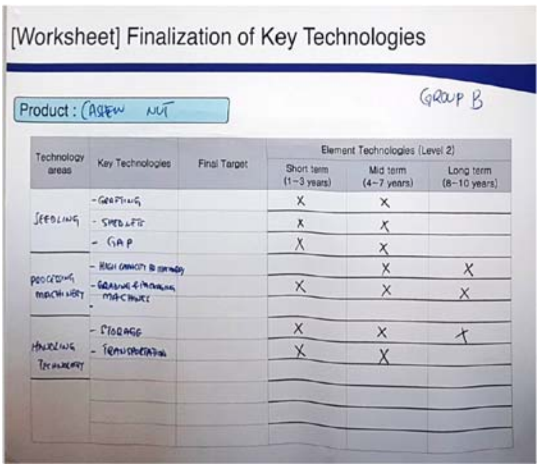

[Worksheet] oo Technology Roadmap (Macro)
Group A
Template : 1
Vision
To have cotton industry that cuts down importation and enhance exportation of cotton based products using appropriate technology by 2030
Objective
Large scale production of clothes
19
20
21
22
23
24
25
26
27
28
Future Prospect
Driver
1st stage Performance Target
2nd stage Performance Target
3rd stage Performance Target
Key Technology 01 & Technology Target
Key Technology 02 & Technology Target
Key Technology 03 & Technology Target
Technology Area
R&D Strategy

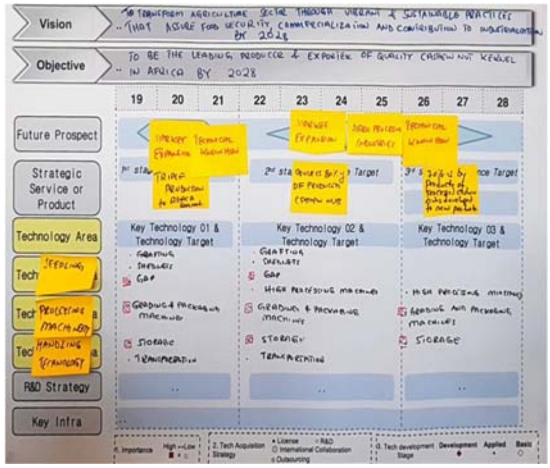
[Figure 5-2] Group B - Agriculture



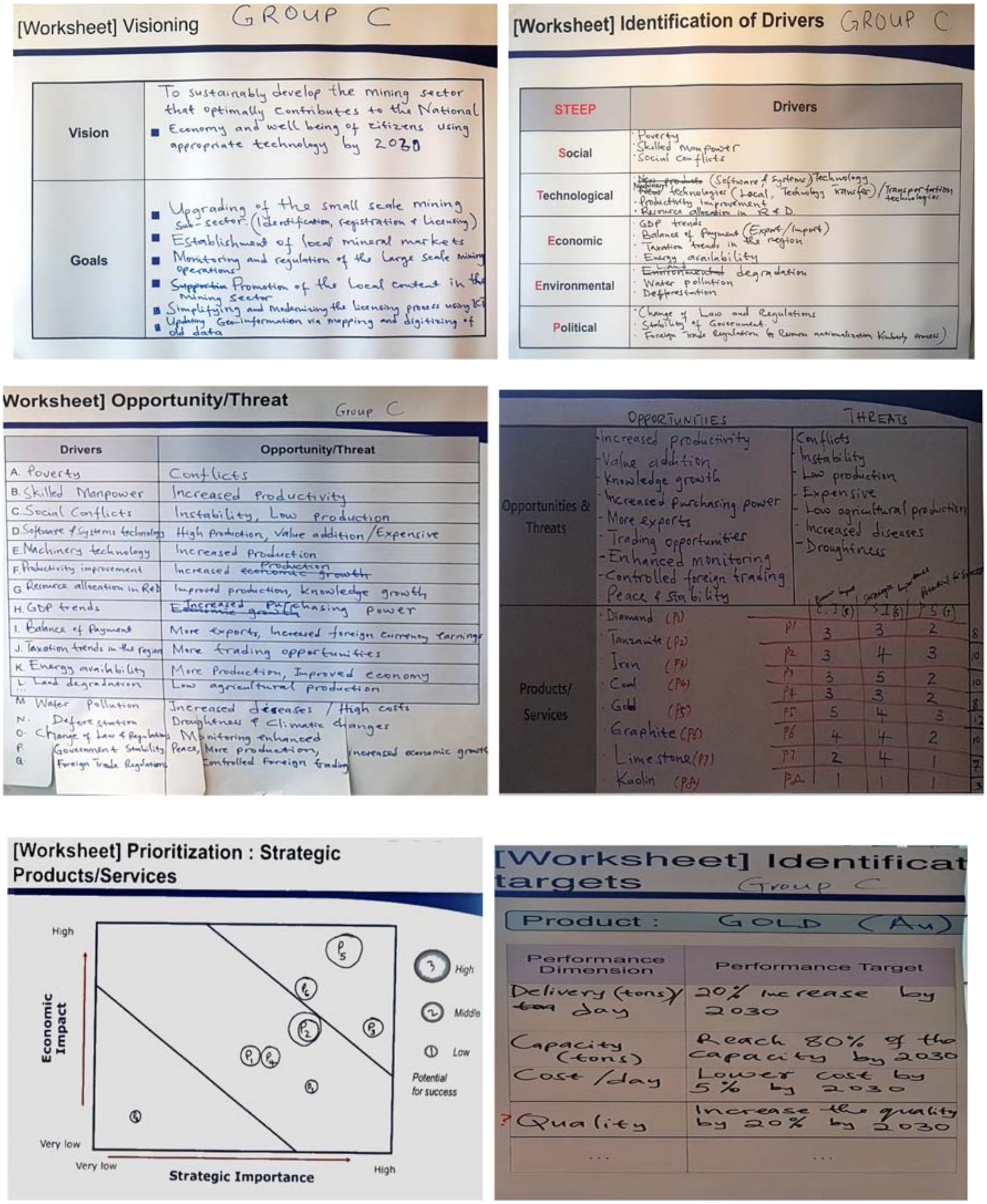

[Worksheet] Technology Tree
GROUP B
Product : CASHEW NUT
Tech Area
Tech Level 1
Tech Level 2
SEEDLING
GRAFTING (1)
SHEDNETS (2)
GAP (3)
PROCESSING MACHINERY
STORAGE (6)
HANDLING TECHNOLOGY
TRANSPORTATION (7)

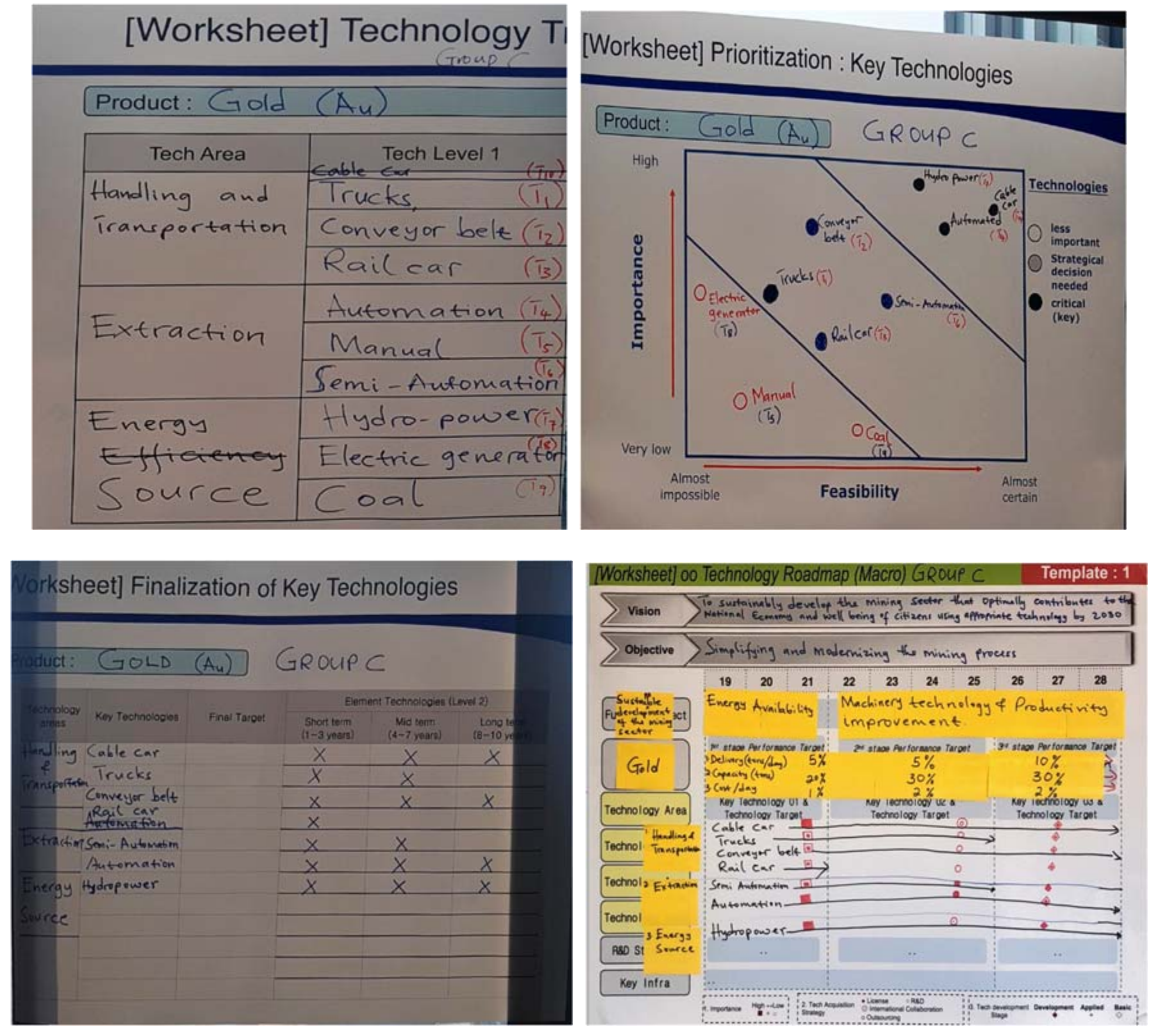

[Worksheet] Prioritization : Key Technologies
GROUP B
Product : CASHEW NUT
High
Importance
Very low
Almost impossible
Feasibility
Almost certain
Technologies
less important
Strategical decision needed
critical (key)


[Worksheet] Finalization of Key Technologies
GROUP B
Product : CASHEW NUT
Technology areas
Key Technologies
Final Target
Element Technologies (Level 2)
Short term (1~3 years)
Mid term (4~7 years)
Long term (8~10 years)
SEEDLING
GRAFTING
GAP
STORAGE


Vision
Objective
TO BE THE LEADING PRODUCER & EXPORTER OF QUALITY CASHEW NUT KERNEL IN AFRICA BY 2028
19
20
21
22
23
24
25
26
27
28
Future Prospect
Strategic Service or Product
Technology Area
R&D Strategy
Key Infra
Key Technology 01 & Technology Target
Key Technology 02 & Technology Target
Key Technology 03 & Technology Target

[Figure 5-3] Group C - Mining





1.2.3 3rd-Day Lecture and Field Trip

a) NTRM as a Policy

The delegation visited the State University of New York, Korea (SUNY Korea) in Songdo International City and attended a lecture by Prof. Johng-Ihl Lee. Prof. Lee introduced the national technology roadmap as a policy by analyzing the relationship between technology and policy, technology and R & D, and technology and forecasting. He emphasized the importance of industry, experts, and appropriate development plans in developing a successful technology roadmap.

b) Field trip in Songdo International City

After attending the lecture, the delegation visited the IFEZ center, the Smart City operation center, the Incheon city history museum, and the global campus in Songdo International City.

1.3 Visit to Kistorium

On the last day, the delegation visited KIST. In its Kistorium (KIST History Museum), the delegation was able to see the history of KIST since its establishment.

1.4 Wrap-up Meeting

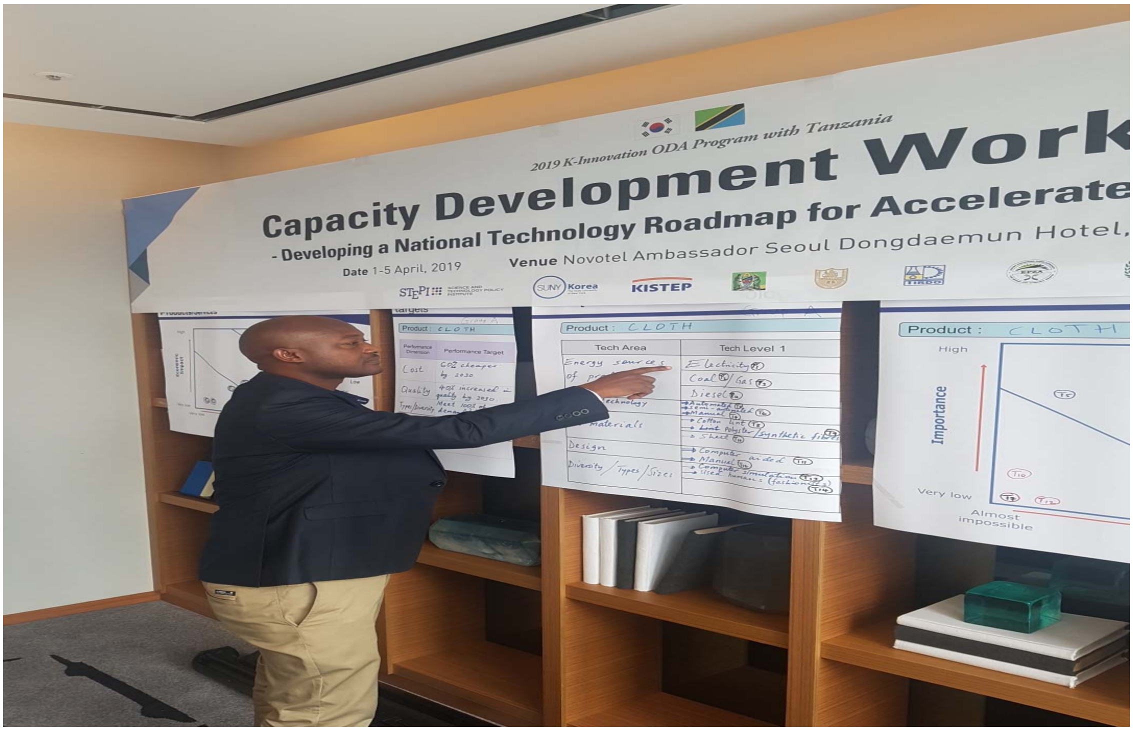
The STEPI research team has agreed to visit Tanzania to hold an intensive workshop in August and assist in the completion of Tanzania's technology roadmap in three sectors.

[Figure 5-4] Invited Capacity Development Workshop

Capacity Development Workshop

Capacity Development Workshop

Capacity Development Workshop



Capacity Development Workshop

2. Local Capacity Development Workshop

2.1 Introduction

2.1.1 Objectives

In relation to the national technology roadmap (NTRM) development project, STEPI held a Local Capacity Development Workshop (IV) in Zanzibar, Tanzania, on August 20-22, 2019. This workshop was organized in line with the capacity development workshop previously held on August 21-24, 2018, October 21-26, 2018, and April 1-5, 2019. The invited Tanzanian delegation, composed of government officials, research officers, and relevant experts, mostly came from the Ministry of Education, Science, and Technology (MoEST), Ministry of Agriculture, Ministry of Minerals, Commission for Science and Technology (COSTECH), Tanzanian Industrial Research and Development Organisation (TIRDO), and Export Processing Zones Authority (EPZA).

2.1.2 Research Team

[Table 5-4] Research Team

Name	Organization	Position & Area of Expertise	Contact Information
Chi Ung Song	STEPI	Vice President / PM	cusong@stepi.re.kr
So Young Cho	STEPI	Researcher/ Project Coordinator	sycho07@stepi.re.kr
Hyun Yim	KISTEP	Senior Research Fellow/ NTRM Methodology	hyim@kistep.re.kr
Johng-Ihl Lee	SUNY Korea	Professor / NTRM Methodology	anothermile @sunykorea.ac.kr

2.1.3 Tanzanian Delegation

[Table 5-5] Tanzanian Delegation

Organization	Name	Position
Ministry of Education, Science, and Technology (MoEST)	Margareth Misheck Komba	Assistant Director for STI
	Rogers Alfayo Msuya	Senior Research Officer
Ministry of Agriculture	Upendo Marko Mndeme	Senior Agriculture Officer
Ministry of Minerals	Reuben Mdoe	Mineral Process Instructor
Tanzania Industrial Research and Development Organisation (TIRDO)	Julius Elias Daud	Research Officer
Commission for Science and Technology (COSTECH)	Gerald Majella Kafuku	Principal Research Officer
	Judith Frances Kadege	Principal Research Officer
Export Processing Zones Authority (EPZA)	Innocent John Umbulla	Senior Officer
	Elias Aloyce Mshomba	Senior ICT Officer

2.1.4 Schedule

[Table 5-6] Workshop Schedule

Date	Time		Place	Key Activity
8/18 (Sun.) - 8/19 (Mon.)	10:25	16:10	Incheon Airport → Istanbul Airport	Departure from Incheon Airport (Turkish Airlines, TK8091)
	19:45	03:05	Istanbul Airport →Zanzibar Airport	Arrival at Zanzibar Airport (Turkish Airlines, TK0567)
	11:30	11:50	Dar es Salaam Airport → Zanzibar Airport	Arrival of Tanzanian Research Team (Hahn Air Systems, H1 6364)
	16:00	17:00	Hotel Verde	Meeting with a Local Consultant
	19:00	21:00	Welcome Dinner	
8/20 (Tues.)	09:00	13:00	Hotel Verde	**[Lecture 1] Review of and Introduction to TRM** - Introduction to Technology Roadmap - Summary of Previous Workshop
	13:00	14:00	Hotel Verde	Luncheon
	14:00	18:00	Hotel Verde	**[Lecture 2]** - Module 1: Identifying the Vision and Goals
	18:30	20:00	Dinner	
8/21 (Wed.)	09:00	13:00	Hotel Verde	**[Lecture 3]** - Module 2: Environment Analysis - Module 3: Identifying Strategic Products
	13:00	14:00	Hotel Verde	Luncheon
	14:00	18:00	Hotel Verde	**[Lecture 4]** - Module 4: Identifying Key Technologies - Module 5: Deploying the Technology Roadmap
	19:00	21:00	Dinner	

Date	Time		Place	Key Activity
8/22 (Thurs.)	09:00	11:00	Hotel Verde	**[Lecture 5]** - Module 6: Group Presentations
	11:00	13:00	Hotel Verde	**[Lecture 6]** NTRM Policy Implementation for the Tanzanian Economy
	13:00	14:30	Farewell Luncheon	
	16:05	16:25	Zanzibar Airport → Dar es Salaam Airport	Departure of the Tanzanian Research Team (Hahn Air Systems, H1 6398)
	17:20	23:50	Zanzibar Airport → Doha (Hamad)	Departure from Zanzibar Airport (Qatar Airways, QR 1352)
8/23 (Fri.)	02:00	16:55	Doha (Hamad) Airport → Incheon Airport	Arrival in Incheon Airport (Qatar Airways, QR0858)

2.2 Preliminary Meeting with a Local Consultant

The Korean research team included Dr. Chi Ung Song, the Vice President of STEPI, Ms. So Young Cho, a researcher at STEPI, Dr. Hyun Yim, a senior research fellow at KISTEP, and Prof. Johng-Ihl Lee of SUNY Korea.

The team chose the meeting venue and prepared for a local capacity-development workshop with local consultants. The local consultant, Dr. Rogers Msuya, confirmed the arrival of the Tanzanian participants.

2.3 Local Capacity Development Workshop in Zanzibar

Since most of the nine (9) participants attended the previous workshop in April, they were assigned to the same group as last time. Therefore, they had the chance to revise the strategic roadmap they have developed during the previous workshop.

2.3.1 1st-day Lecture and Tutorial Workshop

a) Review of the Invited Capacity Development Workshop

Dr. Hyun Yim briefly re-described the definition of technology roadmap and summarized what the group had done in the previous three workshops.
Dr. Yim again explained the five steps of the NTRM Procedure: (1) Vision Building, (2) Environment Analysis, (3) Strategic Products/Services, (4) Key Technologies, and (5) Technology Roadmapping.

b) NTRM Tutorial Workshop

In the practical training session, the participants were divided into three groups according to the strategic industries such as cotton (Group A), agriculture (Group B), and mining (Group C) selected in the previous workshop, and then derived the strategic products/services from the sectors. As a result, Group A, chose clothes, cotton-based medical sanitary products, seed cotton, and cotton lint, whereas Group B chose cashew nut, cotton, maize, and cloves. Group C chose gold, diamond, and tanzanite as strategic products.

2.3.2 2nd-Day Lecture and Tutorial Workshop

a) NTRM Tutorial Workshop II

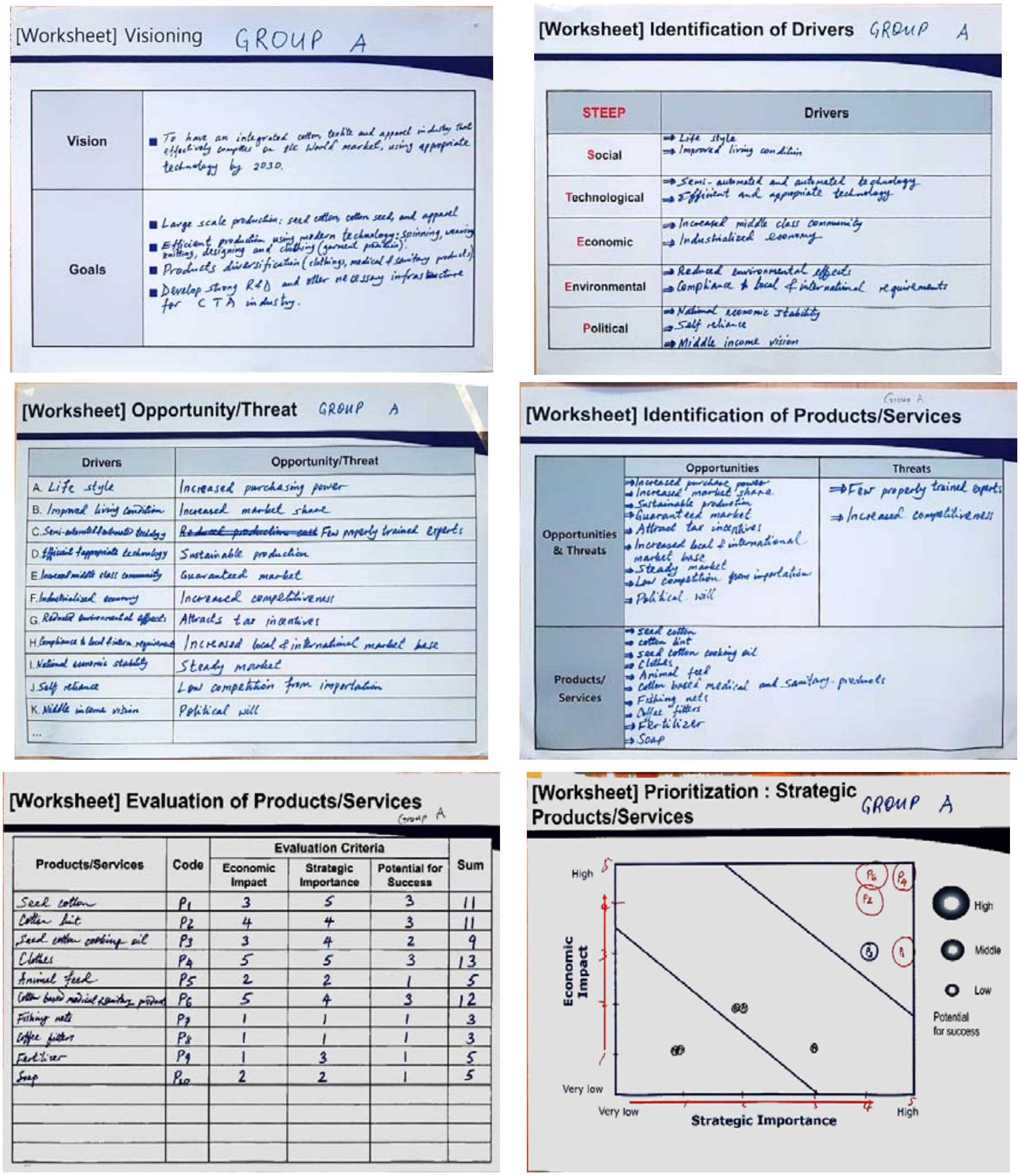
From the selected strategic products above, the three groups chose a strategic product, derived the key technologies, and developed the technology roadmap. Group A selected cloth as strategic product, identified a core technology, and then developed a virtual technology roadmap by 2030. Group B selected cashew nut as strategic product, identified a core technology, and then developed a virtual technology roadmap by 2030. Group C selected gold as strategic product, identified a core technology, and developed a virtual technology roadmap by 2030.

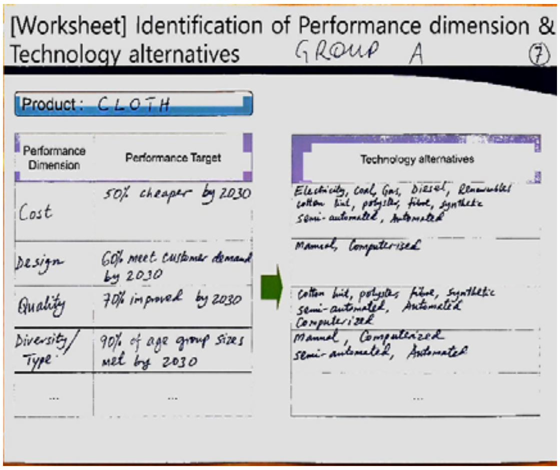
2.3.3 3rd-Day Group Presentations and Lecture

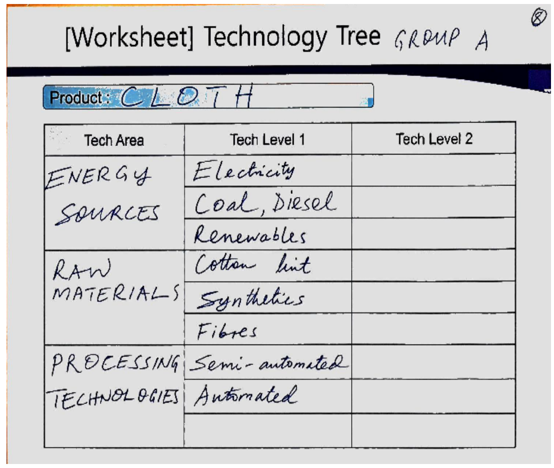
a) Group Presentations

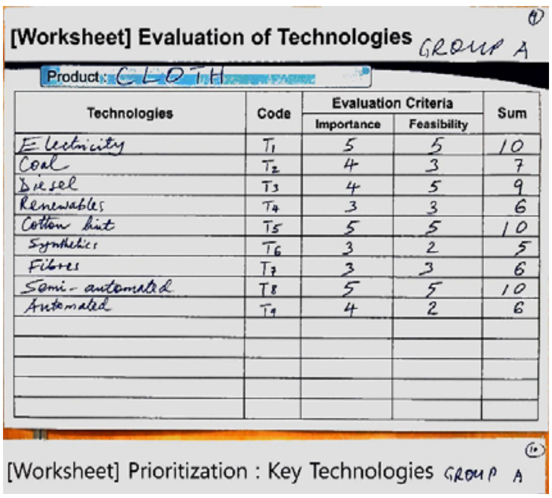
Based on the findings, each group presented its technology roadmap and exchanged opinions.

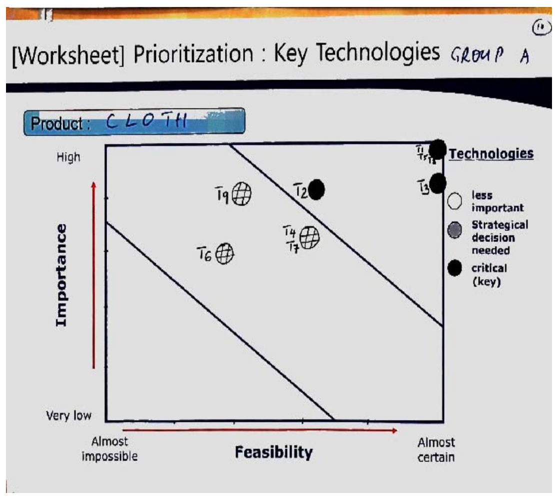
[Figure 5-5] Group A – CTA (Cotton)

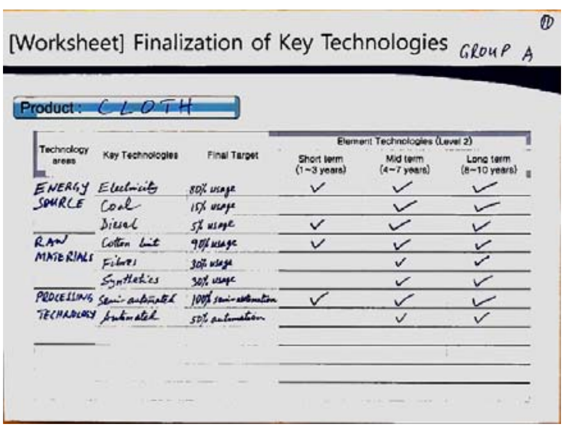


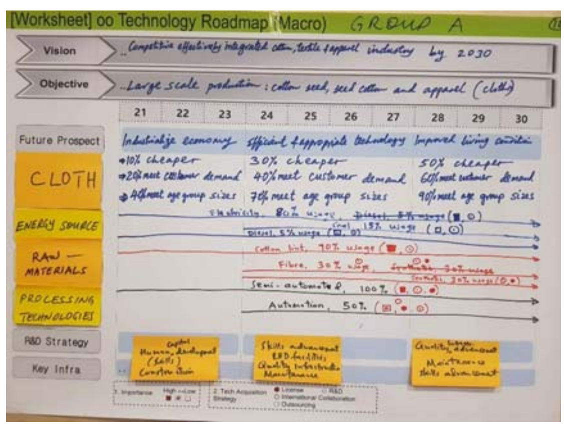

[Worksheet] Identification of Performance dimension & Technology alternatives GROUP A
7
Product : CLOTH
Performance Dimension
Performance Target
Technology alternatives
Cost
50% cheaper by 2030
Electricity, Coal, Gas, Diesel, Renewables
cotton lint, polyster, fibre, synthetic
Semi-automated, Automated
Design
60% meet customer demand by 2030
Manual, Computerised
Quality
70% improved by 2030
cotton lint, polyster, fibre, synthetic
Semi-automated, Automated
Computerized
Diversity/Type
90% of age group sizes met by 2030
Manual, Computerized
Semi-automated, Automated


[Worksheet] Technology Tree GROUP A
8
Product : CLOTH
Tech Area
Tech Level 1
Tech Level 2
ENERGY SOURCES
Electricity
Coal, Diesel
Renewables
RAW MATERIALS
Cotton lint
Synthetics
Fibres
PROCESSING TECHNOLOGIES
Semi-automated
Automated

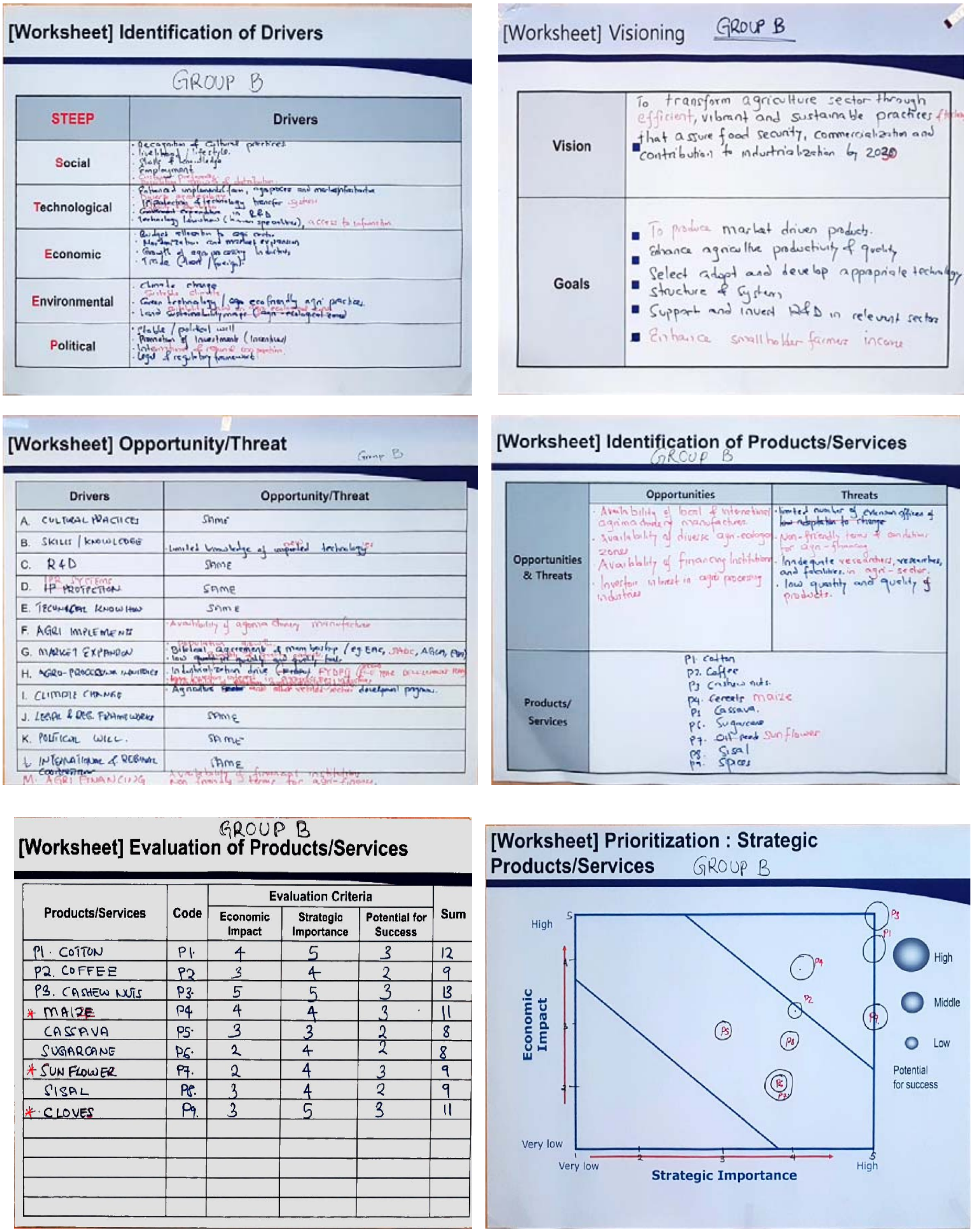

[Worksheet] Evaluation of Technologies GROUP A
9
Product : CLOTH
Technologies
Code
Evaluation Criteria
Importance
Feasibility
Sum
Electricity T1 5 5 10
Coal T2 4 3 7
Diesel T3 4 5 9
Renewables T4 3 3 6
Cotton lint T5 5 5 10
Synthetics T6 3 2 5
Fibres T7 3 3 6
Semi-automated T8 5 5 10
Automated T9 4 2 6
[Worksheet] Prioritization : Key Technologies GROUP A
10

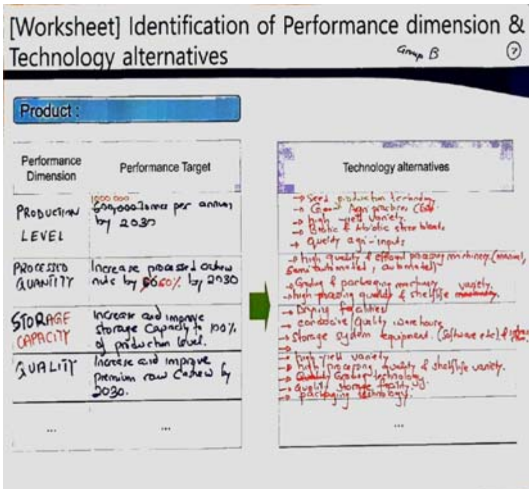

[Worksheet] Prioritization : Key Technologies GROUP A
10
Product : CLOTH
High
Very low
Importance
Almost impossible
Feasibility
Almost certain
T9
T2
T3
T6
T4
T7
Technologies
less important
Strategical decision needed
critical (key)

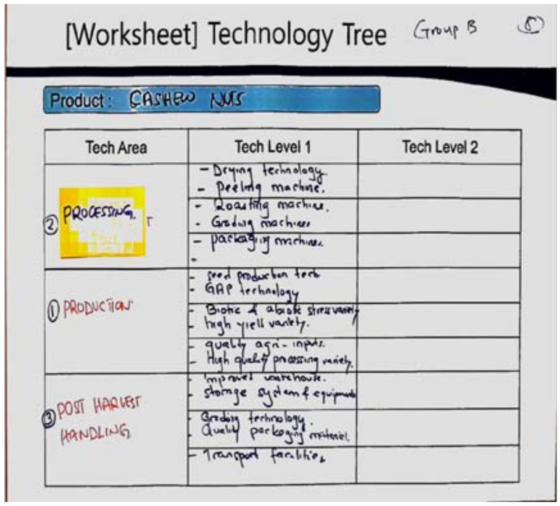

[Worksheet] Finalization of Key Technologies GROUP A
11
Product : CLOTH
Technology areas
Key Technologies
Final Target
Element Technologies (Level 2)
Short term (1~3 years)
Mid term (4~7 years)
Long term (8~10 years)
ENERGY SOURCE
Electricity 80% usage
Coal 15% usage
Diesel 5% usage
RAW MATERIALS
Cotton lint 90% usage
Fibres 30% usage
Synthetics 30% usage
PROCESSING TECHNOLOGY
Semi-automated 100% semi-automation
Automated 50% automation

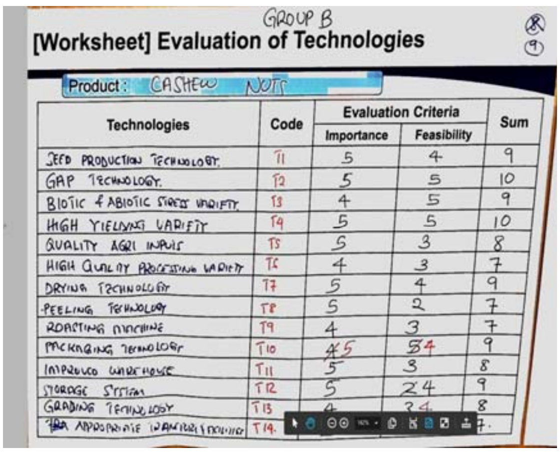

[Worksheet] oo Technology Roadmap (Macro) GROUP A
Vision
Competitive effectively integrated cotton, textile & apparel industry by 2030
Objective
Large scale production : cotton seed, seed cotton and apparel (cloth)
21 22 23 24 25 26 27 28 29 30
Future Prospect
Industrialize economy
Efficient & appropriate technology
Improved living condition
CLOTH
10% cheaper
30% cheaper
50% cheaper
40% meet customer demand
60% meet customer demand
40% meet age group sizes
70% meet age group sizes
90% meet age group sizes
ENERGY SOURCE
RAW MATERIALS
PROCESSING TECHNOLOGIES
R&D Strategy
Key Infra

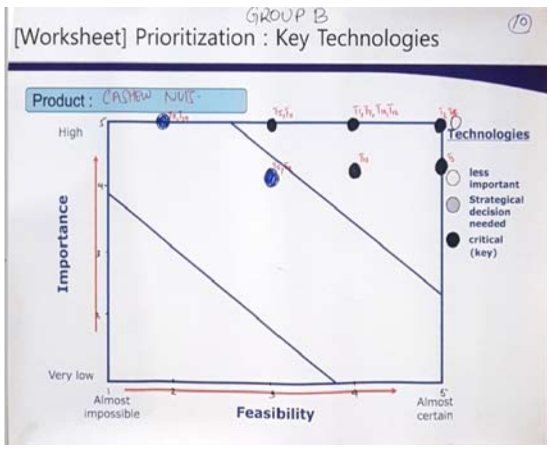
[Figure 5-6] Group B - Agriculture

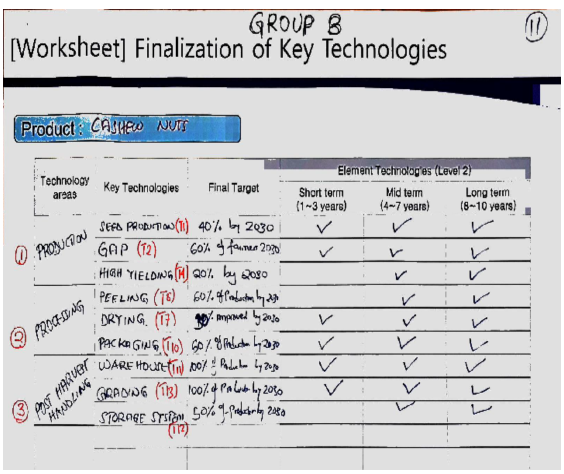


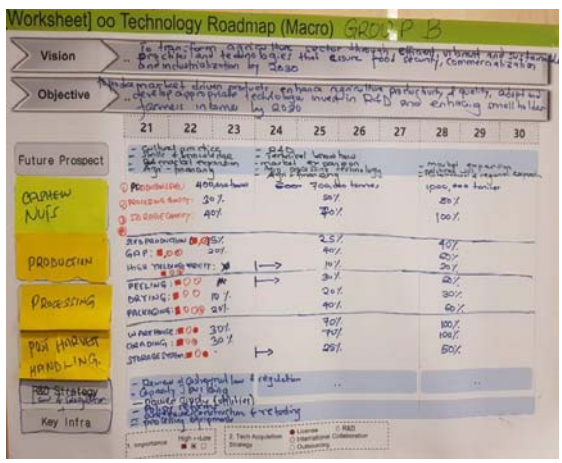

[Worksheet] Identification of Performance dimension & Technology alternatives
Group B
Product :
Performance Dimension
Performance Target
Technology alternatives


[Worksheet] Technology Tree
Group B
Product : CASHEW NUTS
Tech Area
Tech Level 1
Tech Level 2

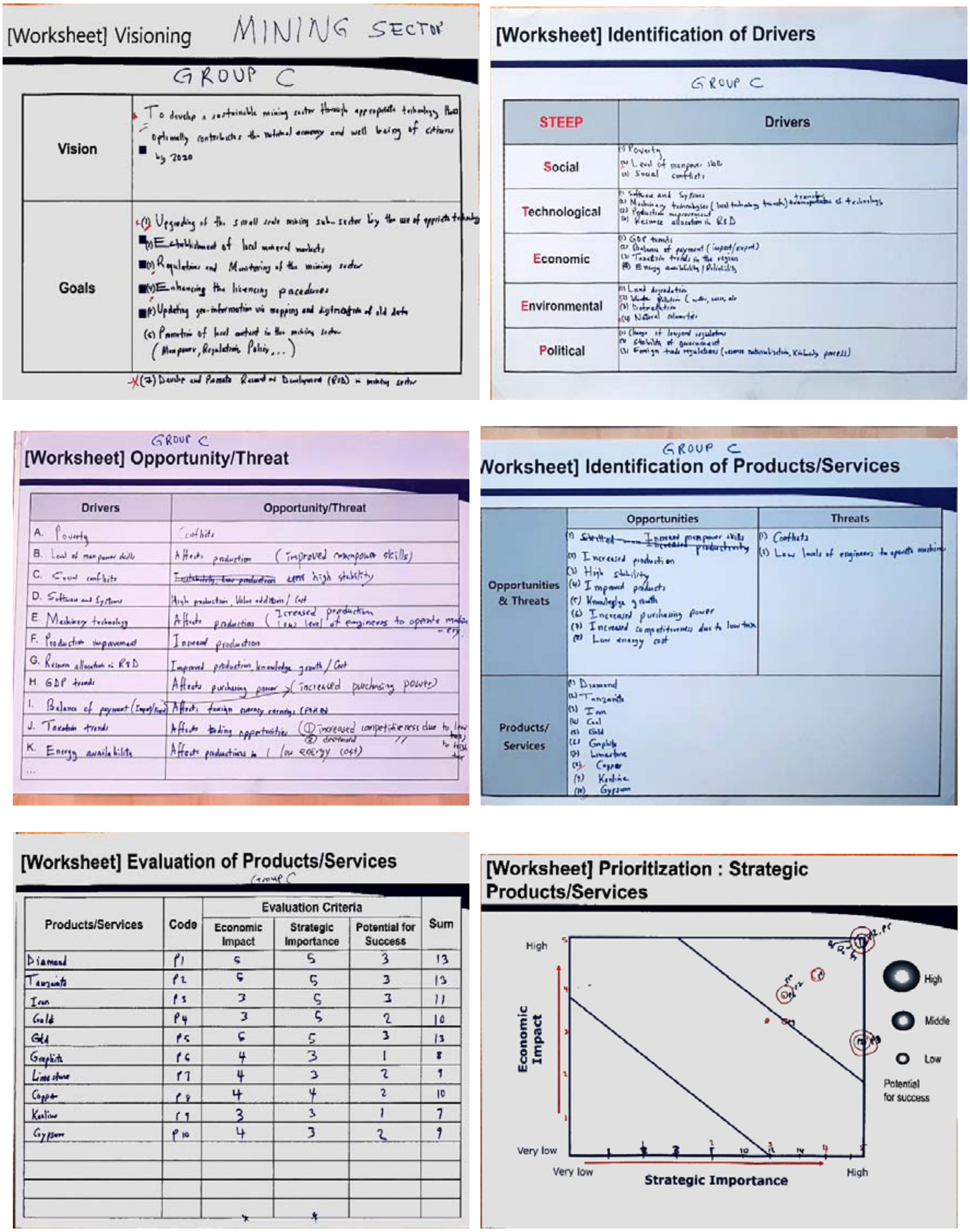

GROUP B
[Worksheet] Evaluation of Technologies
Product : CASHEW NUTS
Technologies
Code
Evaluation Criteria
Importance
Feasibility
Sum

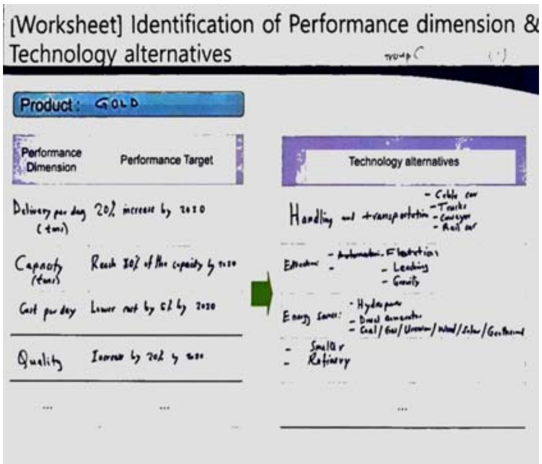

GROUP B
[Worksheet] Prioritization : Key Technologies
Product : CASHEW NUTS
High
Very low
Importance
Almost impossible
Feasibility
Almost certain
Technologies
less important
Strategical decision needed
critical (key)

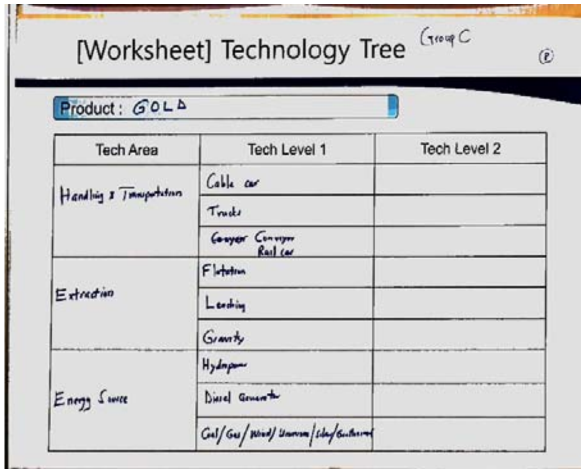

GROUP B
[Worksheet] Finalization of Key Technologies
Product : CASHEW NUTS
Technology areas
Key Technologies
Final Target
Element Technologies (Level 2)
Short term (1~3 years)
Mid term (4~7 years)
Long term (8~10 years)

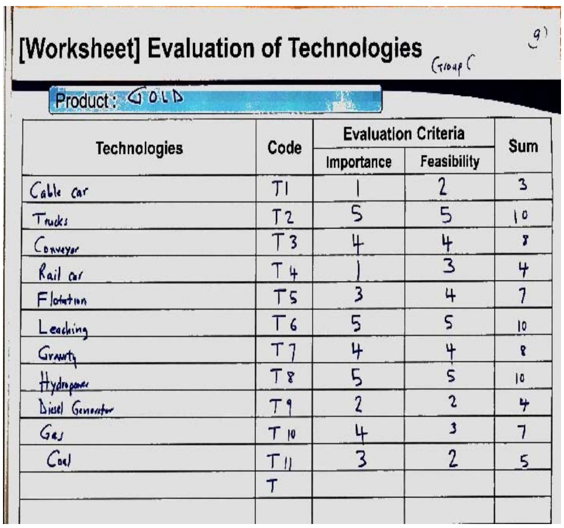

[Worksheet] oo Technology Roadmap (Macro)
GROUP B
Vision
Objective
Future Prospect
CASHEW NUTS
PRODUCTION
PROCESSING
POST HARVEST HANDLING
R&D Strategy
Key Infra

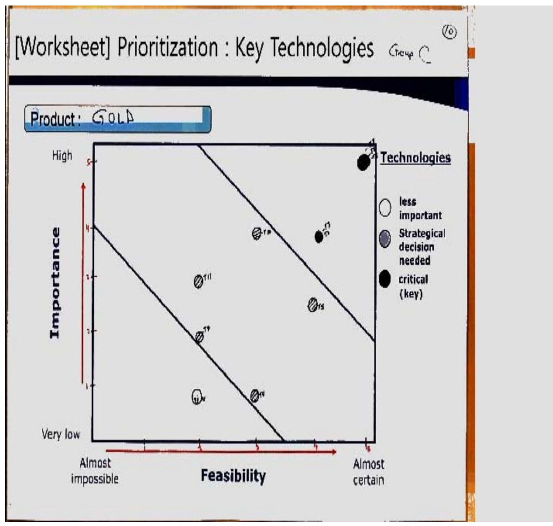
[Figure 5-7] Group C - Mining

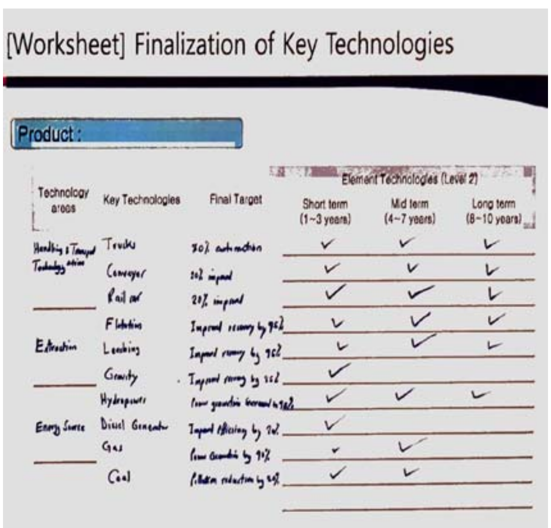


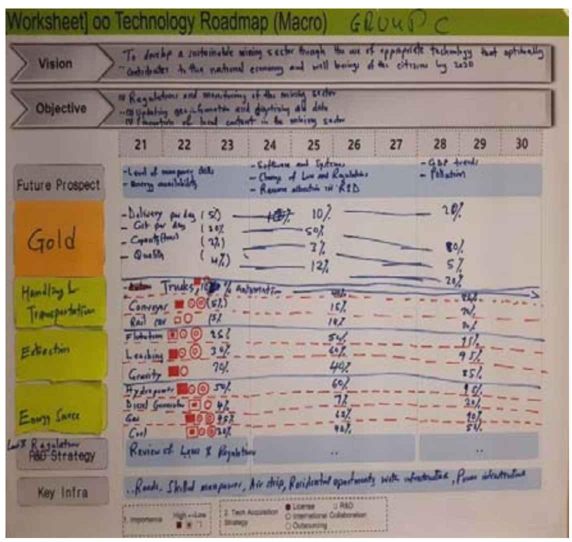

[Worksheet] Identification of Performance dimension & Technology alternatives
Product : GOLD
Performance Dimension
Performance Target
Technology alternatives
Capacity (tons)
Cost per day
Quality


[Worksheet] Technology Tree
Group C
Product : GOLD
Tech Area
Tech Level 1
Tech Level 2
Handling & Transportation
Cable car
Trucks
Extraction
Flotation
Leaching
Gravity
Energy Source
Hydropower
Diesel Generator


[Worksheet] Evaluation of Technologies
Group C
Product : GOLD
Technologies
Code
Evaluation Criteria
Importance
Feasibility
Sum
Cable car T1 1 2 3
Trucks T2 5 5 10
Conveyor T3 4 4 8
Rail car T4 1 3 4
Flotation T5 3 4 7
Leaching T6 5 5 10
Gravity T7 4 4 8
Hydropower T8 5 5 10
Diesel Generator T9 2 2 4
Gas T10 4 3 7
Coal T11 3 2 5


[Worksheet] Prioritization : Key Technologies
Group C
Product : GOLD
High
Importance
Very low
Almost impossible
Feasibility
Almost certain
Technologies
less important
Strategical decision needed
critical (key)


[Worksheet] Finalization of Key Technologies
Product :
Technology areas
Key Technologies
Final Target
Element Technologies (Level 2)
Short term (1~3 years)
Mid term (4~7 years)
Long term (8~10 years)


[Worksheet] oo Technology Roadmap (Macro)
GROUP C
Vision
Objective
21
22
23
24
25
26
27
28
29
30
Future Prospect
Gold
Handling & Transportation
Extraction
Energy Source
R&D Strategy
Key Infra

b) NTRM Policy Implementation for the Tanzanian Economy

Prof. Johng-Ihl Lee gave a lecture on the "NTRM Policy Implementation for the Tanzanian Economy." In the previous lecture, the professor talked about the definition, purpose, and methodology of NTRM. This time, he stressed that the period of accumulation and maturation and minimum power input required over maximum static frictional force are the critical factors for implementing the NTRM. He suggested, as in the case of V-KIST, that Tanzania needs to find a dependable and helpful STI partner and start a collaborative program. It also needs to establish a technology institute, a research or training institute, and then a policy institute such as STEPI.

2.4 Wrap-up Meeting

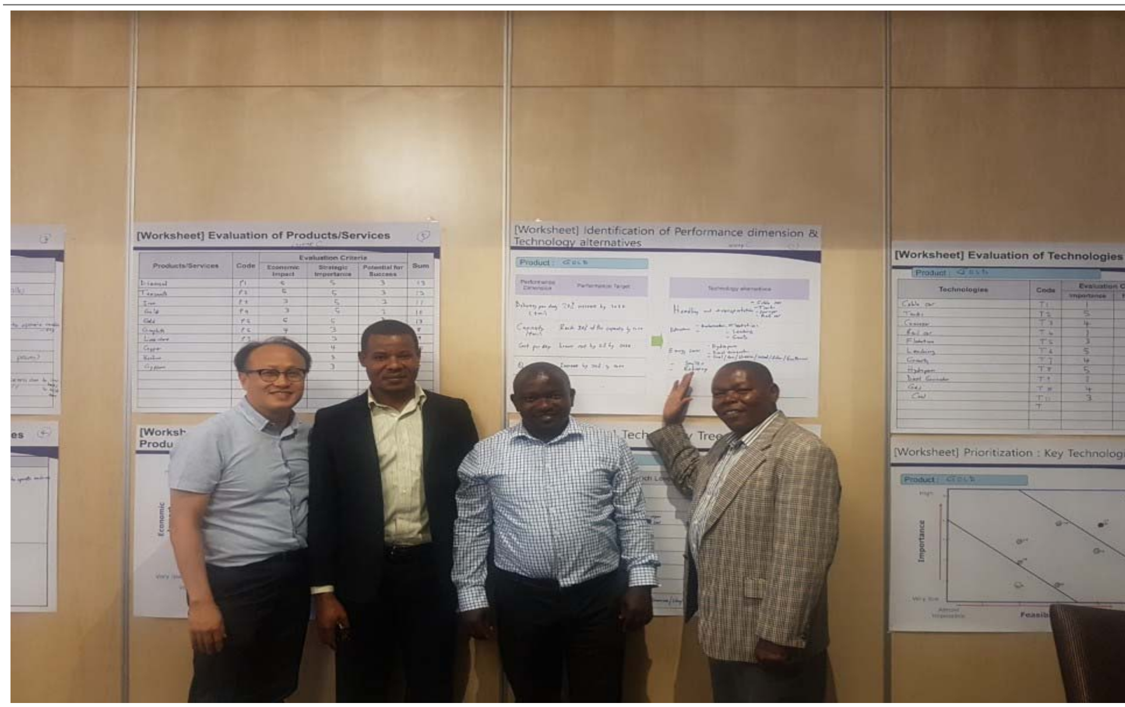
At the wrap-up meeting, the Tanzanian delegation agreed to form a steering committee and a bigger group for experts to develop the NTRM for more strategic products/services and more industrial sectors. Ms. Margareth Misheck Komba, assistant director for STI of MoEST and head of the delegation, expected the mutual partnership with STEPI to continue.

[Figure 5-8] Local Capacity Development Workshop in Zanzibar

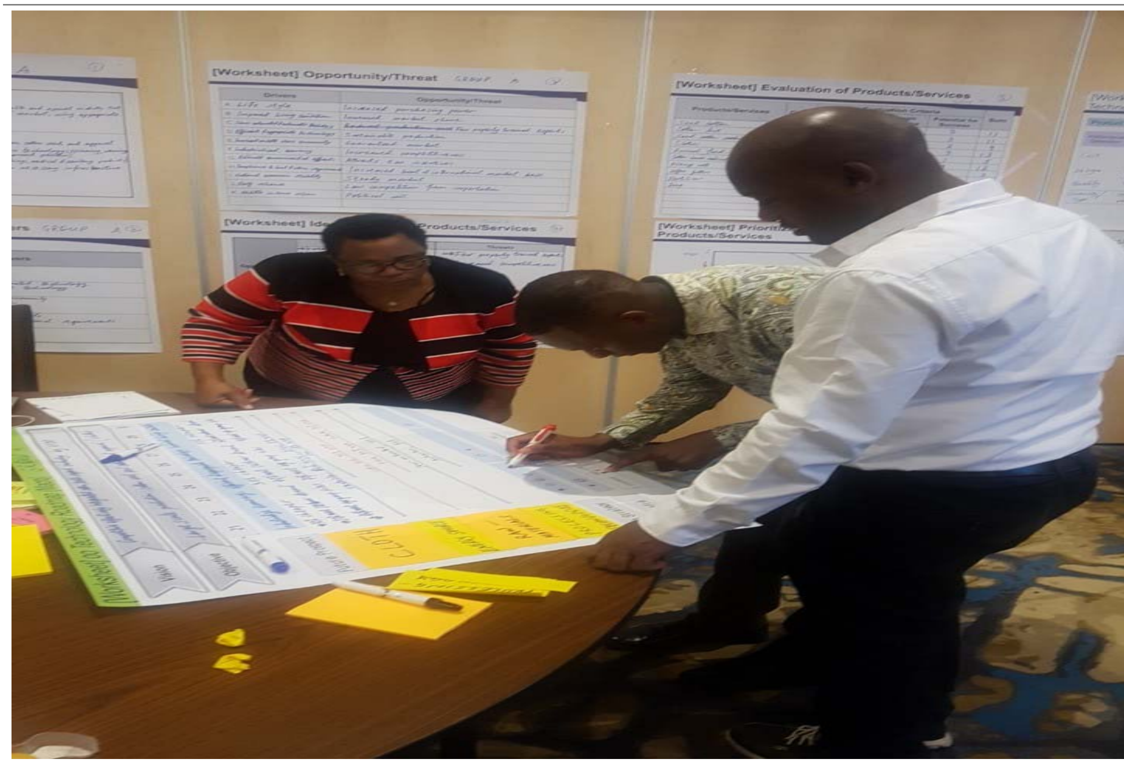
Local Capacity Development Workshop

Local Capacity Development Workshop

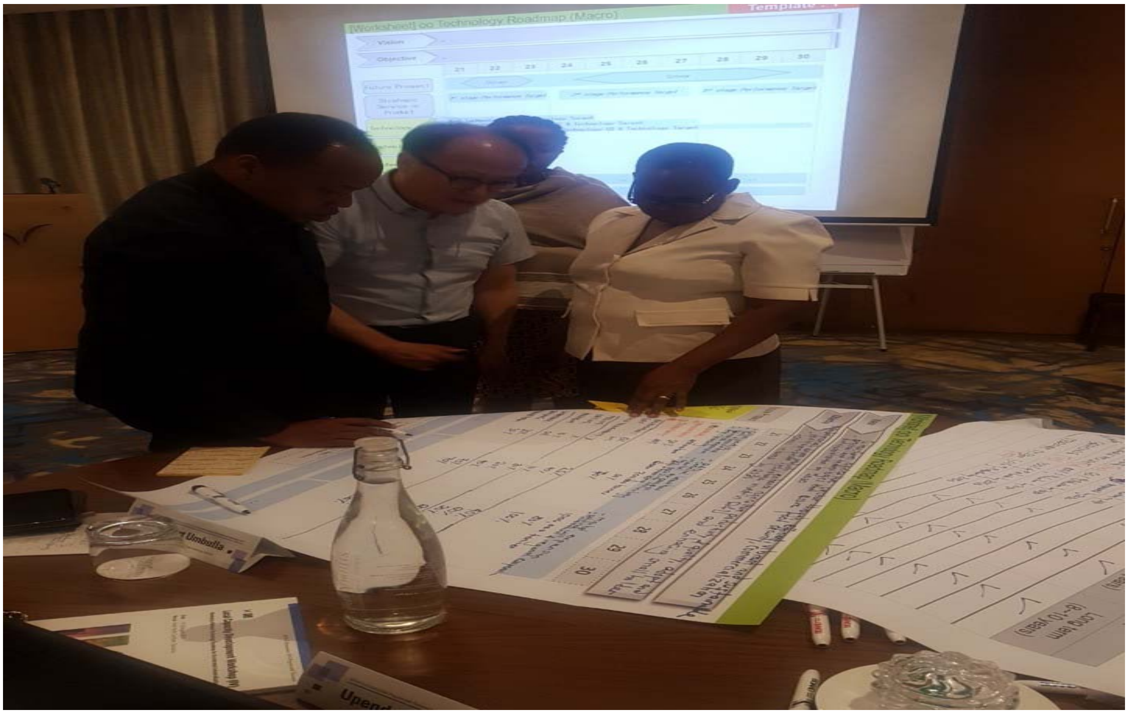
Local Capacity Development Workshop



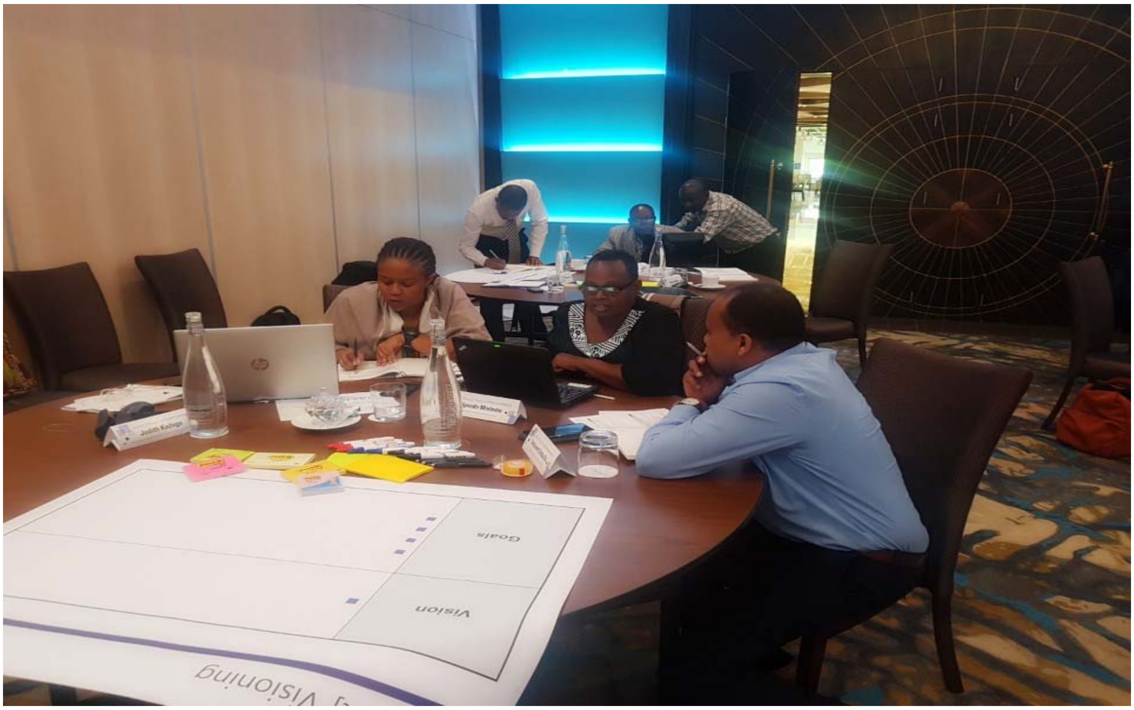
Local Capacity Development Workshop



Local Capacity Development Workshop



Local Capacity Development Workshop



Local Capacity Development Workshop

3. Collaboration Process for Capacity Development

3.1 Evaluation Survey

Seven (7) out of ten (10) attendees at the Invited Capacity Development Workshop gave five(5) points, which is the highest score, in the lectures and training session evaluation (average: 4.6 points). Likewise, for the question “Would you recommend the workshop to others?” eight (8) attendees gave 5 points (average: 4.8 points).

Even in the Local Capacity Development Workshop, the same question of evaluating lectures and training sessions (“How would you evaluate the lecture and practice session?”) garnered 5 points from eight (8) out of nine (9) participants, for an average score of 4.88 points out of 5 points. Surprisingly, the opinion of recommending the workshop to others was given the highest score of 5 by all nine attendees.

The suggestions as to the ways to improve the workshop were very similar to those in the answers from the previous capacity development workshop. Most of the participants answered that the training period was too short. Moreover, there was an opinion that full support from STEPI and Korean experts should be maintained in relation to the development of technology roadmap.

[Figure 5-9] Evaluation Survey Examples from the Invited Capacity Development Workshop

2019 K-Innovation Program with Tanzania

Evaluation Survey

- Capacity Development Workshop (III) on

Developing a National Technology Roadmap for Accelerated Industrialization

Please rate your overall level of satisfaction with the capacity development workshop (III) on the project, 'Developing a National Technology Roadmap for Accelerated Industrialization' held from April 1-4, 2019 in Seoul, Korea.

Answers to the first five close-ended questions should be rated **from 1 (lowest) to 5 (highest)**, whereas the next one open-ended question asks for comments on the overall topic of the capacity development workshop. Your input is valuable to us as we plan future workshops.

Question	Lowest 1	2	3	4	Highest 5
1. To what extent were the workshop objectives achieved?					5 ✓
2. How would you qualify the lecture and practice session?					5 ✓
3. Was the workshop helpful in enhancing your capacity to perform the duties?					✓
4. Were enough participation opportunities provided throughout the workshop?					✓
5. Was the duration of workshop sufficient?				✓	✓
6. Do you think you are able to finish your first NTRM draft by this coming June?				✓	
7. I would recommend this workshop to others.					5 ✓

8. Please give one or two practical suggestions for how we could improve this workshop

More time at least for ~~the~~ 10 working days for a NTRM draft preparation in Korea.

2019 K-Innovation Program with Tanzania

Evaluation Survey

- Capacity Development Workshop (III) on

Developing a National Technology Roadmap for Accelerated Industrialization

Please rate your overall level of satisfaction with the capacity development workshop (III) on the project, 'Developing a National Technology Roadmap for Accelerated Industrialization' held from April 1-4, 2019 in Seoul, Korea.

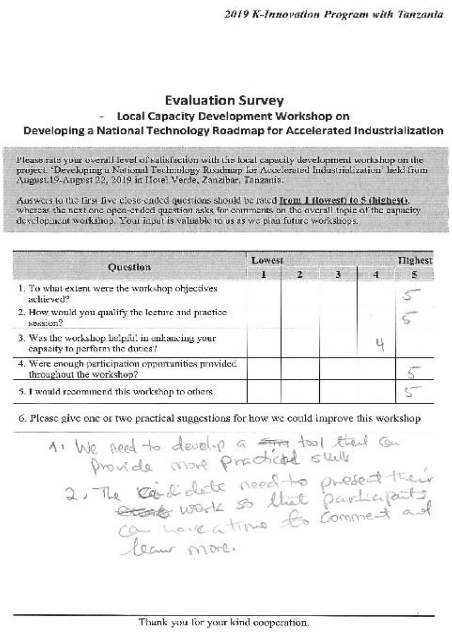
Answers to the first five close-ended questions should be rated **from 1 (lowest) to 5 (highest)**, whereas the next one open-ended question asks for comments on the overall topic of the capacity development workshop. Your input is valuable to us as we plan future workshops.

Question	Lowest 1	2	3	4	Highest 5
1. To what extent were the workshop objectives achieved?					✓
2. How would you qualify the lecture and practice session?					✓
3. Was the workshop helpful in enhancing your capacity to perform the duties?					✓
4. Were enough participation opportunities provided throughout the workshop?					✓
5. Was the duration of workshop sufficient?					✓
6. Do you think you are able to finish your first NTRM draft by this coming June?					✓
7. I would recommend this workshop to others.					✓

8. Please give one or two practical suggestions for how we could improve this workshop

Have more sector specific participants for each group.

[Figure 5-10] Evaluation Survey Examples from the Local Capacity Development Workshop



2019 K-Innovation Program with Tanzania

Evaluation Survey

- Local Capacity Development Workshop on

Developing a National Technology Roadmap for Accelerated Industrialization

Please rate your overall level of satisfaction with the local capacity development workshop on the project. 'Developing a National Technology Roadmap for Accelerated Industrialization' held from August.19-August 22, 2019 in Hotel Verde, Zanzibar, Tanzania.

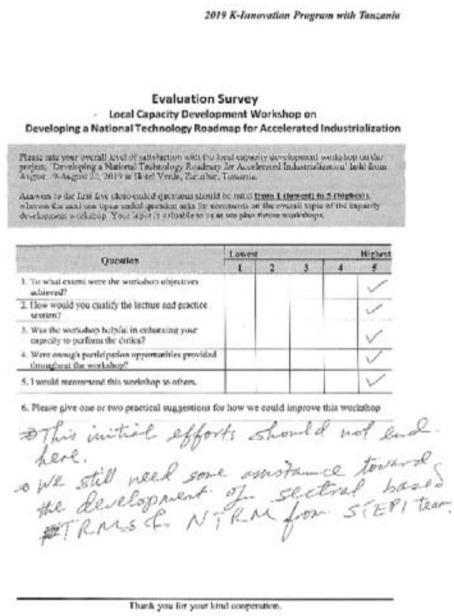
Answers to the first five close-caded questions should be rated **from 1 (lowest) to 5 (highest)**, whereas the next one open-ended question asks for comments on the overall topic of the capacity development workshop. Your input is valuable to us as we plan future workshops.

Question	Lowest 1	2	3	4	Highest 5
1. To what extent were the workshop objectives achieved?					5
2. How would you qualify the lecture and practice session?					5
3. Was the workshop helpful in enhancing your capacity to perform the duties?				4	
4. Were enough participation opportunities provided throughout the workshop?					5
5. I would recommend this workshop to others.					5

6. Please give one or two practical suggestions for how we could improve this workshop

1. We need to develop a tool that can provide more practical skill
2. The candidate need to present their work so that participants can have a time to comment and learn more.

Thank you for your kind cooperation.



2019 K-Innovation Program with Tanzania

Evaluation Survey

- Local Capacity Development Workshop on

Developing a National Technology Roadmap for Accelerated Industrialization

Please rate your overall level of satisfaction with the local capacity development workshop on the project, 'Developing a National Technology Roadmap for Accelerated Industrialization' held from August 19-August 22, 2019 in Hotel Verde, Zanzibar, Tanzania.

Answers to the first five close-ended questions should be rated **from 1 (lowest) to 5 (highest)**, whereas the next one open-ended question asks for comments on the overall topic of the capacity development workshop. Your input is valuable to us as we plan future workshops.

Question	Lowest 1	2	3	4	Highest 5
1. To what extent were the workshop objectives achieved?					✓
2. How would you qualify the lecture and practice session?					✓
3. Was the workshop helpful in enhancing your capacity to perform the duties?					✓
4. Were enough participation opportunities provided throughout the workshop?					✓
5. I would recommend this workshop to others.					✓

6. Please give one or two practical suggestions for how we could improve this workshop

- This initial efforts should not end here.
- We still need some assistance toward the development of sectoral based TRMs & NTRM from STEPI team.

Thank you for your kind cooperation.

3.2 Final Report

The research team has written a final report on "Developing a National Technology Roadmap for Accelerated Industrialization" based on the data and information collected during the two workshops in Korea and Tanzania.

For the final report, Dr. Julius Elias Daud, a Research Officer at TIRDO, has written a chapter on the current status of the Cotton, Textile, and Apparel (CTA) sectors. He wrote part of Chapter 6, explaining the results of Group A's exercises from the previous two workshops held in 2019.

The team asked Dr. Rogers Alfayo Msuya, a Senior Research Officer at MoEST, to analyze the current status of the mineral sector in Tanzania. He also wrote part of Chapter 6 with the results of Group C's exercises from the two workshops.

Moreover, Ms. Upendo Marko Mndeme, a Senior Agriculture Officer at MoA, was requested to analyze the current status of the Cashew industry. She wrote part of Chapter 6 with the results of Group B's exercises from the two workshops.

CHAPTER 6

NTRM Tutorial Workshop (Group Exercises)

Policy Consultation on Developing a National Technology Roadmap for Accelerated Industrialization

Chapter 6. NTRM Tutorial Workshop (Group Exercises)

1. Invited Capacity Development Workshop in Korea

1.1 Introduction

The National Technology Roadmapping(NTRM) for Tanzania was done in three priority sectors during the four workshops held in Tanzania and under the assistance of the Government of Korea through the Science and Technology Policy Institute (STEPI) -- namely cotton, textile, and apparel (CTA), agriculture, and mining sectors -- based on the Vision and Mission of Tanzania's development plan and vision 2025. The capacity-building planning was done in both theoretical and practical training, wherein the participants were divided into three groups according to sectors during practice. The training workshop was attended by participants from the Ministry of Education, Science, and Technology (MoEST), Ministry of Agriculture (MOA), Mineral Resources Institute (MRI), Tanzania Commission for Science and Technology (COSTECH), Tanzania Industrial Research Development Organisation (TIRDO), and Export Processing Zones Authority (EPZA).

The training workshops ended with a guiding NTRM for Tanzania in the three sectors mentioned above, aiming at leading Tanzania to develop the NTRM for all sectors under priorities as shown in the National Development Plan and Vision 2025. In this chapter, the guiding NTRM will be presented based on the description of the roadmap development that includes future prospects, scenarios, analysis of drivers and threats, and identification of strategic products and their technology areas. The roadmap also includes pathways for the three selected strategic sectors of CTA, agriculture, and minerals.

[Table 6-1] Research Team

Group	Name	Organization	Position
Group 1	Julius Elias Daud	TIRDO	Principal Research Officer
	Gerald Majella Kafuku	COSTECH	Principal Research Officer
	Margareth Misheck Komba	MoEST	Principal Research Officer
Group 2	Upendo Marko Mndeme	MoA	Senior Agriculture Officer
	Innocent John Umbulla	EPZA	Senior Officer
	Judith Francis Kadege	COSTECH	Principal Research Officer
Group 3	Rogers Alfayo Msuya	MoEST	Senior Research Officer; Coordinator for RD&I
	Fredrick Martin Mangasini	MRI	Acting Principal
	Veronica Gideon Ngeregeza	EPZA	Senior IT Officer

1.2 NTRM Pilot Project – Group 1 (CTA, Cotton)

[Worksheet] Visioning

Vision	■ To have cotton industry that cuts down importation and enhance exportation of cotton-based products using appropriate technology by 2030
Goals	■ Large scale production: spinning, weaving, etc. ■ Modernise products distribution channels ■ Products diversification (e.g. sanitary, clothes, medical products) ■ Improve the quality of national infrastructure related to cotton (e.g. conformity assessment, standardization, accreditation, methodology, etc.) ■ Develop necessary infrastructure for cotton industry ■ Develop strong R&D and investment

The formulation of this vision and goals benefited from an in-depth assessment of Tanzania's CTA sector. The group performed an evaluation based on the past, present, and future of the CTA sector in line with the National Development Vision (NDV, 2025) priorities and difficulties encountered by the CTA sector in export-oriented products. The setup also took advantage of the country's cotton wealth while addressing the shortage of cotton-based products in the local and international markets by considering crucial products, market demand, technologies involved, and pathways for the next 10 years. The identified goals were increased large-scale production, CTA-based products diversification, and improved quality of CTA products through the application of modern technologies and Research and Development (R&D).

[Worksheet] Identification of Drivers

STEEP	Drivers
Social	• Life style • Poverty reduction • Livelihood
Technological	• Semi-automated technology • Efficient technology
Economic	• Economic growth and development • Industrialized economy
Environmental	• Minimized pollution and climate change
Political	• Self reliance • National economic stability • Middle income

The textile (CTA) sector's technology drivers were identified based on social, technological, economic, environmental, and political (STEEP) analysis according to their future importance. Eleven (11) drivers were identified by Group "A" covering four social drivers, two technological drivers, two economic drivers, one environmental driver, and three political drivers. The drivers were found to wield great impact on the future growth of the CTA sector in Tanzania.

[Worksheet] Opportunity/Threat

Drivers	Opportunity/Threat
A. Life Style	Increased market demand
B. Poverty reduction	High purchase power
C. Livelihood	Increased market demand
D. Semi-automated tech.	Available labour force
E. Efficient technology	Sustainable production
F. Economic growth & dev.	Increased market share
G. Industrialized economy	Readily available intermediary industries (easy supply of tools)
H. Minimized pollution	Increased cost to control pollution
I. Self reliance	Low competition from importation
J. National economy stability	Steady market
K. Middle income	Increased market share

The identified potential opportunities and threats from the set drivers reflect the vision and goals of the CTA sector. Ten potential opportunities and one threat were identified based on the current situational analyses of the CTA sector in and outside the country, and they are the key enhancement factors for the success of the sector. On other hand, the identified threat is one of the competitive constraints of the sector.

[Worksheet] Identification of Products/Services

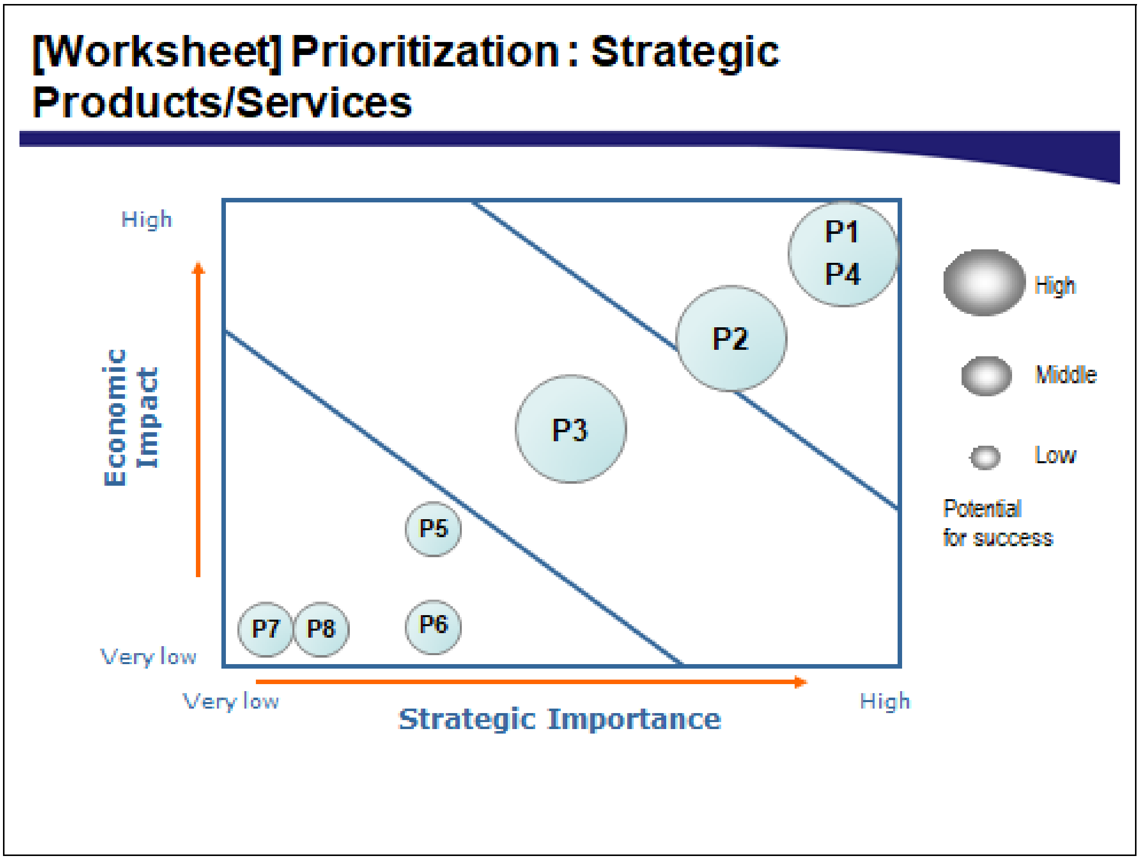
Opportunities & Threats	[Opportunity] • Increased market demand of share • High purchase power • Available labour force • Sustainable production • Readily available intermediary industries • Low competition from importation • Steady market [Treat] • Increased cost to control pollution
Products/ Services	• P1: Clothes • P2: Cotton based medical products, e.g. bandages, gauze • P3: Cotton based sanitary products, e.g. pads • P4: Cotton lint • P5: Cotton seeds cooking oil • P6: Animal feeds • P7. Fishing nets • P8. Coffee filters

Group A members identified eight (8) products along the CTA value chain based on their economic importance, strategic importance, and potential for success. These include: clothes, cotton-based medical and sanitary products, cotton lint, cotton seeds' cooking oil, animal feeds, fishing nets, and coffee filters. These products will have significant impact on the farmers and the national economy if the identified opportunities are enhanced while mitigating the threats.

[Worksheet] Identification of Products/Services

	Economic Impact	Strategic Importance	Potential Success	Sum
P1	5	5	3	13
P2	4	4	3	11
P3	3	3	3	9
P4	5	5	3	13
P5	2	2	1	5
P6	2	1	1	4
P7	1	1	1	3
P8	1	1	1	3

The eight identified products (coded P1-P8) along the CTA value chain were evaluated based on economic impact, strategic importance, and potential for success using point scale. The point scale score was considered to rate the products in order to identify strategic products. The products were rated on a 3-point scale for their potential for success and on a 5-point scale under economic impact and strategic importance factors. The total score of each product was calculated by summing up the point scale score of each product based on its economic impact, strategic importance, and potential success factors. However, only half of the identified products scored more than 50%.



The strategic products that occupied the highest position for consideration compared to other products were selected to lead the CTA sector for the next ten years. Economic impact scores for the eight coded products (P1-P8) were plotted against strategic importance scores while considering their potential for success to identify strategic products. Clothes (P1), cotton lint (P4), and cotton-based medical products (P2) were the identified strategic products with high economic impact and strategic importance and high potential for success compared to other products along the CTA value chain. The rest (P5, P6, P7 and P8) were plotted on the low zone and were consequently rejected.

[Worksheet] Identification of Technology drivers & targets

Product : Cloth

Performance Dimension	Performance Target
Cost	60% cheaper by 2030
Quality	40% increased in quality by 2030
Types / Diversity	Meet 100% of age demand,/ sizes by 2030
Style of design	Meet 60% of market styles produced by 2030

Technology areas	Technology drivers & targets	Technology alternatives
...	...	...

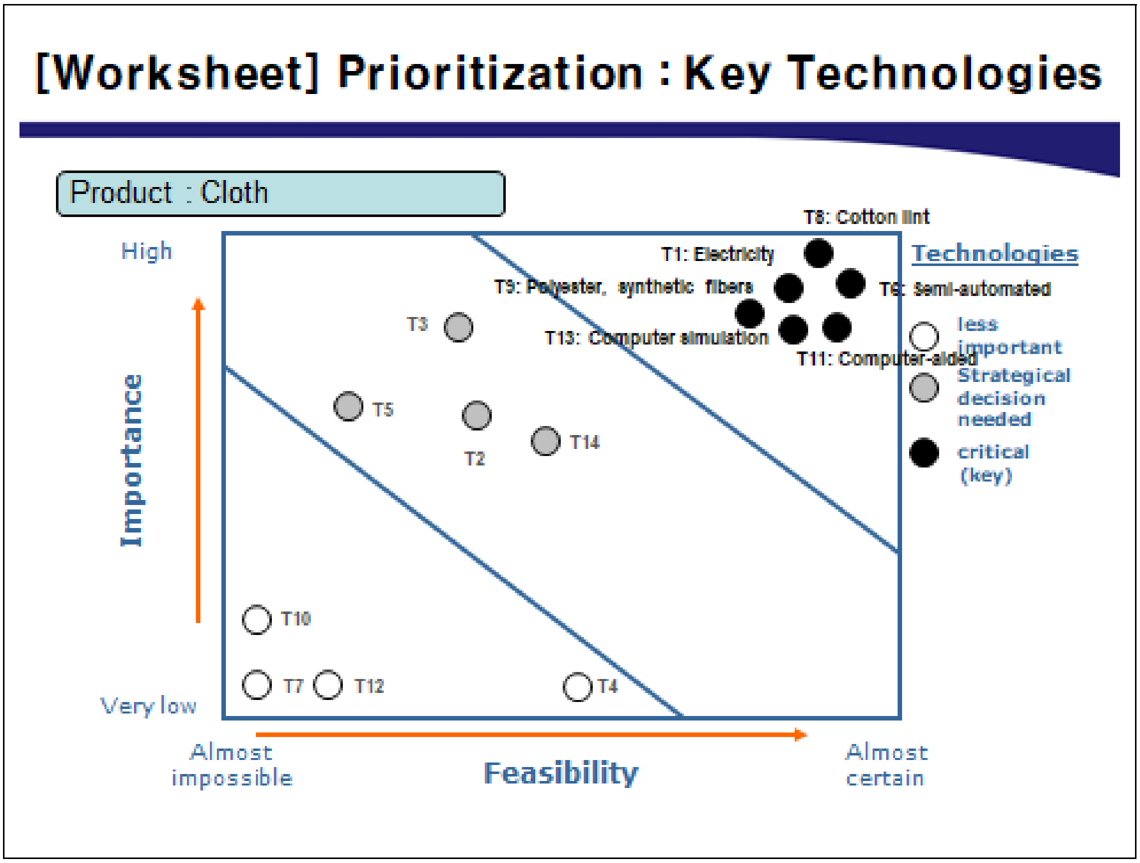
The performance dimension and performance target associated with performance dimension for cloth as strategic product were identified. The performance dimensions with strong impact on the cost, quality, types or diversity, and style of design of the produced cloth were analyzed to set the targets from which performance dimensions will be evaluated. The year 2030 is the set timeline for accomplishing the performance targets that are aligned with the overall vision and goals of the sector.

[Worksheet] Technology Tree

Product : Cloth

Tech Area	Tech Level 1	Tech Level 2
Energy sources of production	T1: Electricity	
	T2: Coal	
	T3: Gas	
	T4: Diesel	
Process technology	T5: Automated	
	T6: Semi-automated	
	T7: Manual	
Raw materials	T8: Cotton lint	
	T9: Polyester, Synthetic fibers	
	T10: Sheet	
Design	T11: Computer aided	
	T12: Manual	
Diversity / Types / Sizes	T13: Computer simulation	
	T14: Used humans (fashionists)	

A technology tree was plotted wherein a number of technologies were assigned to the technology area identified. Five technology areas were identified as key factors to enhance cloth production in the CTA sector based on the criteria such as feasibility and importance. Fourteen technologies were identified and coded (T1 to T14) from five technology areas. Such shortlist is a key for making the right decision in technology selection, showing the direction of research and development of the future technology need of the CTA sector.



The identification of key technologies was done from the list of identified technologies in the technology tree. These alternative technologies were assessed based on their importance (economic, social, and environmental benefits, scientific and technological opportunities) and feasibility (research and technology potential and potential to absorb economic and social benefits) and ranked on a 5-point scale. The 5-point scale was considered to rate all identified technologies and rank them based on their importance and feasibility. The highest score shows the possibility of success, whereas the lowest score indicates less importance or lack of feasibility.

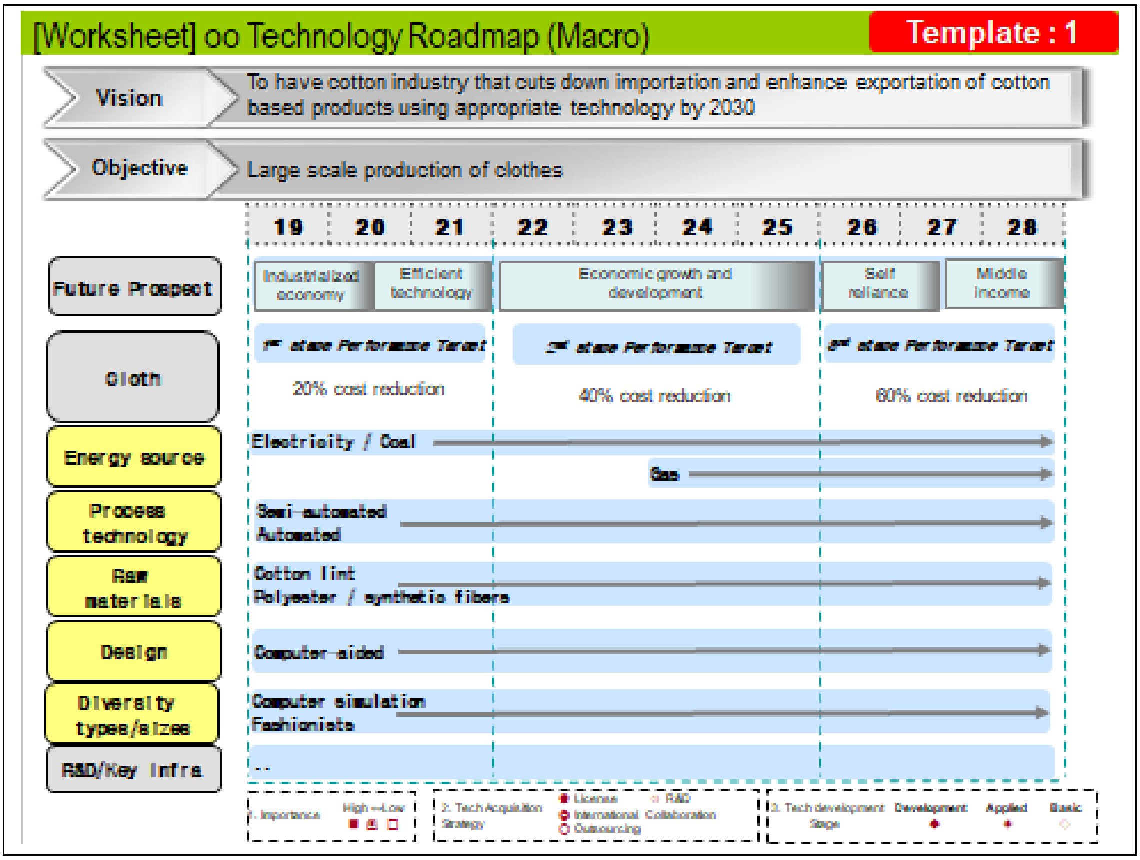
The calculated importance scores of the 14 coded alternative technologies (T1-T14) were plotted versus the feasibility scores in order to select the key technologies. Electricity (T1), cotton lint (T5), and semi-automated (T8) were selected as key technologies based on their higher importance and feasibility compared to other identified technologies along the CTA value chain. In general, only 6 out of 14 technologies scored above 70%, whereas 4 (T4, T7, T12 & T14) scored less than 50 %.

[Worksheet] Finalization of Key Technologies

Product : Cloth

Technology areas	Key Technologies	Final Target	Element Technologies (Level 2)		
			Short term (1 –3 years)	Mid term (4 –7 years)	Long term (8 –10 years)
Energy sources of production	Electricity		V		
	Coal		V		
	Gas			V	V
Process technology	Semi-automated		V	V	V
	Automated		V	V	V
Raw materials	Cotton lint		V	V	V
	Polyester / synthetic fibers		V	V	V
Design	Computer aided		V	V	V
Diversity Types Sizes	Computer simulation		V	V	V
	Humans (fashionists)		V		

The selected key technologies were tabulated against their respective technology areas, and the time frame (i.e., short-term (covering 1-3 years), mid-term (4-7 years) and long-term (8-10 years)) for their implementation was set. All technologies require short implementation time except gas. These key technologies will bring about structural change in the CTA sector as macro level strategies.



The macro road map for the Textile (CTA) sector was developed, showing the vision, objectives, overall target, and key technologies and future prospects for the next 10 years. The future prospect, technology area, and their respective selected key technologies, R&D strategies, and key infrastructure reflect the vision and objective of CTA TRM. The selected key technologies were tabulated against their respective technology areas. Key technologies should meet the targeted 20% cost reduction by 2021, 40% cost reduction by 2025, and 60% cost reduction by 2028.

1.3 NTRM Pilot Project – Group 2 (Agriculture, Cashew Nut)

[Worksheet] Visioning

Vision	■ To transform agriculture sector through vibrant and sustainable practices that assure food security, commercialization and contribution to industrialization by 2028
Goals	■ Market driven products ■ Enhanced agriculture productivity and quality ■ Select, adopt, and develop appropriate technology, infrastructure, and system. ■ Support and invest R&D in relevant sectors

The focus is to transform the agriculture sector from subsistence to commercialization level through the application of improved technologies and available resources so as to contribute significantly to the industrial economy

In order to attain the aforesaid vision, agriculture production and breeding should be market-oriented, and technologies that increase production and productivity should be selected, developed, and adopted. Investment in Research and Development is crucial for the growth of the sector.

[Worksheet] Identification of Drivers

STEEP	Drivers
Social	• Recognition and cultural practices • Livelihood / life style • Skills and knowledge • Employment
Technological	• Enhanced implements (farm, agro-process and market infrastructure) • IP protection and technology transfer • Government expenditure in R&D • Technology know-how (human specialties)
Economic	• Budget allocation to agricultural sector • Modernization and market expansion • Growth of agro-processing industry • Trade (foreign / local)
Environmental	• Climate change • Green technology / Eco-friendly agricultural practices • Land sustainability maps (agro-ecological zones)
Political	• Stable / political wil • Investment, incentives • International and regional cooperation • Legal and regulatory frameworks

The team identified 19 drivers under Social (4), Technological (4), Economic (4), Environmental (3), and Political (4). The listed drivers influence agricultural growth in various dimensions from production to consumption of the products.

[Worksheet] Opportunity/Threat (1)

Drivers	Opportunity/Threat
A. Cultural practices	• Availability of existing indigenous technology • Slow adoption / adaptation to change
B. Skills / Knowledge	• Availability of agriculture experts, specialists, and institutions • Limited number of extension officers • Limited knowledge of imported technology by farmer
C. R&D	• Adequate R&D institutions • Inadequate research specialists in agriculture • Inadequate research facilities and resources
D. IP Protection	• Creativity and innovation: availability of institutions and specialists • Availability of legal frameworks on IP and TT protection • Absence of national IP policy
E. Technical know-how	• Availability of improved technologies (e.g. seeds, processing equipment) • Technology dissemination challenges to farmers • Limited product development technology (to be improved)
F. Agriculture implements	• Inadequate agriculture implements • Low purchasing power, Expensive equipment • Availability of agro-machinery, production, institutions

The team selected some of the listed drivers and identified the opportunities and threats. Understanding the existing cultural practices and knowledge is considered important as it serves as a stepping stone for new innovation. Nonetheless, some cultural practices can hinder the adoption to change; hence the need for sensitization and mobilization before introducing new innovation to the community.

Research & development is a foundation for innovations and protection of indigenous knowledge that can be utilized for research purposes. The technology know-how of stakeholders along the value chain, existence of improved technologies, dissemination channels, and availability of agricultural implements are vehicles for agricultural growth.

[Worksheet] Opportunity/Threat (2)

Drivers	Opportunity/Threat
G. Market expansion	• Growing demand for agro-produce (locally, regionally) • Bilateral agreements and membership (e.g. EAC, AGOA, EBA) • Low in quantity and quality of food
H. Agro-processing industries	• Industrialization drive (FTDPU) • Less competitive products for international markets
I. Climate change	• Draught tolerant crops • Agro ecological zone (availability of wide range) • Crop failure, pests and diseases
J. Legal and regulatory frame works	• Agricultural sector development programme (strong strategic direction) • Science and technology policy (which is outdated) • Agriculture policy
K. Political will	• National commitment in agriculture (high) • Availability of relevant institutions • Change of priority in government
L. International and regional cooperation	• Participation and membership to international and regional bodies • Markets / Knowledge sharing • Regional conflict

The growing demand for agricultural products at all levels, existence of bilateral agreement (EAC, SADEC, etc.), and growing population serve as market opportunities for agricultural products. Agro-industrial growth also provides market and employment and ultimately income earnings.

Additionally, the country is politically stable, and the government is willing to support the agriculture sector through agricultural sector programs, conducive policies and regulations, and incentive packages for agricultural investment. The country has strong linkages with Regional and International organizations for knowledge sharing.

There are seven (7) agro-ecological zones suitable for various crops. The underperformance of some crops is attributable to the impact of climate change, among others.

[Worksheet] Identification of Products/Services (1)

Opportunities & Threats	[Opportunity] • Available indigenous technology • Availability of agri-institutions (R&D, universities) • Improved technologies • Growing demand of agri-produce • Industrialization drive • Available policy statement and programmes • Regional and international cooperation (to agreements) • Political will through government commitment [Treat] • Limited number of extension officers of low adaptation to change • Inadequate researches and research facilities • Low purchasing power of farmers (agro-machinery) • Low quality and quantity • Change of government priority • Regional conflict • Weak, outdated policies and laws

The selected opportunities and threats were summarized, showing that there are a wide range of knowledge from indigenous to advanced technologies in the country, growing demand for agro-products, reputable Research Institutions, presence of improved technologies, industrial growth, political will, and government commitment to agricultural developments. There is also strong Regional and International cooperation that provides a platform for knowledge sharing.

On the threat side are the inadequate number of extension officers, limited number of research specialists and facilities, low purchasing power of farmers, and low quantity and quality of the products; hence the inconsistency of supply and less competitiveness in the market.

[Worksheet] Identification of Products/Services (2)

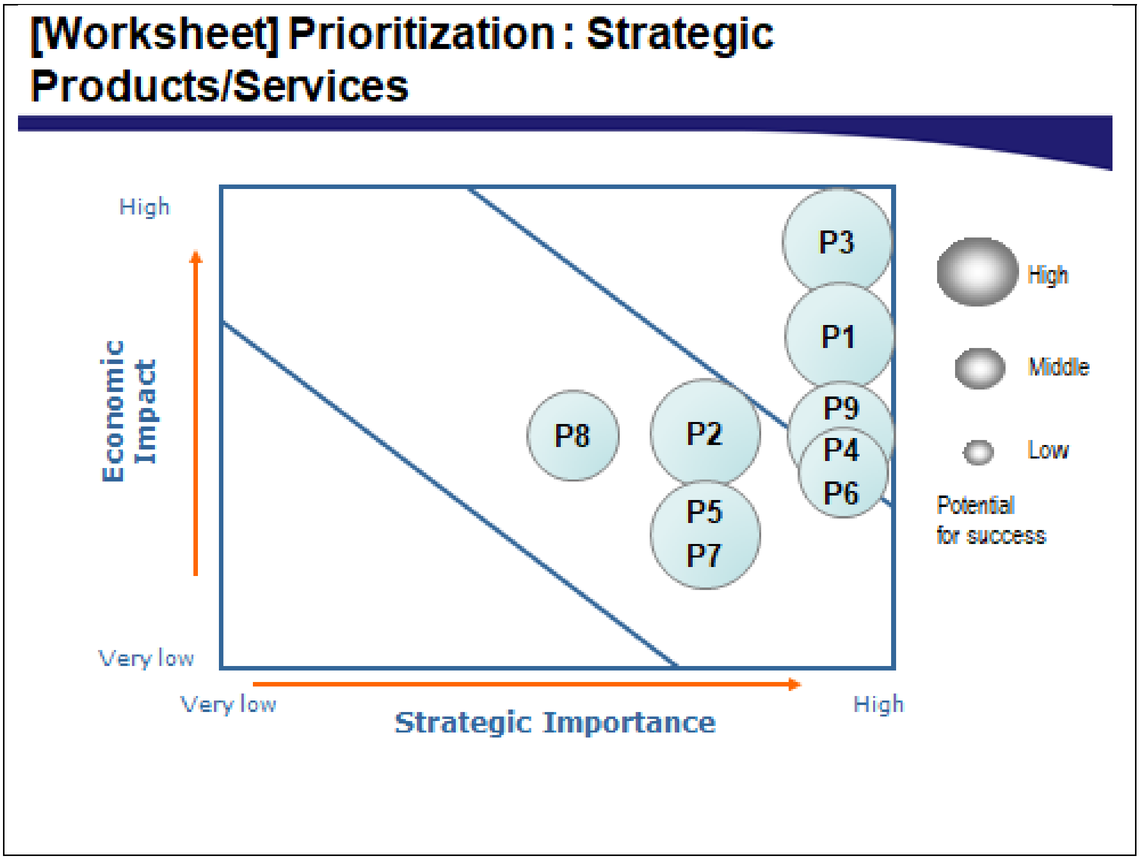
Products/ Services	• P1: Cotton • P2: Coffee • P3: Cashew nuts • P4: Cereals (maize, grains) • P5: Cassava • P6: Sugarcane • P7. Oil seeds (sunflower, palm) • P8. Sisal • P9. Spices

Based on the national priority of focusing on transforming the country's economy into an industrial economy, nine (9) products were identified, five of which (P1, P2, P3, P6, P8) are traditional cash crops, two (P4, P5) are food and cash crops, one (P7) is an oil seed crop, and one (P9) is a horticultural crop. All of the selected crops are raw materials for the growing industries, food security, and edible oils and spices for a number of dishes and beverages.

[Worksheet] Identification of Products/Services

	Economic Impact	Strategic Importance	Potential Success	Sum
P1	4	5	3	12
P2	3	4	3	10
P3	5	5	3	13
P4	3	5	2	10
P5	2	4	3	9
P6	3	5	2	10
P7	2	4	3	9
P8	3	3	2	8
P9	3	5	3	11

The products were evaluated based on criteria such as economic impact, strategic importance, and potential for success. P3 (cashew nut) ranked high followed by P1, P9, P2, and P6. However, all of the identified products scored more than 50%.



As the resources are limited, the selected products were prioritized. P3 (cashew nut), P1 (cotton) and P9 (Spices) were found to occupy the highest position for consideration compared to other products. The rest occupied the medium zone, and none of the products were plotted on the rejection zone.

[Worksheet] Identification of Technology drivers & targets

Product : Cashew nut

Performance Dimension	Performance Target
Production level	600,000Tn per annum by 2028
Processed cashew nuts	Increase processed cashew nuts by 80% by 2028
Quality	Increase and improve premium raw cashew nuts by 60% by 2028
By-products development and production	70% of production of processed by-products by 2028 per annum
Cost	Decrease cost of production by 10% per annum by 2028

Technology areas	Technology drivers & targets	Technology alternatives
...	...	...

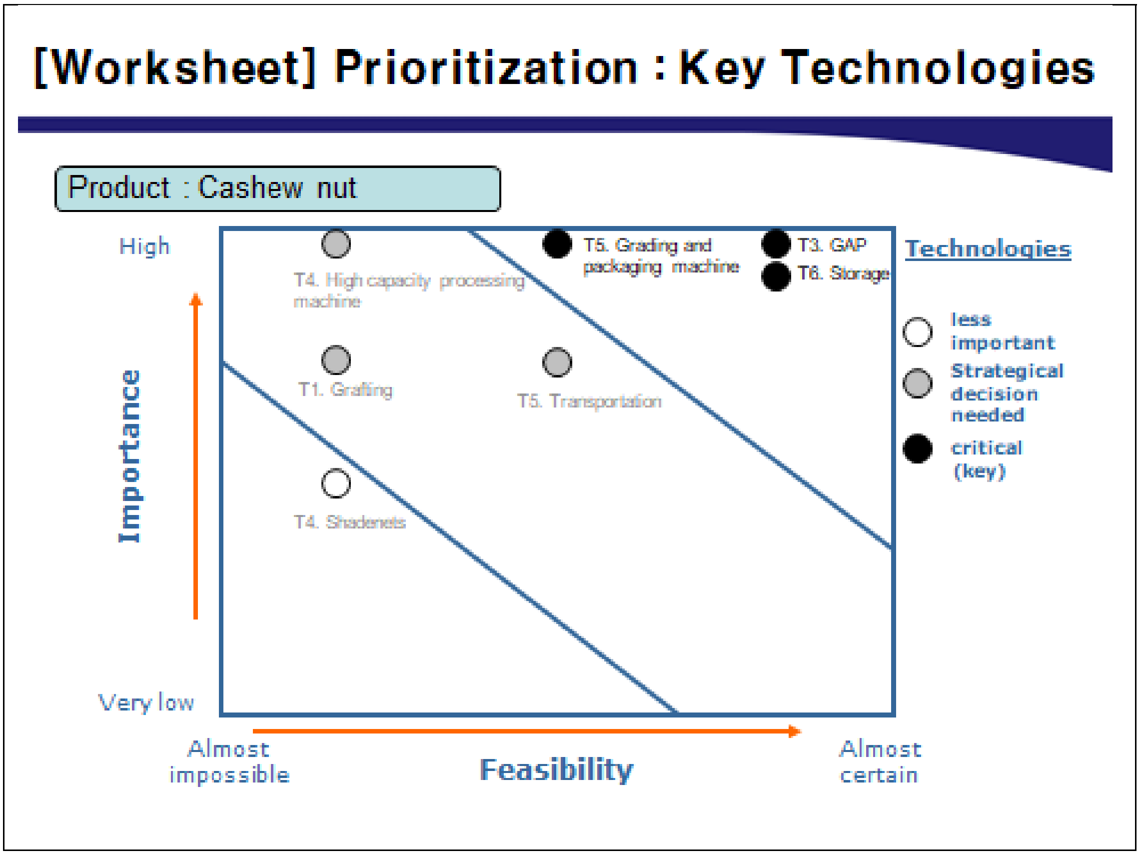
After identification of the products, performance dimension was identified including the target for each by 2028. The agreed-upon targets include increase of production from 300,000 to 600,000, increase of processed products to 80%, increase of premium grade of RCN by 60%, 70 % of the byproducts developed into new products, and reduction of costs by 10%.

[Worksheet] Technology Tree

Product : Cashew nut

Tech Area	Tech Level 1	Tech Level 2
Seedling	T1. Grafting	
	T2. Shadenets	
	T3. GAP	
Processing machinery	T4. High capacity Processing machines	
	T5. Grading and packaging machines	
Handling technology	T6. Storage	
	T7. Transportation	

A technology tree was plotted wherein a number of technologies were assigned to the technology area identified. It was pointed out that there is a need to improve seed production technologies and equip producers with good agricultural practices (GAP). In processing and post-harvest technology, high-capacity processing machine and quality storage facilities are key technologies of importance. In addition, quality and appropriate transport facilities were found to be a prerequisite for cashew subsector growth.



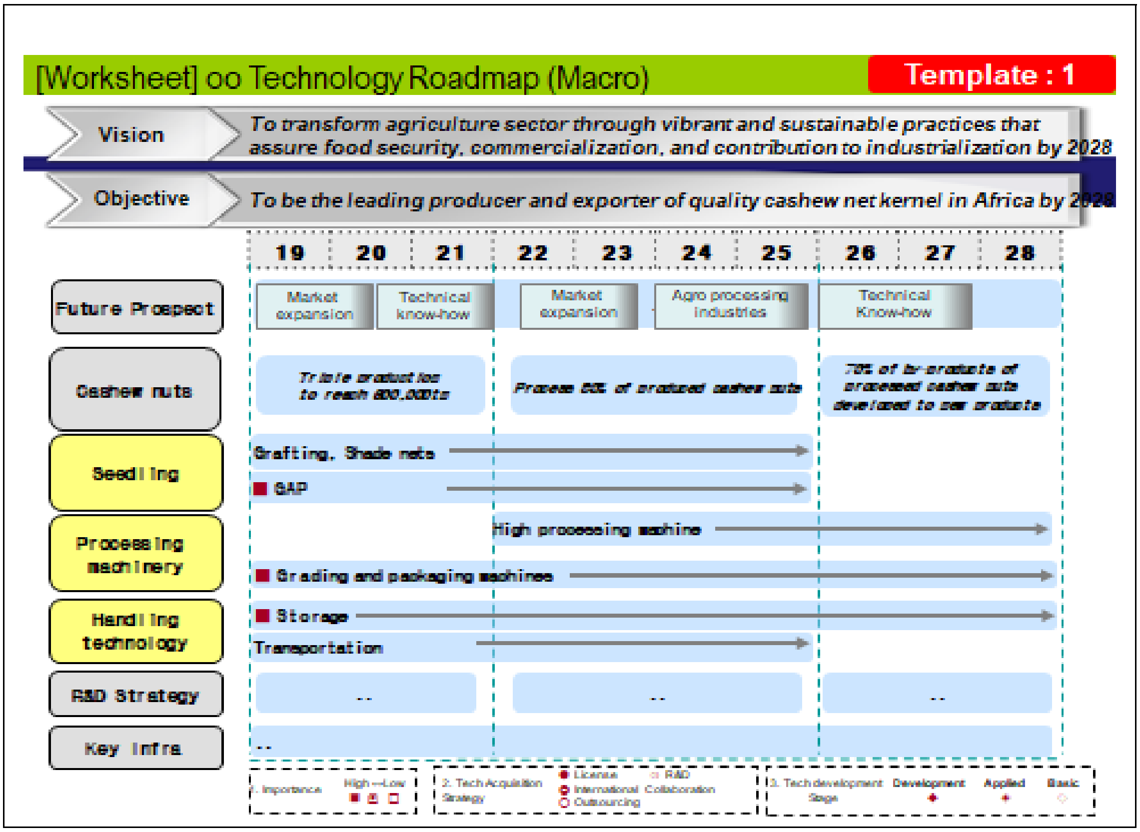
The key technologies were then assessed based on importance and feasibility criteria where T5 (Grading and packaging), T3 (GAP), and T6 (quality Storage facilities) were found to be critical in cashew subsector development. Strategic decision making on high-capacity processing machine and appropriate transport facilities is required. T4 (shed net) was less important and can consequently be rejected.

[Worksheet] Finalization of Key Technologies

Product : Cashew nut

Technology areas	Key Technologies	Final Target	Element Technologies (Level 2)		
			Short term (1 – 3 years)	Mid term (4 – 7 years)	Long term (8 – 10 years)
Seedling	Grafting		V	V	
	Shade nets		V	V	
	GAP		V	V	
Processing machinery	High capacity processing machinery			V	V
	Grading and packaging machines		V	V	V
Handling technology	Storage		V	V	V
	Transportation		V	V	

The priority technologies were summarized, and the time frame for their implementation was set. Majority of the technologies were short- and medium-term, whereas three were medium- and long-term. However, the target for each technology was not established.



At the end of the exercise, the macro road map was developed showing the vision, overall target, and key technologies. The target was based on market expansion, growing agro-processing industries, and technical know-how. Among the technologies identified, GAP, Grading and packaging, and quality storage facilities were deemed important. High-quality processing machine is a medium- to long-term investment. Quality storage facilities can be short-, medium-, and long-term through the rehabilitation of the existing warehouses (short-term) and construction of new ones (medium- to long-term). The methods of acquiring the identified technologies were not pointed out.

1.4 NTRM Pilot Project – Group 3 (Mining, Gold)

[Worksheet] Visioning

Vision	■ To sustainably develop the mining sector that optimally contributes to the national economy and well being of citizens using appropriate technology by 2030
Goals	■ Upgrading of the small scale mining sub-sector (identification, registration and licensing) ■ Establishment of local mineral markets ■ Monitoring and regulation of the large scale mining operations ■ Promotion of the local content in the mining sector ■ Simplifying and modernizing the licensing process using ICT ■ Updating geo-information via mapping and digitizing of old data

The slide shows the Vision and Objective to be achieved during the NTRM development. This Vision has been drawn from National Development Vision 2025 and Development Programs for the Country to become a middle-income country by 2025 with respect to the mining sector for Tanzania. The objectives and possible targets should be well-explained and understood before starting the NTRM development.

This Visioning was developed during the mineral NTRM capacity development workshops for the Mineral Sector in Tanzania. This drew focus to the need to transform the mineral sector to commercialization using improved technologies and available resources so as to achieve the industrial economy.

[Worksheet] Identification of Drivers

STEEP	Drivers
Social	• Poverty • Skilled manpower • Social conflicts
Technological	• Software and systems • Machinery technologies (local, tech. transfer), transporation technology • Production improvement • Resource allocation in R&D
Economic	• GDP trends • Balance of payment (export/import) • Taxation trends in the region • Energy availability
Environmental	• Land degradation • Water pollution • Deforestation
Political	• Change of law and regulations • Stability of government • Foreign trade regulation (resource nationalization, Kimberly process)

The slides analyzed the drivers for the development of NTRM for the mineral sector in Tanzania. These Drivers were analyzed by considering factors that can influence or support the technology roadmap development. For example, at present, mineral sector employment is becoming significant for the local communities where there are mineral development activities in their vicinity, which consider both skilled and semi-skilled citizens; thus socially, politically, environmentally, etc. influencing NTRM development one way or another.

[Worksheet] Opportunity/Threat (1)

Drivers	Opportunity/Threat
A. Poverty	Conflicts
B. Skilled manpower	Increased productivity
C. Social conflicts	Instability, Low production
D. Software and systems tech.	High production, value addition / Expensive
E. Machinery technology	Increased production
F. Production improvement	Increased production
G. Resource allocation in R&D	Improved production, knowledge growth
H. GDP trends	Increased purchase power
I. Balance of payment	More export, Increase foreign currency earnings
J. Taxation trends	More trading opportunities
K. Energy availability	More production, improved economy

[Worksheet] Opportunity/Threat(2)

Drivers	Opportunity/Threat
L. Land degradation	Low agricultural production
M. Water pollution	Increased diseases / High costs
N. Deforestation	Droughtness, climate changes
O. Change of law & regulations	Monitoring enhanced
P. Government stability	Peace, more production, increased economic growth
Q. Foreign trade regulations	Controlled foreign trading

The slide shows the opportunities and threats that can occur in the sector, since some factors can support the development of the mining sector and others may humper the mining activities. Thus, the team pointed out the opportunities and threats per driver as shown in the slide. After all drivers were identified, explanation followed as to how it can positively influence the economy of the society and country at large.

From that, it can be noted openly where to put more effort for the success of the technology and which area to avoid so that the technology can progress to foster the economy of the country.

[Worksheet] Identification of Products/Services

Opportunities & Threats	[Opportunity] • Increased productivity • Value addition • Knowledge growth • Increased purchasing power • More exports • Trading opportunities • Enhanced monitoring • Controlled foreign trading • Peace and stability [Treat] • Conflicts • Instability • Low production • Expensive • Low agricultural production • Increased diseases • Droughtness

The slide lists the opportunities and threats that should be considered before developing NTRM for the mineral sector. This can help in determining which one to avoid and which factor to consider for successful NTRM for the mineral sector. The team selected opportunities and threats as summarized in the slide, and it was noted that knowledge and skills should be used to examine carefully the threats and opportunities when identifying the technology for a certain product by adding value to the market.

[Worksheet] Identification of Products/Services

Products/ Services	• P1: Diamond • P2: Tanzanite • P3: Iron • P4: Coal • P5: Gold • P6: Graphite • P7. Limestone • P8. Kaolin

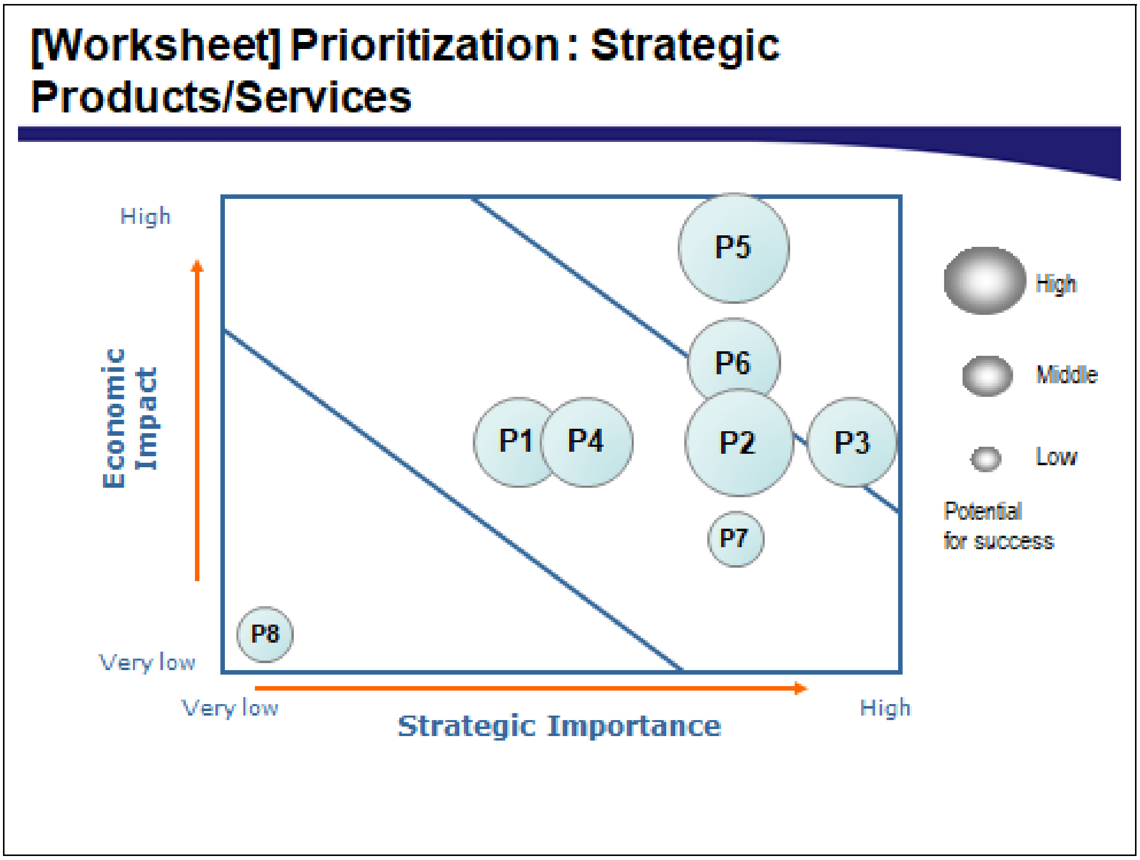
The slide shows the priority products that the country should start with in harvesting in the mineral sector. When developing an NTRM, the type of products to be achieved should be considered together with the country's priority. As we brainstormed during the training and after consulting the Government document, we ended up selecting the key products to be considered in the development of NTRM for Tanzania in the mineral sector, with 8 products selected from P1 to P8.

[Worksheet] Identification of Products/Services

	Economic Impact	Strategic Importance	Potential Success	Sum
P1	3	3	2	8
P2	3	4	3	10
P3	3	5	2	10
P4	3	3	2	8
P5	5	4	3	12
P6	4	4	2	10
P7	2	4	1	7
P8	1	1	1	3

The slide shows how to identify strategic mineral commodities in the analysis of strategic sectorial products and services that should be focused on, and 8 mineral commodities were selected considering their importance. The products were evaluated based on criteria such as economic impact, strategic importance, and potential for success. The important criteria being considered were expounded on as follows:

- Economic Importance: Job opportunity, increasing annual income.
- Strategic Significance: Export value, Import substitution, Input for industries
- Potential for Success: Reserve amount of deposit, Technology, and potential for private investment.



The slide shows the analysis of the economic impact of the product against the strategic importance of the product. The data is plotted on a graph to identify the relative significance of the commodities for future development. Accordingly, the precious mineral gold was identified as strategic mineral with high economic significance and great potential for development (P5, P6, and P3). For simplicity, the most Important and most Impactful was denoted by a big circle at the upper right corner like P1 and P4, but the less important and less impactful is denoted by a small circle like P8 plotted on the lower left corner.

[Worksheet] Identification of Technology drivers & targets

Product : Gold

Performance Dimension	Performance Target
Delivery (tons) per day	20% increase by 2030
Capacity (tons)	Reach 80% of the capacity by 2030
Cost per day	Lower cost by 5% by 2030
Quality	Increase the quality by 20% by 2030
...	...

Technology areas	Technology drivers & targets	Technology alternatives
...	...	...

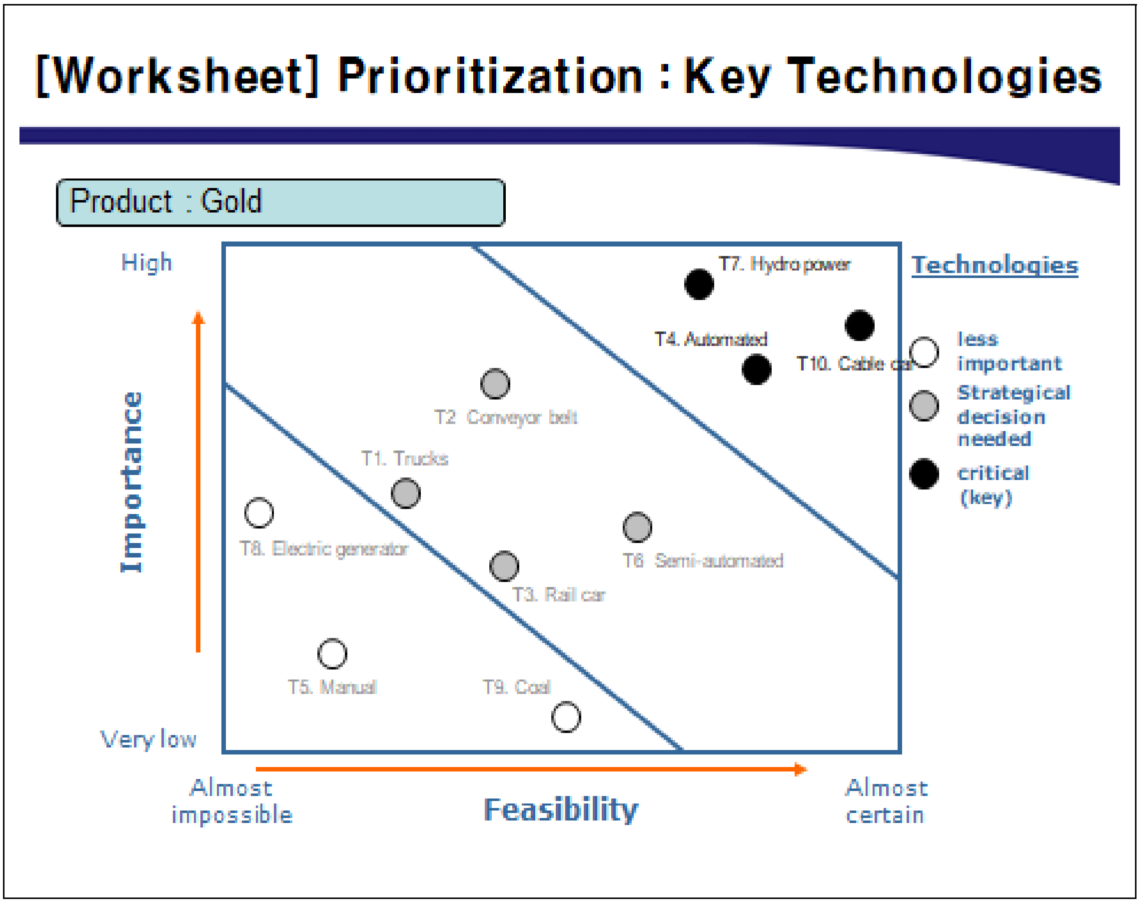
In order to have the Product and add value to it to meet market demand, it is important to Identify the appropriate technologies to be used. Thus, the technology is selected according to the expected finished products -- for example, from raw gold to well-designed chain or earrings or rings. The targets include increased production and increased premium grade of quality.

[Worksheet] Technology Tree

Product : Gold

Tech Area	Tech Level 1	Tech Level 2
Handling and transportation	T10: Cable car	
	T1: Trucks	
	T2: Conveyor belt	
	T3: Rail car	
Extraction	T4: Automation	
	T5: Manual	
	T6: Semi-automation	
Energy source	T7: Hydro power	
	T8: Electric generator	
	T9: Coal	

The slide shows the evolution of the technologies wherein selection was made by type of product and technologies alternative were already identified; hence the need for analysis of the technologies identified by considering their importance and feasibility. In doing so, it can be easy to decide which technology you choose first and which one is the last technology to be considered in the given product. Therefore, for each technology area chosen, the level of technologies was considered.



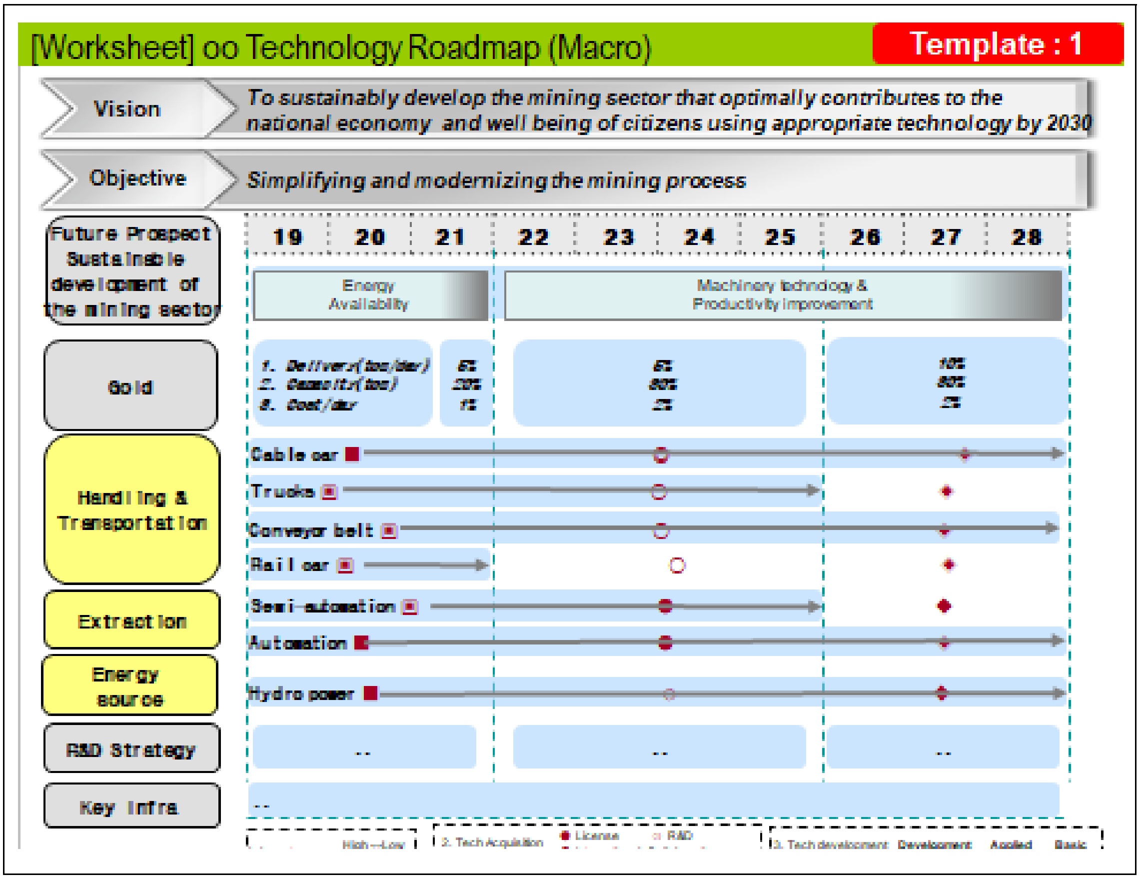
The strategic product for which the roadmap is to be developed was selected, taking into account economic impact and strategic importance. The most wanted technology will appear on the upper right part of the chart and denoted by a black circle (T7, T4, and T10), the strategic importance will appear in the middle (T2, T1, T3 and T6), and the less important technology will appear on the lower right, denoted by a non-shaded circle (T5, T9, and T8). Therefore, the best technologies to be considered by the NTRM development will be those shown in a black circle placed on the upper right.

[Worksheet] Finalization of Key Technologies

Product : Gold

Technology areas	Key Technologies	Final Target	Element Technologies (Level 2)		
			Short term (1~3 years)	Mid term (4~7 years)	Long term (8~10 years)
Handling and transportation	Cable car		V	V	V
	Trucks		V	V	
	Conveyor belt		V	V	V
	Rail car		V		
Extraction	Semi-automation		V	V	
	Automation		V	V	V
Energy source	Hydro power		V	V	V

The team prioritized technologies as summarized in the slide, and the time frame for their implementation was allocated accordingly. Most of the technologies are short- and medium-term, whereas four are long-term. In the group, however, the target for each technology was not considered.



The slide shows the Macro Technology Roadmap, where you start from visioning to the real product. The vision was developed by the team based on the country's development program and targets as given in the Vision 2025 for Tanzania. By doing so, the considered objectives and future prospect of product were established. Thus, the macro roadmap was developed showing the vision, overall target, and key technologies to achieve the product.

2. Local Capacity Development Workshop

2.1 Introduction

The fourth and final capacity development workshop held by STEPI-Korea for Tanzanian government officials was held in Zanzibar on August 19 ~ 22, 2019. The workshop on "Developing NTRM for Accelerated Industrialization" aimed at building the capability of participants from various sectors in developing the technology roadmap. The workshop covers tutorial exercises on NTRM development and presentations of the developed TRM by creating three groups: A, B, and C. At the end, Group "A" managed to develop TRM for the Textile (CTA) sector, Group "B" developed the TRM for the Mining sector, and Group "C" was for the Agriculture sector.

[Table 6-2] Research Team

Group	Name	Organization	Position
Group 1	Julius Elias Daud	TIRDO	Research Officer
	Gerald Majella Kafuku	COSTECH	Principal Research Officer
	Margareth Misheck Komba	MoEST	Assistant Director for STI
Group 2	Upendo Marko Mndeme	MoA	Senior Agriculture Officer
	Innocent John Umbulla	EPZA	Senior Officer
	Judith Francis Kadege	COSTECH	Principal Research Officer
Group 3	Rogers Alfayo Msuya	MoEST	Senior Research Officer, Coordinator RD&I
	Reuben Josephat Mdoe	MoM	Mineral Process Instructor
	Elias Aloyce Mshomba	EPZA	Senior ICT Officer

2.2 NTRM Tutorial Workshop-Group 1(CTA, Cotton)

[Worksheet] Visioning

Vision	■ To have integrated cotton, textile and apparel industry that effectively competes on the world market, using appropriate technology by 2030
Goals	■ Large scale production: seed cotton, cotton seed and apparel ■ Efficient production using modern technology: spinning, weaving, knitting, designing and clothing (garment production) ■ Products diversification (clothings, medical & sanitary products) ■ Develop strong R&D and other necessary infrastructure for CTA (cotton, textile and apparel) industry.

As the building blocks of any technology roadmap, the vison and goals should reflect the national priorities, strategies, policies, and/or master plan of the sector involved. The vision and goals were set based on the criteria in line with the National Development Vision (NDV, 2025) priorities and difficulties encountered by the CTA sector in export-oriented products. The setup also took advantage of the country's cotton wealth while addressing the shortage of cotton-based products in the local and international markets by considering the crucial products, market demand, technologies involved, and pathways for the next 10 years.

[Worksheet] Identification of Drivers

STEEP	Drivers
Social	• Life style • Improved living condition
Technological	• Semi-automated and automated technology • Efficient and appropriate technology
Economic	• Increased middle class community • Industrialized economy
Environmental	• Reduced environmental effects • Compliance to local and international requirements
Political	• National economic stability • Self reliance • Middle income vision

The textile (CTA) sector's technology drivers were identified based on social, technological, economic, environmental, and political (STEEP) analysis according to their future importance. The identification work started by assessing the current situation of Tanzania's Textile (CTA) sector in line with the existing technologies and the shortcomings of the current technology in the sector to indicate the existing gaps from the viewpoint of technologically advancement. These include, but are not limited to, the implication on quality, quantity, and efficiency of existing technologies as well as market demand. The identified drivers include: lifestyle, poverty reduction, and livelihood for the social aspect; efficient and semi-automated technologies for the technological aspect; industrialized economy and economic growth and development under the economic aspect; minimization of pollution and climate change for the environmental aspect; and self-reliance, national economic stability, and middle income for the political aspect.

[Worksheet] Opportunity/Threat

Drivers	Opportunity/Threat
A. Life Style	Increased purchasing power
B. Improved living condition	Increased market share
C. Semi-automated and automated technology	Few properly trained experts
D. Efficient and appropriate technology	Sustainable production
E. Increased middle class community	Guaranteed market
F. Industrialized economy	Increased competitiveness
G. Reduced environmental effects	Attracts tax incentives
H. Compliance to local and internat ional requirements	Increased local and international market base
I. National economy stability	Steady market
J. Self reliance	Low competition from importation
K. Middle income vision	Political will

The identified potential opportunities and threats from the set drivers reflect the vision and goals of the CTA sector. The group members assess and identify eleven (11) potential opportunities and threats that were found to have greater impact on CTA sector growth. The ten identified potential opportunities are key enhancement factors for the success of the sector. Note, however, that the identified threat was viewed as a competitive constraint of the sector.

[Worksheet] Identification of Products/Services

	Opportunities	Threats
Opportunities & Threats	• Increased purchase power • Increased market share • Sustainable production • Guaranteed market • Attract tax incentives • Increased local and international market base • Steady market • Low competition from importation • Political will	• Few properly trained experts • Increased competitiveness
Products/ Services	• Seed cotton • Cotton lint • Seed cotton cooking oil • Clothes • Animal feed • Cotton based medical and sanitary products • Fishing nets • Coffee filters • Fertilizer • Soap	

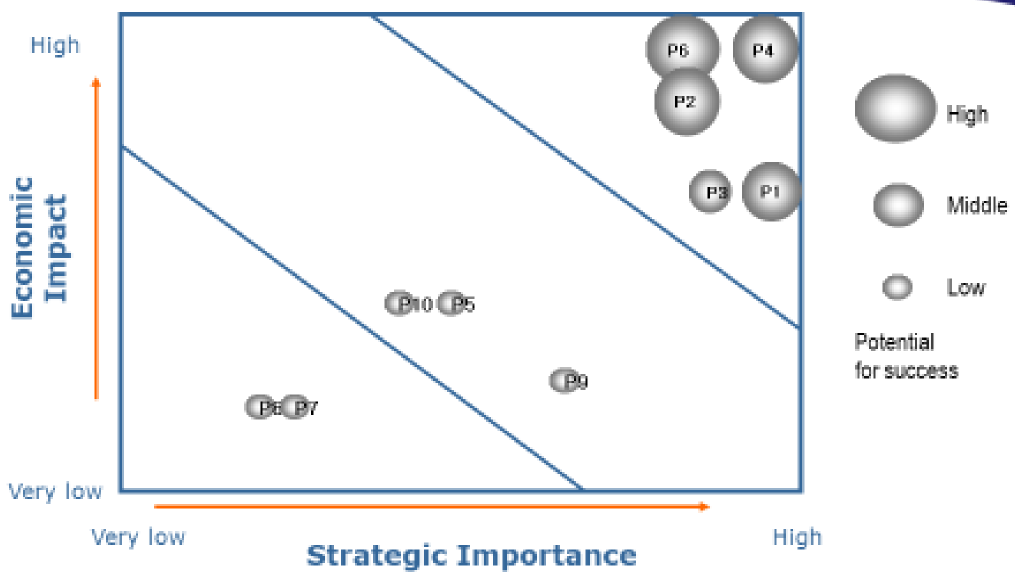
Ten products (clothes, cotton-based medical and sanitary products, cotton lint, cotton seeds' cooking oil, animal feeds, fishing nets, coffee filters, fertilizer, and soap) in line with the national priority were identified along the CTA value chain based on their economic importance, strategic importance, and potential success factors. These products are deemed to have significant impact on both farmers and national economy if the identified opportunities will be enhanced and their threats are mitigated.

[Worksheet] Evaluation of Products/Services

Products/Services	Code	Evaluation Criteria			Sum
		Economic Impact	Strategic Importance	Potential for Success	
Seed Cotton	P1	3	5	3	11
Cotton lint	P2	4	4	3	11
Seed cotton cooking oil	P3	3	4	2	9
Clothes	P4	5	5	3	13
Animal feed	P5	2	2	1	5
Cotton based medical and s anitary products	P6	5	4	3	12
Fishing nets	P7	1	1	1	3
Coffee filters	P8	1	1	1	3
Fertilizer	P9	1	3	1	5
Soap	P10	2	2	1	5

The ten products (coded P1-P10) were identified along the CTA value chain and evaluated against their economic impact, strategic importance, and potential success using point scale. The point scale score was considered to rate the products in order to identify strategic products. The products were rated on a 3-point scale for their potential success and on a 5-point scale under economic impact and strategic importance factors. The total score of each product was calculated by summing up the point scale score of each product based on economic impact, strategic importance, and potential success factors. Based on the findings, half of the identified products scored more than 50%. Therefore, investment in and/or application of improved technologies to the products that scored above 50% will contribute significantly to the growth of the national economy.

[Worksheet] Prioritization : Strategic Products/Services



The identification of strategic products to lead the CTA sector for the next ten years was done from among the major products. Economic impact scores for the ten coded products (P1-P10) were plotted against strategic importance scores while considering their potential for success to identify strategic products. Clothes (P4), cotton-based medical products (P6) cotton lint (P2), and seed cotton (P1) were the identified strategic products with high economic impact and strategic importance and high potential for success compared to other products along the CTA value chain. The rest occupied the lowest position, thereby failing to qualify as strategic products.

[Worksheet] Identification of Technology drivers & targets

Product : Cloth

Performance Dimension	Performance Target
Cost	50% cheaper by 2030
Design	60% meet customer demand by 2030
Quality	70% improved by 2030
Diversity/ Type	90% of age group sizes met by 2030
...	...

Technology alternatives
Electricity, Coal ,Gas, Diesel, Renewables, Cotton lint, Polyester, Fibre, Synthetic, Semi-automated, Automated
Manual, Computerized
Cotton lint, Polyster, Fibre, Synthetic, Semi-automated, Automated, Computerized
Manual, Computerized, Semi-automated, Automated

Cloth was the most strategic product selected along the CTA value chain. The performance dimensions with strong impact on cost, quality, types or diversity, and style of design for cloth were analyzed. Performance targets from which performance dimensions will be evaluated were set while considering 2030 as the timeline for accomplishing performance targets. Moreover, twenty-four (24) technologies alternatives were also identified in order to meet the stipulated performance targets.

[Worksheet] Technology Tree

Product : Cloth

Tech Area	Tech Level 1	Tech Level 2
Energy Sources	Electricity	
	Coal , Diesel	
	Renewables	
Raw Materials	Cotton Lint	
	Synthetics	
	Fibres	
Processing Technologies	Semi-Automated	
	Automated	

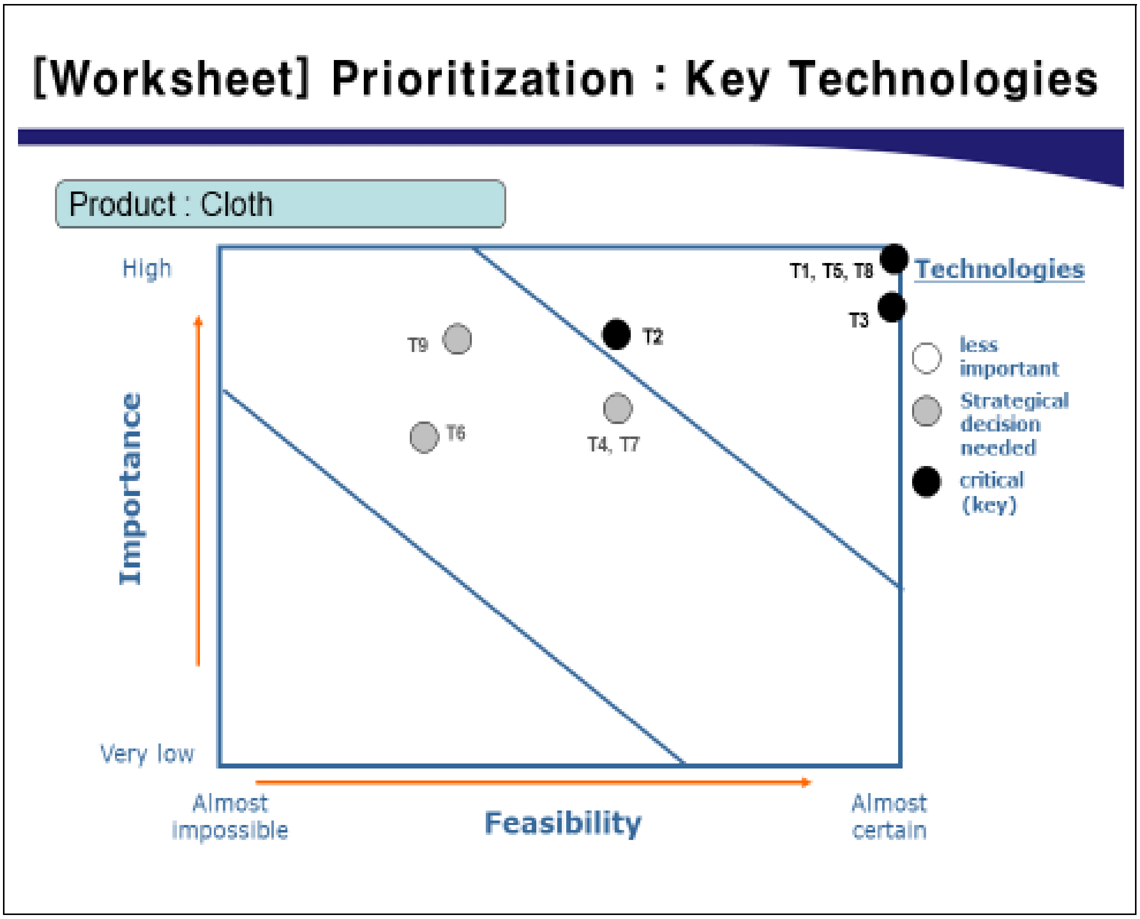
A technology tree was developed by grouping the identified technology alternatives based on their respective technology areas. Nine out of twenty-four identified alternative technologies were used to plot the technology tree based on criteria like feasibility and importance. Such shortlist is a key for making the right decision in technology selection, showing the direction of research and development of the future technology need of the CTA sector.

[Worksheet] Evaluation of Technologies

Product : Cloth

Technologies	Code	Evaluation Criteria		Sum
		Importance	Feasibility	
Electricity	T1	5	5	10
Coal	T2	4	3	7
Diesel	T3	4	5	9
Renewables	T4	3	3	6
Cotton Lint	T5	5	5	10
Synthetics	T6	3	2	5
Fibres	T7	3	3	6
Semi-automated	T8	5	5	10
Automated	T9	4	2	6

The nine selected alternative technologies were then evaluated based on their importance (economic, social and environmental benefits, scientific and technological opportunities) and feasibility (research and technology potential and potential to absorb economic and social benefits) and ranked on a 5-point scale. The 5-point scale was considered to rate all identified technologies and rank them based on their importance and feasibility. Eight out of nine technologies scored above 50% except synthetics, which scored 50 %. The highest score showed the possibility of success, whereas the lowest score indicates less importance or lack of feasibility.



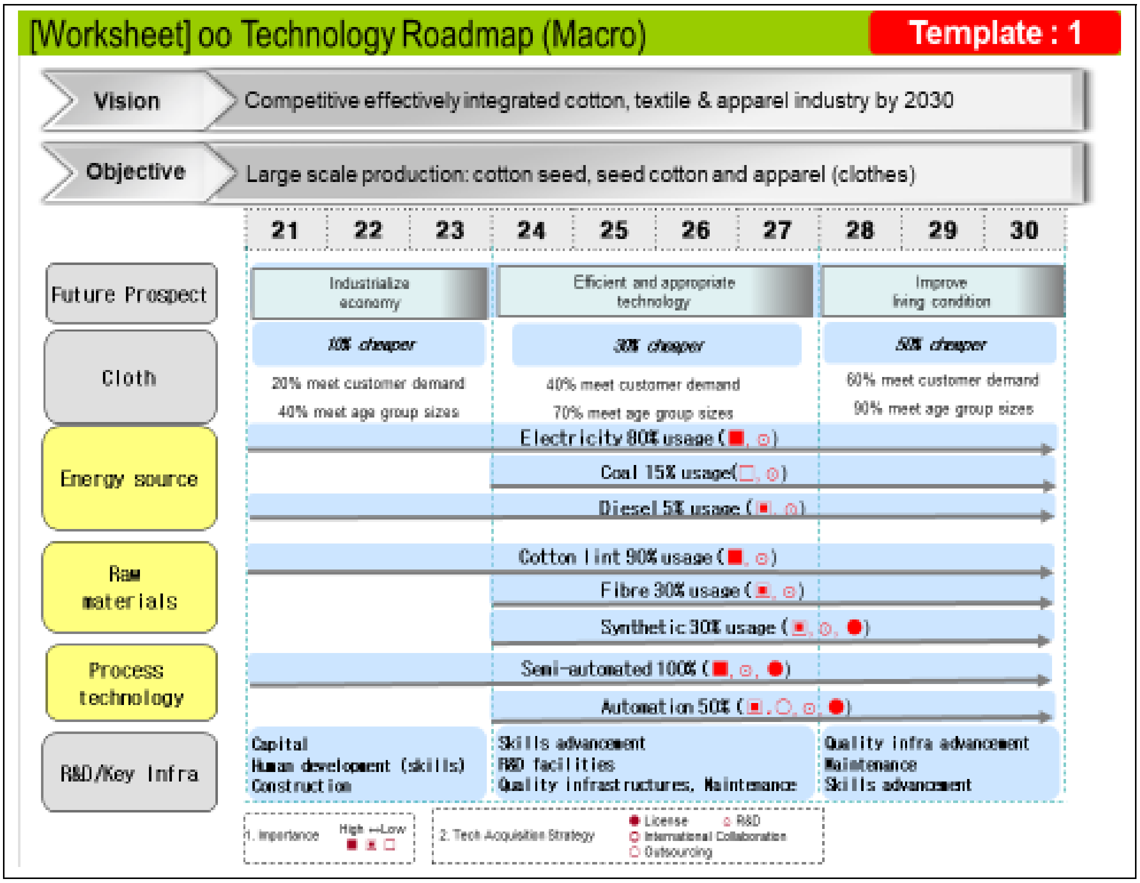
Key technologies were identified by plotting the calculated importance scores of the nine coded alternative technologies (T1-T9) versus feasibility scores on the graph. Electricity (T1), cotton lint (T5), Semi-automated (T8), Coal (T2), and Diesel (T3) were selected as key technologies based on their higher importance and feasibility compared to other identified technologies along the CTA value chain.

[Worksheet] Finalization of Key Technologies

Product : Cloth

Technology areas	Key Technologies	Final Target	Element Technologies (Level 2)		
			Short term (1~3 years)	Mid term (4~7 years)	Long term (8~10 years)
Energy sources	Electricity	80% usage	V	V	V
	Coal	15% usage		V	V
	Diesel	5% usage	V	V	V
Raw materials	Cotton lint	90% usage	V	V	V
	Fibres	30% usage		V	V
	Synthetics	30% usage		V	V
Process technology	Semi-automated	100% semi-automation	V	V	V
	Automated	50% automation		V	V

Key technologies were tabulated against their respective technology areas wherein the implementation timeframe together with the final target was set. Full deployment of all selected key technologies requires short-term, mid-term, or long-term implementation time to reach their final targets. Electricity, diesel, cotton lint, and semi-automated technologies will attain their final target in a short-term timeframe between 2021 and 2023.



The textile (CTA) TRM development process is finalized by this slide, which includes the vision and objective for the sector for the next 10 years. Future prospect, technology area, and their respective selected key technologies, R&D strategies, and key infrastructure reflect the vision and objective of CTA TRM. The selected key technologies were tabulated against their respective technology areas. Key technologies should meet the targeted 10% cost reduction target by 2021/23, 30% cost reduction by 2024/27, and 50% cost reduction by 2028/30. The customer demand should be met by 20% in 2021/23, 40% in 2024/27, and 60% in 2028/30. The demand of age group sizes should be met by 40% in 2021/23, 70% in 2024/27, and 90% in 2028/30.

2.3 NTRM Tutorial Workshop-Group 2 (Agriculture, Cashew Nut)

[Worksheet] Visioning

Vision	■ To transform agriculture sector through efficient, vibrant and sustainable practices & technology that assure food security, commercialization and contribution to industrialization by 2030
Goals	■ To produce market driven products ■ Enhance agriculture productivity and quality ■ Select, adopt, and develop appropriate technology structure, and system. ■ Support and invest R&D in relevant sectors ■ Enhance small holder farmer income

Agriculture contributes 28.2 percent of the national economy, employs 65.5% of the population, and contributes 65% of industrial raw materials. It also makes the country food self-sufficient. The country's vision is to commercialize Agriculture and transform the economy into middle-income through an industrial economy.

The National vision was one of the driving forces for the group to come up with the stipulated vision. The team discussed and analyzed the current situation in the Agriculture sector and projected the future of the sector. The goals developed focused on ensuring increased production, productivity, and quality of agricultural products through the application of improved technologies and enhanced Research and Development (R&D).

[Worksheet] Identification of Drivers (1/2)

STEEP	Drivers
Social	• Recognition and cultural practices • Livelihood / life style • Skills and knowledge • Employment • Customer preferences • Population growth & distribution
Technological	• Enhanced implements (farm, agro-process and market infrastructure) • Diverse agroecology • IP protection and technology transfer systems • Capacity & Capabilities in R&D • Technology know-how (human specialties) • Access to information

The group of trainees in roadmap development identified a number of drivers that influence or contribute to the growth of the sector. The growing population, Customer preference, and lifestyle have potential influence on agriculture transformation. Population growth provides labor force and market for agricultural products. The community's reluctance to change may have a negative impact on the intended goals, however.

The existence of the technologies along the value chain at the local, regional, and international levels provides an opportunity for the growth of the sector. Low use of improved technology and research findings contributes to a less efficient agriculture sector. Therefore, critical analysis of social and technological drivers is compulsory in roadmap development.

[Worksheet] Identification of Drivers (2/2)

STEEP	Drivers
Economic	• Budget allocation to agricultural sector • Modernization and market expansion • Growth of agro-processing industries • Trade (local/foreign)
Environmental	• Climate change • Suitable climate • Green technology / agro-, eco-friendly agricultural practices • Land suitability based on agro-ecological environment
Political	• Stable / political will • Promotion of investment (incentives) • International and regional cooperation • Legal and regulatory frameworks

The Government has committed to investing in agriculture through its annual budget allocations and other programs, and the growing market for agricultural products at local, regional, and International levels is one of the potential drivers for agricultural growth. The growing agro-processing industries provide both market for produced crops and employment opportunities; hence the income generation for the communities.

In Tanzania, there are seven agro-ecological zones with suitable climate that favor the production of variety of crops from those grown in temperate to tropical climates. Additionally, there is enough suitable land for agriculture expansion, and climate smart agriculture guidelines are in place to support the application of eco-friendly farming practices.

Political will and stability create a conducive environment for both public and private sector involvement in agriculture investments. The Nation has a strong relationship with regional and international organizations as well as reputable legal frameworks to provide the direction toward attaining the National vision and goals.

[Worksheet] Opportunity/Threat (1/3)

Drivers	Opportunity/Threat
A. Cultural practices	• Availability of existing indigenous technology • Slow adoption / adaptation to change
B. Skills / Knowledge	• Availability of agriculture experts, specialists, and institutions • Limited number of extension officers • Limited knowledge of new technologies
C. R&D	• Adequate R&D institutions • Inadequate research specialists in agriculture • Inadequate research facilities and resources
D. IPR System	• Creativity and innovation: availability of institutions and speci alists • Availability of legal frameworks on IP and TT protection • Absence of national IP policy
E. Technical know-how	• Availability of improved technologies (e.g. seeds, processing equipment) • Technology dissemination challenges to farmers • Limited product development technology (to be improved)
F. Agriculture implements	• Inadequate agriculture implements • Low purchasing power, Expensive equipment • Availability of agro-machinery manufacture

The team analyzed and selected some of the drivers that were found to have greater impact on the Agriculture Sector growth. These include, among others, cultural practices, skills and knowledge, R&D, IPR System, technical know-how, and agriculture implements.

Opportunities and threats for each driver were pointed out. Fully utilizing the identified opportunities and addressing the stipulated threats will improve the overall performance of the sector. Tanzania has a number of improved technologies and a reputable research institute (Tanzania Agricultural Research Institute - TARI) whose mandate is to develop and evaluate technologies. The availability of technologies and dissemination channels as well as Intellectual Property Right (IPR) system serves as another opportunity for success of the Agriculture Sector.

[Worksheet] Opportunity/Threat (2/3)

Drivers	Opportunity/Threat
G. Market expansion	• Population growth • Bilateral agreements and membership (e.g. EAC, SADC, AGOA, EBA) • Low quantity and quality products
H. Agro-processing industries	• Industrialization drive (FYDPII, second five year development plan) • Investors' interest in agro-processing industries • Less competitive products for international markets • Availability of agro-pressing industries
I. Climate change	• Draught tolerant crops • Agro ecological zone (availability of wide range)
J. Legal and regulatory frameworks	• Agricultural and other related sectors development programme (strong strategic direction) • Science and technology policy (which is outdated) • Agriculture policy
K. Political will	• National commitment in agriculture (high) • Availability of relevant institutions • Change of priority in government

Other drivers include market expansion, existence of agro-processing industries, strong legal and regulatory framework, political will, and climate changes. In Tanzania, there is a growing market as a result of population growth and change in preference. In addition, the geographical position provides the country with an opportunity to access other markets such as East African Countries, SADEC Countries and beyond.

Climate change has greater influence on agriculture as it dictates the type of crop to be grown and the crop management required. Note, however, that the incumbent Government's priorities vary with time, and the current Government is in favor of Agriculture Sector Development.

[Worksheet] Opportunity/Threat (3/3)

Drivers	Opportunity/Threat
L. International and regional cooperation	• Participation and membership to international and regional bodies • Markets / Knowledge sharing • Regional conflict
M. Agri Financing	• Availability of financial institutions • Non-friendly terms for agri-finance

The existence of International and Regional platforms provides a window for the sharing and exchange of technologies. They also provide market information and data, thereby serving as a tool for the planning of agricultural development programs. The country has entered into a bilateral agreement for business terms and technical cooperation at various levels.

Agricultural financing is done through banks and other financial institutions to support Farmers and service providers in acquiring agricultural inputs. In addition, due to the unfriendly terms and conditions of some financial institutions for farmers, the Government established the Agricultural Input Trust Fund (AGITF) to help the farmers access agricultural machines (tractor, combined harvester, weeder, etc.), seed, agro-chemical, and post-harvest technologies under minimal conditions and affordable interest rate.

[Worksheet] Identification of Products/Services

	Opportunities	Threats
Opportunities & Threats	• Availability of local and international agri-machinery manufacturers • Availability of diverse agri-ecological zones • Availability of financing institutions • Investors interest in agro processing industries	• Limited number of extension offices • Non-friendly terms & conditions for agri-financing • Inadequate researchers, researches and facilities in agri-sector • Low quality and quantity of products
Products/ Services	• P1: Cotton • P2: Coffee • P3: Cashew nuts • P4: Maize • P5: Cassava • P6: Sugarcane • P7. Sunflower • P8. Sisal • P9. Spices	

The group of experts identified a number of critical opportunities such as diverse agro-ecological zones with suitable agricultural land, presence of agro-machinery manufacturers at the local and international levels, presence of a number of investors interested in agro-processing industries, and reputable research Institutions as drivers for success. In addition, the presence of the Tanzania Agricultural Development Bank (TADB) and other financial institutions that are willing to support agricultural development is another window for agricultural growth.

Despite the existing opportunities, there are also threats that contribute significantly to the inefficiency of the sector. Some terms and conditions of financial institutions are not friendly to the low-income farmers as they lack collateral and other bank requirements. Climate change, small number of qualified personnel, and unsteadiness and low quality of products impair agricultural growth.

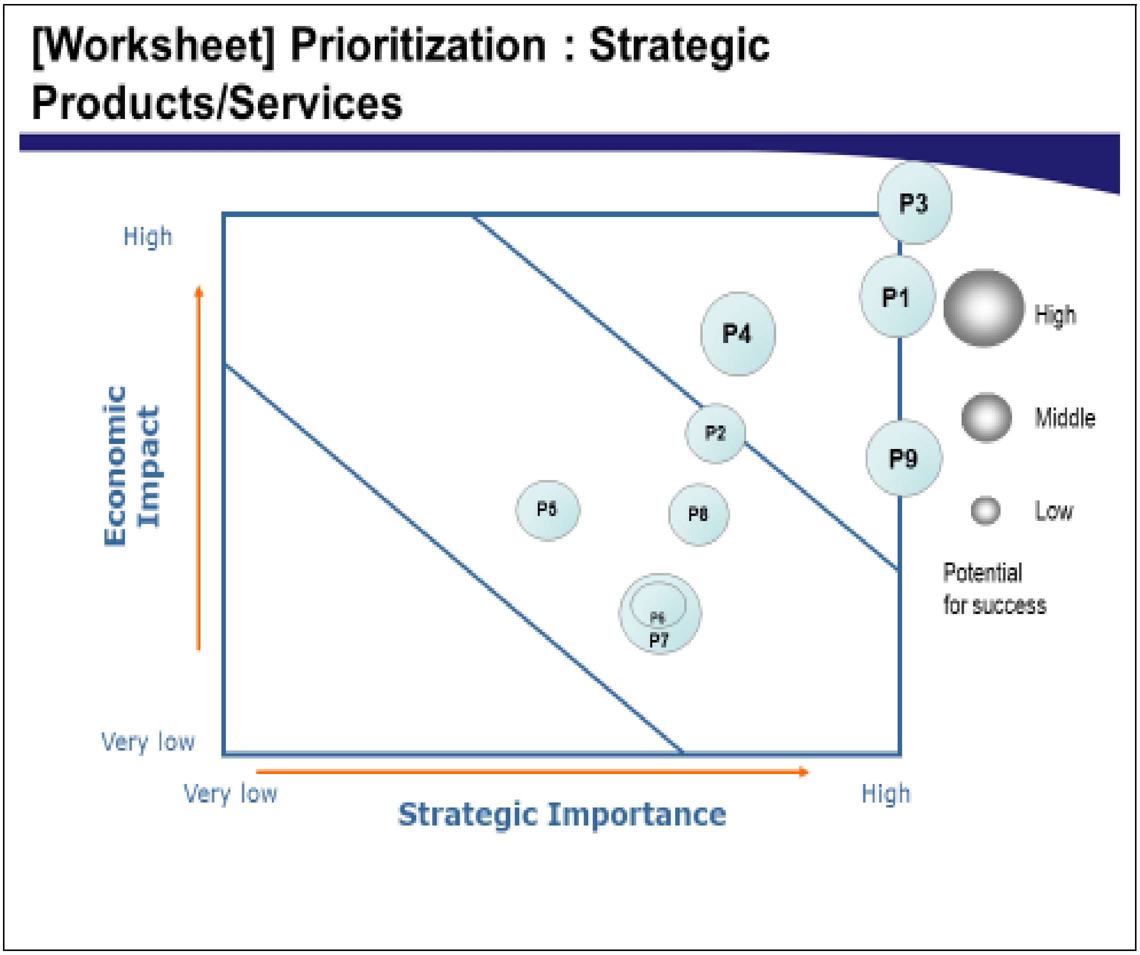
The team identified nine (9) products deemed to have significant impact on both farmers and National economy. The selection was done based on national priorities where the strategic cash crops (cotton, Sisal, Cashew, sugarcane) were listed. Oil seed crops (sunflower) were selected based on the country's focus to ensure self-sufficiency in edible oil, whereas Maize and Cassava were selected as important Food Security crops.

[Worksheet] Identification of Products/Services

Products/Services	Code	Evaluation Criteria			Sum
		Economic Impact	Strategic Importance	Potential for Success	
Cotton	P1	4	5	3	12
Coffee	P2	3	4	2	9
Cashew nuts	P3	5	5	3	13
Maize	P4	4	4	3	11
Cassava	P5	3	3	2	8
Sugarcane	P6	2	4	2	8
Sunflower	P7	2	4	3	9
Sisal	P8	3	4	2	9
Spices	P9	3	5	3	11

The team assessed the listed products and ranked them based on criteria such as economic impact, strategic importance, and potential for success, with cashew nut ranking highest followed by cotton, maize, and spices. Note, however, that all products scored above 50%. Thus, they should be given attention. Through the application of improved technologies, all products can perform and contribute significantly to the growth of the National economy.

The selected product (cashew) contributes significantly to foreign exchange earnings, having contributed about 529.6 million USD in 2017. It has great potential for success as it can be grown in a number of regions. Currently, there are 20 regions engaged in cashew production. Additionally, there are 54 improved varieties, 16 of which are polyclonal, 22 are hybrid, and 16 are dwarf varieties. The Tanzania Research Institute is a center of excellence for cashew research in Africa.



The strategic products were plotted on the chart for prioritization. Cashew was plotted on the highest zone followed by cotton, maize, and spices, whereas coffee, Sisal, cassava, sugarcane, and sunflower were plotted at the medium zone. None of the products fall under the lower zone or rejection zone. This means that all products are potential candidates for improvement during the agriculture sector transformation.

[Worksheet] Identification of Technology drivers & targets

Product : Cashew nuts

Performance Dimension	Performance Target	Technology alternatives
Production level	1,000,000 Tonnes per annum by 2030	• Seed production technology • Good agri practices • High yield variety • Biotic & Abiotic stress variety • Quality agri-inputs
Processed Quantity	Increase processed cashew nuts by 60% by 2030	• High quality and efficient processing machinery (manual, semi-automated, automated) • Grading & packaging machinery • High processing quality & shelf life variety
Storage Capacity	Increase and improve storage capacity to 100% of production level	• Drying facilities • Conducive quality warehouses • Storage system equipment (software etc) & infrastructure
Quality	Increase and improve premium raw cashew by 2030	• High yield variety • High processing quality & shelf life variety • Grading technology • Quality storage facility • Packaging technology

The team discussed and identified the performance dimension, target, and technologies associated with the identified performance dimension. The group assessed the current production and processing of cashew and set the target to be reached by 2030. Four performance dimensions -- production level, processed level, storage capacity, and quality products -- were identified. To meet the stipulated targets, a total of 16 technology alternatives were identified.

[Worksheet] Technology Tree

Product : Cashew nuts

Tech Area	Tech Level 1	Tech Level 2
1. Processing	- Drying technology - Peeling machine	
	- Roasting machine - Grading machine	
	- Packaging machine	
2. Production	- Seed production test - GAP technology	
	- Biotic and abiotic stress variety - High yield variety	
	- Quality agri-inputs - High quality processing variety	
3. Post harvest handling	- Improves warehouse - Storage system & equipment	
	- Grading technology - Quality packaging material	
	- Transport facilities	

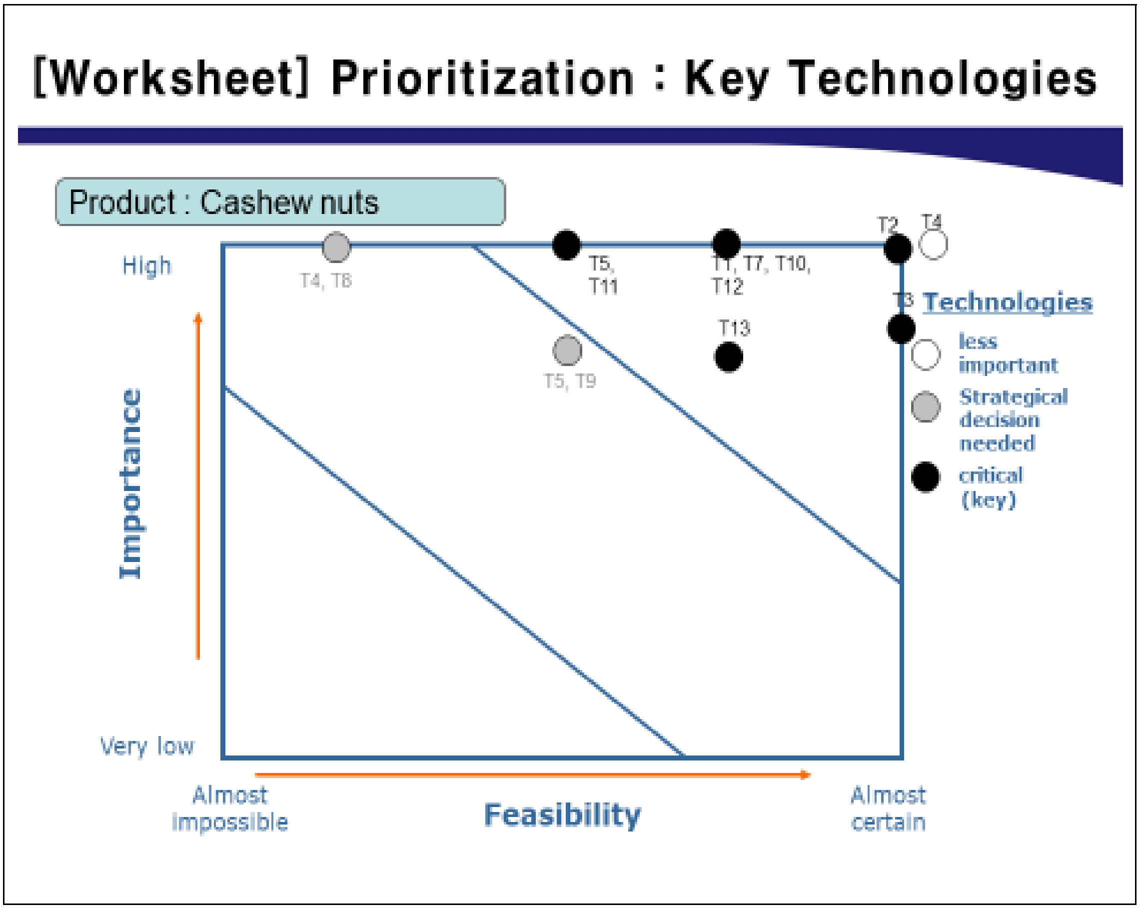
After the identification of the technology alternatives, the group identified the technology areas and plotted the technology tree. It was pointed out that each technology area requires a number of efficient and cost-effective technologies to enable performance. Six (6) technologies were plotted on the production side, five (5) technologies, in processing and five (5) technologies, in post-harvest handling.

[Worksheet] Evaluation of Technologies

Product : Cashew Nuts

Technologies	Code	Evaluation Criteria		Sum
		Importance	Feasibility	
Seed production technology	T1	5	4	9
GAP technology	T2	5	5	10
Biotic & Abiotic stress variety	T3	4	5	9
High yielding variety	T4	5	5	10
Quality agri inputs	T5	5	3	8
High quality processing variety	T6	4	3	7
Drying technology	T7	5	4	9
Peeling technology	T8	5	2	7
Roasting machine	T9	4	3	7
Packaging technology	T10	5	4	9
Improved warehouse	T11	5	3	8
Storage system	T12	5	4	9
Grading technology	T13	4	4	8
Appropriate transportation facilities	T14	5	2	7

The team evaluated the identified technologies based on importance and feasibility criteria by scoring them from 1 to 5. The highest score showed the possibility of success, whereas the lowest score indicated less importance or lack of feasibility. In general, out of 14 technologies, 10 scored above 50% except four (High processed quality varieties, Transport facilities, roasting machines, and peeling technologies), which scored less than 50 % (46.6%).



The team plotted the evaluated technologies based on the score in a chart, which showed that all the identified technologies need to be considered for the improvement of the cashew subsector to attain the stipulated goals and objectives. Improved technologies in the production and processing chain are very crucial for the sector.

To meet market demand and quality product, post-harvest handling technologies such as quality storage facilities and grading and packaging as well as appropriate transport facilities should not be overlooked.

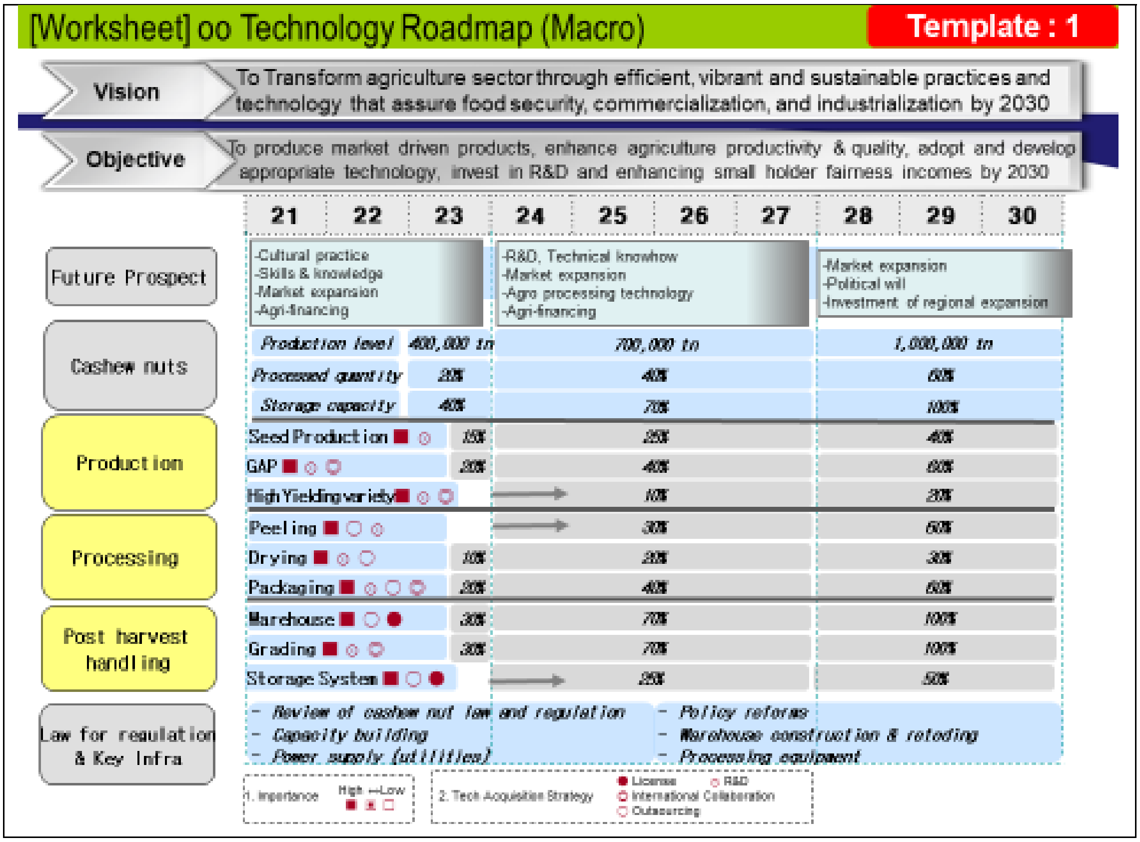
[Worksheet] Finalization of Key Technologies

Product : Cashew nuts

Technolog y areas	Key Technologies	Final Target (By 2030)	Element Technologies (Level 2)		
			Short term (1~3 years)	Mid term (4~7 years)	Long term (8~10 years)
Production	Seed production (T1)	40%	V	V	V
	GAP (T2)	60% of fairness	V	V	V
	High Yielding (T4)	20%		V	V
Processing	Peeling (T8)	60% of production		V	V
	Drying (T7)	30% improved	V	V	V
	Packaging (T10)	60% of production	V	V	V
Post Harvest Handling	Warehouse (T11)	100% of production	V	V	V
	Grading (T13)	100% of production	V	V	V
	Storage System (T12)	50% of production		V	V

During the finalization of the exercise, the team discussed and set the target for each technology to be reached at the specified time frame. They are: 60% of cashew growers (famers) equipped with Good Agricultural Practices (GAP); and increase in the number of high yielding varieties by 20% and seed production technology by 40%. The team agreed to give special attention to improved peeling technologies (T8), regardless of its score (46.6%), as the peeling of cashew is labor-intensive; thus, it is of crucial importance to invest in semi-automated and automated peeling machines. The target is to ensure that 60% of the produced cashews are peeled using improved technologies.

The stipulated target for warehouse technologies and grading aim at ensuring that everything that is produced is stored in appropriate storage facilities based on their respective grades. By 2030, the drying technologies for both raw cashew nut and processed nuts are expected to be improved by 30%.



The team summarized the process by preparing the macro roadmap for the cashew subsector. All technologies were found important, with seven to be acquired through research and development, two (2) to be licensed, five (5) outsourced, and four (4) through international collaboration. These indicate that no one method of acquiring technologies is satisfactory. Most of the identified technologies can be acquired through the involvement of a number of interested parties and various approaches.

The roadmap aims at increasing production to 1,000,000 tons annually, 60% of which is processed and 100% is stored in appropriate storage facilities. Nonetheless, the identified technologies and target need to be presented to a bigger group of professionals and users so as to evaluate and modify them according to the need of the stakeholders along the value chain and Nation at large.

2.4 NTRM Tutorial Workshop - Group 3 (Mining, gold)

[Worksheet] Visioning

Vision	■ To develop a sustainable mining sector through appropriate technology that optimally contributes the national economy and well being of citizens by 2030
Goals	■ Upgrading of the small scale mining sub-sector by the use of appropriate technology ■ Establishment of local mineral markets ■ Regulations and monitoring of the mining sector ■ Enhancing the licensing procedures ■ Updating geo-information via mapping and digitizing of old data ■ Promotion of local content in the mining sector (manpower, regulation, policy, etc) ■ Develop and promote research and development (R&D) in mining sector

While considering the Mining sector in Tanzania from small-scale to heavy miners, it is necessary to set the target and goal of having improved technologies and required resources for the social economic development of the country and achieving Vision 2025.

To attain the stated vision, through the Mineral sector, the proper technology should be used, but there should be NTRM for the sector, which should start by considering the vision and goal to be attained.

[Worksheet] Identification of Drivers

STEEP	Drivers
Social	• Poverty • Level of manpower skills • Social conflicts
Technological	• Software and systems • Machinery technologies (local tech. transfer), transfer of technology • Production improvement • Resource allocation in R&D
Economic	• GDP trends • Balance of payment (import/export) • Taxation trends in the region • Energy availability / Reliability
Environmental	• Land degradation • Pollution (water, noise, air) • Deforestation • Natural calamities
Political	• Change of law and regulations • Stability of government • Foreign trade regulation (resource nationalization, Kimberly Process)

Five drivers were identified for the mineral sector by brainstorming and shifting the national focus toward the sector. Through the given STEEP, the group identified 3 drivers under social, technological (4), economic (4), environmental (4), and political (3).

[Worksheet] Opportunity/Threat (1)

Drivers	Opportunity/Threat
A. Poverty	Conflicts
B. Level of manpower skills	Affects production (improved manpower skills)
C. Social conflicts	High stability
D. Software and systems	High production, value addition / Cost
E. Machinery technology	Affects production (increased production, low level of engine ers to operate machinery)
F. Production improvement	Increased production
G. Resource allocation in R&D	Improved production, knowledge growth / Cost
H. GDP trends	Affects purchasing power (Import/Export, increased purchas ing power)
I. Balance of payment	Affects foreign currency earnings
J. Taxation trends	Affects trading opportunities (increased competitiveness due to low tax / decreased competitiveness due to high tax)
K. Energy availability	Affects productions (low energy cost)

The slide shows the drivers against the identified opportunities and threats. In this case, the team analyzed the opportunity to be enhanced and the threat to be avoided to achieve the proper technology selection for NTRM. The threats and opportunities were critically discussed so as to make sure that the technology roadmap will be successfully developed.

[Worksheet] Identification of Products/Services

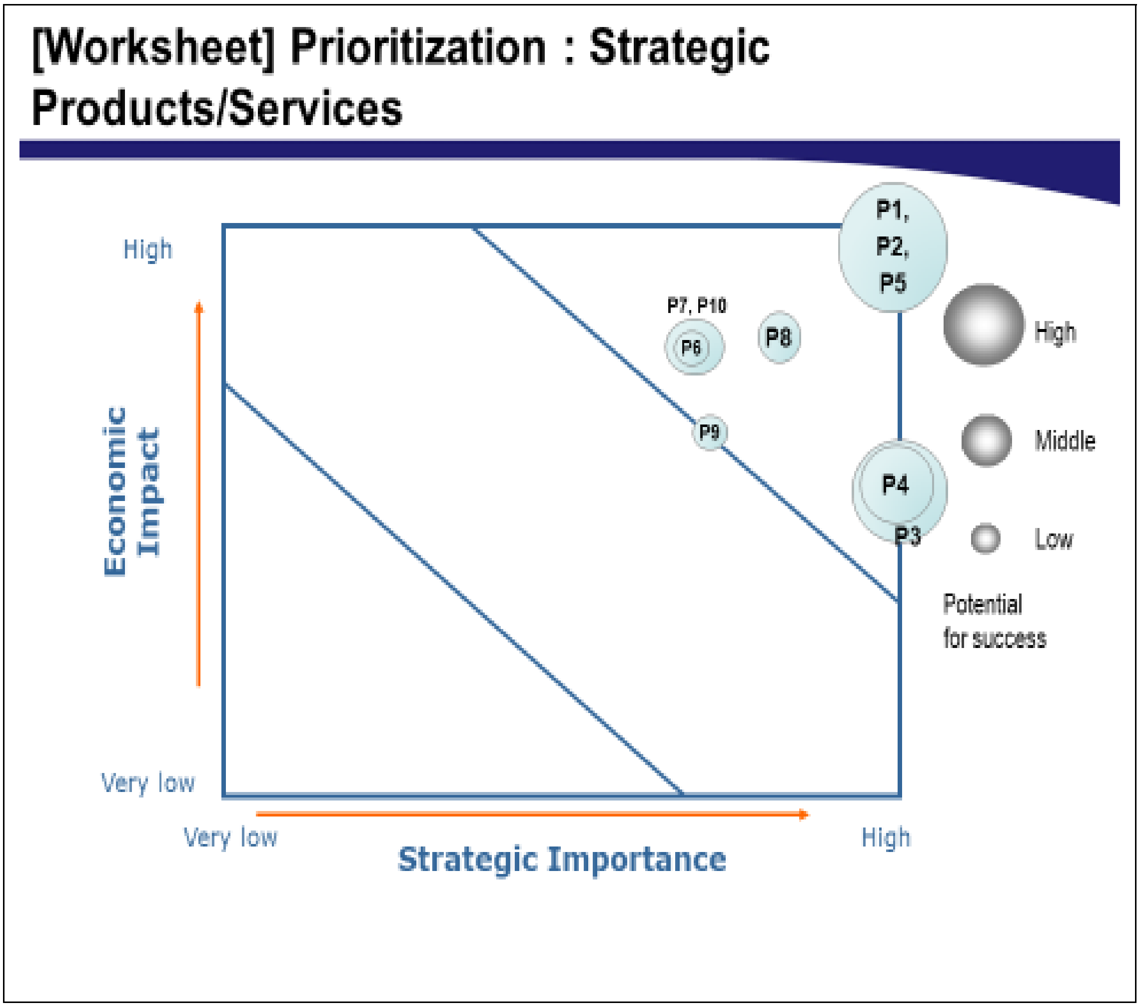
	Opportunities	Threats
Opportunities & Threats	• Increased manpower skills • Increased production • High stability • Improved products • Knowledge growth • Increased purchasing power • Increased competitiveness due to low tax • Low energy cost	• Conflicts • Low levels of engineers to operate machine
Products/ Services	1. Diamond 2. Tanzanite 3. Iron 4. Coal 5. Gold 6. Graphite 7. Limestone 8. Copper 9. Kaolin 10. Gypsum	

For the identified opportunities, the group analyzed them to check for threats. Two threats were identified, i.e., conflict from improved products and low level of engineers in operation of machines and knowledge growth. The slide also analyzed the products and services as far as the mineral sector is concerned.

[Worksheet] Identification of Products/Services

Products/Services	Code	Evaluation Criteria			Sum
		Economic Impact	Strategic Importance	Potential for Success	
Diamond	P1	5	5	3	13
Tanzanite	P2	5	5	3	13
Iron	P3	3	5	3	11
Coal	P4	3	5	2	10
Gold	P5	5	5	3	13
Graphite	P6	4	3	1	8
Limestone	P7	4	3	2	9
Copper	P8	4	4	2	10
Kaolin	P9	3	3	1	7
Gypsum	P10	4	3	2	9

The key products were categorized and analyzed by considering Economic Impact, Strategic Importance, and Potential for success. P1 (Diamond) ranked high, whereas P10 (Gypsum) ranked low.



For efficient use of available resources, it is necessary to select the products according to priority. P11, P2, P3, P4, and P5 were the priority products, and they need to be considered. Other strategic products had lower priority settings but were also important for the technology development for the sector.

[Worksheet] Identification of Technology drivers & targets

Product : Gold

Performance Dimension	Performance Target
Delivery per day (tons)	20% increase by 2030
Capacity (tons)	Reach 80% of the capacity by 2030
Cost per day	Lower cost by 5% by 2030
Quality	Increase by 20% by 2030
…	…

Technology alternatives
Handling and transportation - Cable car - Trucks - Conveyor - Rail car
Extraction - Flotation - Leaching - Gravity
Energy Source - Hydro power - Diesel generator - Coal / Gas / Uranium / Wind / Solar / Geothermal
- Smelter - Refinery

When the product is identified, the performance dimension was also identified with the target for each by 2030. The agreed-upon targets were set after some detailed discussion, including increase from 20 to 80% in performance in processing the products.

[Worksheet] Technology Tree

Product : Gold

Tech Area	Tech Level 1	Tech Level 2
Handling and transportation	Cable car	
	Trucks	
	Conveyor, Rail car	
Extraction	Flotation	
	Leaching	
	Gravity	
Energy source	Hydro power	
	Diesel generator	
	Coal / Gas / Wind / Uranium / Solar/ Geothermal	

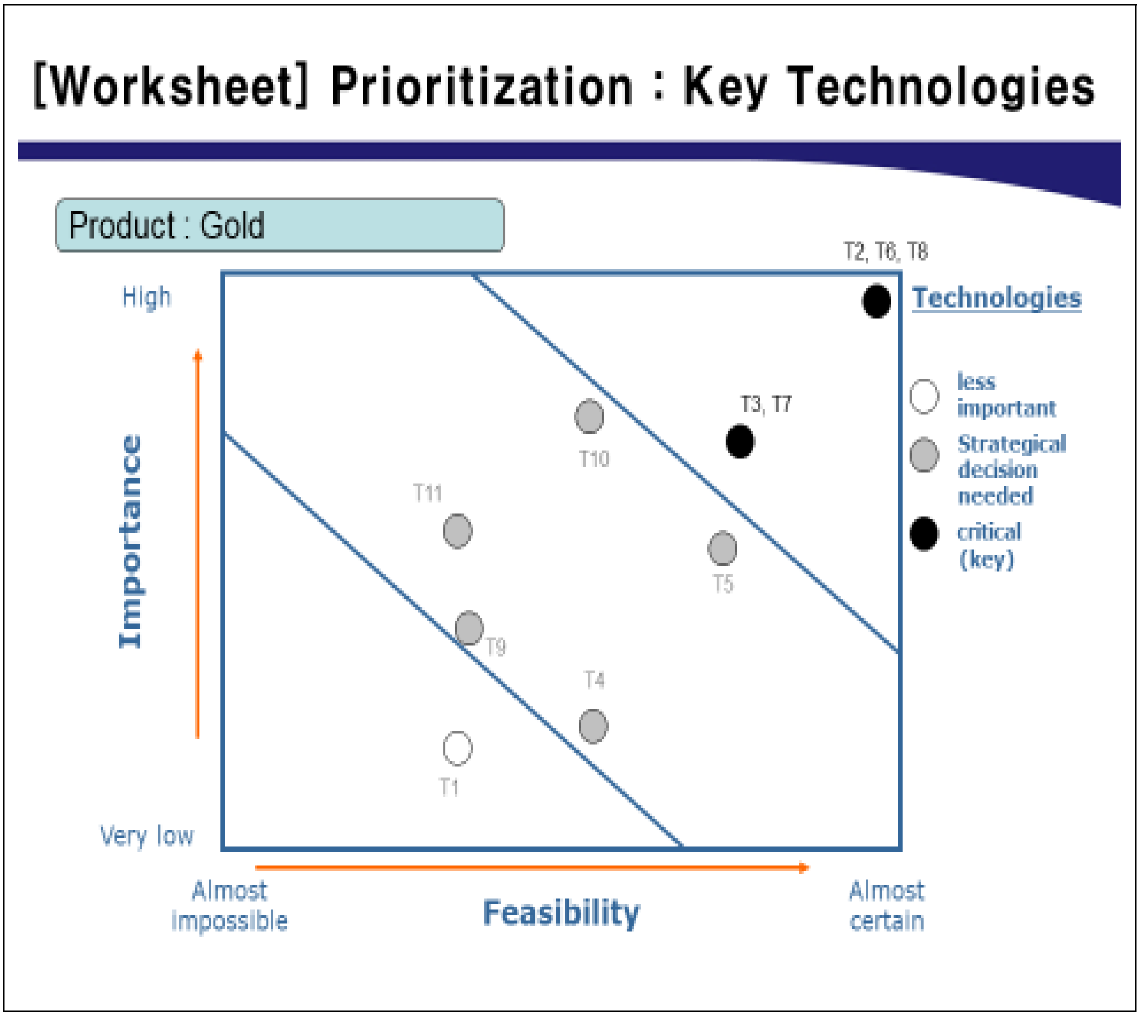
Technology areas were plotted with respect to the levels. The level given was the alternative technologies. The focus was to maximize production, so the most appropriate and efficient technology was chosen for the product.

[Worksheet] Evaluation of Technologies

Product : Gold

Technologies	Code	Evaluation Criteria		Sum
		Importance	Feasibility	
Cable car	T1	1	2	3
Trucks	T2	5	5	10
Conveyor	T3	4	4	8
Rail car	T4	1	3	4
Flotation	T5	3	4	7
Leaching	T6	5	5	10
Gravity	T7	4	4	8
Hydropower	T8	5	5	10
Diesel Generator	T9	2	2	4
Gas	T10	4	3	7
Coal	T11	3	2	5

The technology was evaluated according to the set criteria, which are importance and feasibility. Through the criteria of technologies, T2, T4, and T8 ranked high, whereas T1 ranked low.



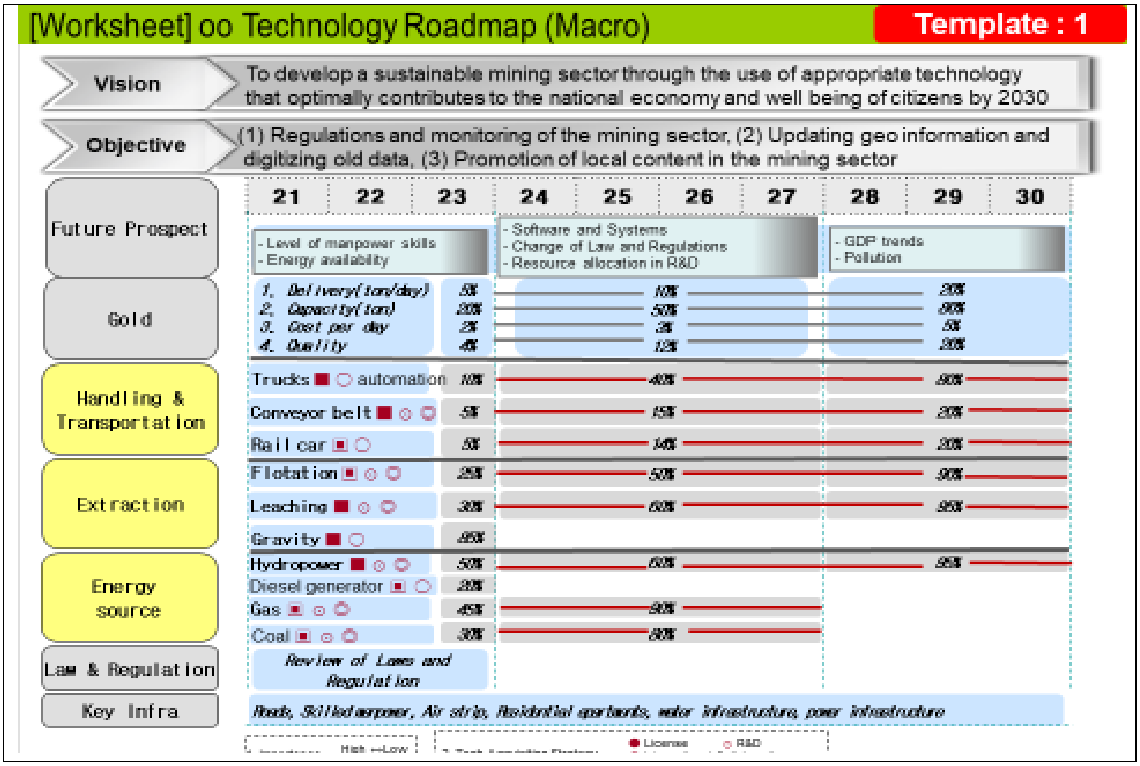
The key technologies were analyzed according to Feasibility against importance, and T2, T8, and T6 ranked very high compared to T3, which ranked low. The technology that ranked high is the one to be considered in the development of NTRM for the country, and the low-ranking one is normally left out.

[Worksheet] Finalization of Key Technologies

Product : Gold

Technology areas	Key Technologies	Final Target	Element Technologies (Level 2)		
			Short term (1~3 years)	Mid term (4~7 years)	Long term (8~10 years)
Handling and transportation	Trucks	80% automation	V	V	V
	Conveyor	20% improve	V	V	V
	Railcar	20% improve	V	V	V
Extraction	Flotation	Improved by 95%	V	V	V
	Leaching	Improved by 95%	V	V	V
	Gravity	Improved by 85%	V		
Energy source	Hydropower	Power generation increase to 90%	V	V	V
	Diesel generator	Improved efficiency by 20%	V		
	Gas	Power generation by 90%	V	V	
	Coal	Pollution reduction by 80%	V	V	

The technology areas were listed, and then key technologies were analyzed together with the final targets. The priority technologies were summarized based on short term, medium term, and long term. The time frame for their implementation was set, with many technologies falling under short- and medium-term.



The slide shows the guiding principle in developing the NTRM by following all basic steps starting from Visioning to Products while using the properly selected technology. The slide also shows the technology alternatives from which the best one can be selected and applied.

CHAPTER 7

Conclusion

Policy Consultation on Developing a National Technology Roadmap for Accelerated Industrialization

Chapter 7. Conclusion

1. Implications and Suggestions

The technology roadmap is defined as a dynamic set of various requirements such as technology, policy, law, finance, market, and organization as identified by stakeholders who are involved in its development.[1] The technology roadmap development will improve information sharing among stakeholders and enhance collaboration among them. Thus, the input and participation of stakeholders are crucial to the successful development of the technology roadmap. In the case of the business technology roadmap, different tiers of business organization need to participate and provide input. At the same time, there should be related experts who really know the products and understand the market very well among the participants.

In the case of the national technology roadmap, different parts of the government as well as various types of experts such as university professors, businesspersons, engineers, and policy specialists need to join the process and share the information. In order to promote the participation of government officials in the road mapping process, leaders such as ministers need to understand the importance of technology roadmap and commit to joining the technology road mapping process. This is because government officials have to allot time and effort in order to join the roadmapping job. If there is no commitment from the top level, government officials from different levels may not be able to commit their time and effort to the roadmapping task.

There is another reason why the government needs to join the national technology roadmap process. The government should play the role of roadmap maker as well as the audience of the roadmap outcome. This is because the national technology roadmap will be used in the decision making of financial resource allocation. Therefore, leaders from the central government as well as local government need to understand the purpose of the technology roadmap and make a commitment to the task. The government can then

[1] Energy Technology Roadmaps "A Guide to Development and Implementation" (2014), International Energy Agency

support the vision from the roadmapping process and utilize the outcome in planning the R&D program, allocating the financial resources, and setting up the policy.

In order to set up the technology roadmap, participants need to verify the social, technological, economic, environment, and political (STEEP) factors that can affect national development. Subsequently, they need to identify the target industry and specific products based on the STEEP analysis. At the same time, they need to specify the technology areas and technology drivers and targets. Finally, the milestone and timeline need to be specified.

In this context, three conclusions can be suggested on the K-Innovation ODA Program for Tanzania titled “Policy Consultation on Developing a National Technology Roadmap for Accelerating Industrialization.” First, there is a certain limit in participation from various stakeholders in the technology roadmapping process. There were several sessions held in Korea and Tanzania, but only a small group composed of approximately ten people could participate. Still, the participants made a strong commitment to the task and actively engaged in the task with strong passion. Nonetheless, more people with more diversified background are certainly needed.

Second, the ministry in charge of Science and Technology along with EPZA (Export Processing Zones Authority) provided solid commitment to the roadmapping task, so the participants could successfully finished the job and have a draft of technology roadmap in their hands. Still, there should be more people from different ministries in order to have a more complete national-level technology roadmap. If consensus can be established within the Tanzanian government regarding the importance and usefulness of technology roadmap, there should be more participants from different ministries as well as private sectors.

Finally, more industries need to be identified at the next step. In this project, only three industries were selected because this is a pilot project for a more developed technology roadmap. At the next level, at least more than 12 industries need to be identified and analyzed. In order to do this, strong government commitment and more participants from both public and private sectors are required once again. This would be the next step and the future project that both the Korean government and the Tanzanian government need to discuss together.

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<List of Institutions Consulted in Tanzania>

- Ministry of Industry and Trade (MIT)
- Tanzania Cotton Board (TCB)
- Textile Development Unit (TDU)
- Textile and Garment Manufacturers' Association of Tanzania (TEGAMAT)
- Tanzania Agricultural Research Institute (TARI)

<List of People Consulted in Tanzania>

- Mr. Marco Mtungi (Tanzania Cotton Board)
- Mr. Prosper Kulaya (Tanzania Cotton Board)
- Ms. Edith Mtinda (Textile Development Unit)
- Dr. Kapinga Fotunus (Tanzania Agricultural Research Institute)

APPENDICES

Appendix 1: Top Twenty Cashew Nut-Producing Countries in 2017

No.	COUNTRY	PRODUCTION (TONS)	% SHARE
0	WORLD	3,971,046	
1	Vietnam	863,060	21.734
2	India	745,000	18.76
3	Cote d'Ivoire	711000	17.90
4	Philippines	222,541	5.60
5	Tanzania	164,245	4.14
6	Guinea Bissau	155,953	3.93
7	Benin	151,836	3.82
8	Mozambique	139,000	3.500
9	Brazil	133,465	3.36
10	Indonesia	131,685	3.32
11	Burkina Faso	99,027	2.49
12	Nigeria	98,253	2.47
13	Ghana	90,000	2.27
14	Mali	72,009	1.81
15	Colombia	70,000	1.76
16	Kenya	25,625	0.65
17	Thailand	22,057	0.56
18	Malaysia	15,661	0.39
19	Guinea	10,500	0.26
20	Togo	9,897	0.25

Source: FAOSTAT

Appendix 2a: Factories Built by Private Companies

NO.	FACTORY NAME	INSTALLED CAPACITY (MT)	CURRENT PROCESSING CAPACITY	LOCATION	EMPLOYMENT (WORKERS)	CURRENT STATUS
1.	Perfect Cashew Kernels	2,000	1500	Masasi	250	Operating. Given 91 Mt by the Government
2	Kitama Farmers' group	1,000	750	Tandahimba	350	Operating. Given 105Mt by the Government
3.	Demros Women Group	300	0 No raw material	Tanga City	0	Not Operating
4.	Southern Jumbo Ltd.	1,000	1000	Dar es Salaam	230	Not Operating
5	Mbagala Cashew Project	2,000	2000	Dar es Salaam	400	Not Operating
6	HAWTE Investment Ltd.	2,100	810	Mtwara Municipal	490	Operating. Up to now, given 553 Mt by the Government
7	Masasi High Quality	300	0	Masasi Town	0	Not processing
8.	Kibaha AMCOS	300	0	Kibaha Town	0	Not Operating
9.	Amama Farms Ltd.	2000	1600	Tandahimba	250	Operating. Up to now, given 900Mt by the Government. Started 23 April, 2019

NO.	FACTORY NAME	INSTALLED CAPACITY (MT)	CURRENT PROCESSING CAPACITY	LOCATION	EMPLOYMENT (WORKERS)	CURRENT STATUS
10	Yalin Cashew Processing Factory	10,000	0	Mtwara Municipal	18	Not Operating. Trial production using machine from China.
11	Biotani Group Ltd.	2,000	2000	Dar es Salaam - Mbagala	230	Operating (Processing own cashew later to process Government Cashew)
12	Mkemi Agrix Ltd.	3,000	1,000	Pwani - Mkuranga	200	Operating, processing own cashew.
13	Fuzzy International Ltd.	2,000	2,000	Pwani - Mkuranga	200	Operating – Given Government cashew, 30Mt
14	Tan Cashew	3,000	1500	Mtwara	250 (When operational)	Not Operating
	TOTAL	**20,000**	**6,500**			

Appendix 2b: List of Government Privatized Factories

NO.	OLD NAME	CURRENT NAME	CURRENT INSTALLED CAPACITY (MT)	CURRENT PROCESSING CAPACITY (MT)	LOCATION	EMPLOYMENT (workers)	CURRENT STATUS
1.	Likombe Cashew Nut Factory	Micronix System Ltd.	2,500	2000	Mtwara	325	Operating
2.	Newala II Cashew Nut Factory	Micronix System Ltd.	4,000	2500	Newala	650	Operating, given Government korosho
3.	Tunduru Cashew Nut Factory	Korosho Africa Ltd.	4,500	3500	Tunduru	825	Operating, given Government korosho
4.	Mtwara Cashew Factory	CC 2005 Ltd.	2,000	0 (no RCN)	Mtwara	14	Service Provider for SIDO Mtwara small groups
5.	Mtama Cashew Nut Factory	Lindi Farmers Ltd.	2,000	0	Lindi Rural	0	Not Operating
6.	Newala I Cashew Factory	Agro Focus (T) Ltd.	10,000	0	Newala	0	Not Operating
7.	Lindi Cashew Factory	Bucco Investment Holdings (T)	12,000	400 0	Lindi Municipal	250	Operating. Given 150 Mt of korosho by the Government
8.	Kibaha Cashew Factory	Safa Petroleum and Minerals Co. Ltd. (TERRA Cashew Factory)	5,000	300 0	Kibaha Town	150	Operating

NO.	OLD NAME	CURRENT NAME	CURRENT INSTALLED CAPACITY (MT)	CURRENT PROCESSING CAPACITY (MT)	LOCATION	EMPLOYMENT (workers)	CURRENT STATUS
9.	Nachingwea Cashew Factory	Lindi Farmers	5,000	0	Nachingwea	0	Not Operating
10.	Masasi Cashew Factory	Bucco Investment Holdings (T)	0	0	Masasi Town	0	Not working
11.	Tanita I Cashew Factory	Export Trading Ltd.	0	0	Dar es Salaam	0	Warehouse
12.	Tanita II Cashew Factory	Mohamed Enterprises Ltd.	0	0	Dar es Salaam	0	Not Operating
	TOTAL		**47,000**	**7000**			

Appendix. 3: Cashew Cropping Calendar for Major Producing Countries

	JAN.	FEB.	MAR.	APR.	MAY	JUN.	JUL.	AUG.	SEP.	OCT.	NOV.	DEC.
Tanzania	x	x								x	x	x
Vietnam			x	x	x	x						
Nigeria		x	x	x	x							
India			x	x	x	x						
Ivory Coast			x	x	x	x						
Guinea Bisau					x	x	x	x	x			
Benin		x	x	x	x							
Indonesia	x	x						x	x	x	x	x

Appendix 4: Regional Production Trend from Season 2008/2009 to 2017/2018

S/N	REGION	2008/09	2009/10	2010/11	2011/12	2012/13	2013/14	2014/15	2015/16	2016/17	2017/18
1	MTWARA	50,396.223	49,830.954	85,137.858	108,947.351	87,285.730	86,070.288	131,781.799	106,940.222	172,771.109	195,481.941
2	LINDI	21,588.263	15,866.866	21,970.688	32,523.569	34,968.240	26,647.980	46,101.346	27,936.734	61,739.316	76,018.616
3	PWANI	3,005.535	4,022.285	9,007.785	9,801.198	2,400.000	5,217.376	9,627.314	8,081.727	13,257.861	20,091.360
4	DAR ES SALAAM	542.857	173.812	-	4.284	-	64.355	1,814.200	256.591	138.670	228.376
5	RUVUMA	2,706.831	3,478.055	4,135.042	7,271.969	3,178.340	11,572.793	8,262.229	10,528.716	15,429.183	20,912.569
6	TANGA	575.245	1,974.692	883.600	170.480	107.053	339.986	164.080	1,151.655	682.058	709.524
7	SINGIDA	-	-	-	-	-	-	-	-	207.748	-
8	NJOMBE	-	20.000	-	-	-	-	6.533	5.000	80.429	81.000
9	MBEYA	-	-	-	-	8.110	211.000	175.000	344.000	358.000	303.000
10	MOROGORO	-	-	-	-	-	-	-	-	102.590	-
11	IRINGA	-	40.000	-	-	16.220	422.000	363.066	349.000	450.360	-
12	DODOMA	-	-	-	-	-	-	-	-	19.521	-
TOTAL		**78,814.954**	**75,406.664**	**121,134.973**	**158,718.851**	**127,963.693**	**130,545.778**	**198,295.567**	**155,593.645**	**265,236.845**	**313,826.386**